

OPERATIONS PROCESS MANAGEMENT

MANAGEMENT ENGINEERING

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1. OPERATIONS AND PROCESSES

1.1 OPERATIONS AND PROCESS MANAGEMENT

- ^{ACTIVITY} **OPERATIONS AND PROCESS MANAGEMENT** = activity of managing the resources and processes that produce products and services, for internal and external customers.
- ^{EXAMINATION} **OPERATIONS MANAGEMENT** = examines how the operations function produces products and services for external customers.
- **PROCESS** = arrangement of resources and activities that transform inputs into outputs that satisfy internal or external customer needs.
- **OPERATIONS** = transform inputs into outputs. All operations are composed of processes.

All orgs have operations → all orgs produce PRODUCTS / SERVICES / MIX

Survive in a turbulent environment

→ Steady improvement in its performance

ALL ENTERPRISES HAVE OPERATIONS

The operations function is the part of the organization that produces products or services. All organizations have operations because all organizations produce some mix of products and services. It's a central and important activity for any organization.

An effective operations function has the potential to survive in a turbulent environment and the ability to maintain a steady improvement in its performance.

Big differences WITHIN ECONOMIC SECTORS than BETWEEN THEM → how operations are managed related to the market pos.

But not all operations are the same.

All have operations function that produce the thing that their customers are willing to pay for. But there are often bigger differences within economic sectors than between them. The main difference between how their operations activities need to be managed is more closely related to the market position that they occupy.

The implication of this is important. It means that the surface appearance of a business and its economic sector are less important to the way its operations should be managed than the intrinsic characteristics of what it is trying to achieve, such as the volume of its output, the variety of the products and services it needs to produce and, above all, how it is trying to compete in its market.

Ex: hotels produce accommodation services, financial services invest and sell money.

• Appearance of a business and its economic sector LESS IMPORTANT to the way its operations should be managed than the intrinsic characteristics of what it is trying to achieve.

Expansion on all the functions and orgs + interactions in S.N.



HYSTORY OF OPERATIONS MANAGEMENT

The scope of operations management has expanded from focusing on processes in manufacturing organizations to include service organizations, processes in other functions (such as marketing, finance and HRM) and how operations interact with each other in supply networks and chains.

1. PRODUCTION

Initially operations management was associated with the manufacturing sector.

2. OPERATIONS MANAGEMENT → including the non core activities

2 trends:

PURCHASING
DELIVERY
AFTER SALES, ...

- Many of the ideas, approaches and techniques traditionally used in the manufacturing sector could be equally applicable in the much large service sector
- Expand the scope of production in manufacturing companies to include non core, but important production – related processes such as purchasing, physical distribution, after sale services, product development and so on

3. OPERATIONS AND PROCESS MANAGEMENT

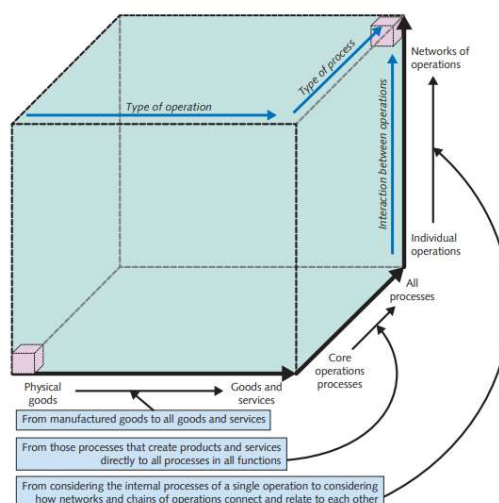
Extend the management to include the whole org.
→ Every function is organized in order to achieve its objectives

The shift in the scope of the subject to include the whole organization. In any business, the production of products and services is not confined to the operations function. Every part of any business achieves its objectives by organizing its resources such as people, information systems, buildings and equipment into individual processes

4. OPERATIONS AND SUPPLY CHAIN MANAGEMENT

The acceptance that operations outside the conventional boundary of an enterprise would have an important effect on how it performed. The suppliers of goods and services to an operation could have as much impact on its activities as how it managed its internal resources and processes. Moreover, the boundary of an enterprise is not fixed or immutable, it is determined by decisions made by the operation itself.

Outside the boundary of the org. → Managing suppliers to understand how internal resources and processes should be managed.



1.2 PROCESS PERSPECTIVE

COLLECTION OF INTERRELATED PROCESSES.
↓
Some contribute directly to the production
others will be in other functions

Does the organization take a process perspective?

There are many valid approaches to describing organizations. The process perspective is a particularly valuable one.

PROCESS PERSPECTIVE means understanding that all parts of the business can be seen as processes, and that all processes can be managed using operations management principles.

The process perspective analyzes businesses as a collection of interrelated processes. Some of these processes will be within the operations function and will contribute directly to the production of its products and services; other processes will be in the other functions of the business, but will still need managing using similar principles to those within the operations function

- **ORGANIZATION** = collection of processes, interconnecting and hopefully all contributing to fulfilling its strategic aims. IMP.

use of operations management aims to improve any processes in any part of the orgs.

All functions have operations with their own collection of processes
↑
PLANS
POLICIES
REPORTS
SERVICES

OPM RELEVANT TO ALL PARTS OF BUSINESS

All part of the business manages processes, so all parts of the business have an operations role and need to understand operations management.

Each function will have its technical knowledge, but also will have an operations role that entails using its processes to produce plans, policies, reports and services. So, all functions are operations with their own collection of processes. COMPORTA

So every manager, in all parts of an organization, is to some extent an operations manager. They all should want to give good service to their internal customers and want to do this efficiently.

Operations management must be relevant for all functions, units and groups within the organization. And the concepts, approaches and techniques of operations management can be used to help improve any process in any part of the organization.

Ex: in marketing it includes the market expertise needed for designing and shaping marketing plans, but also the marketing function has processes with inputs of market information, staff, computers and so on. Its staff transforms the information into outputs such as marketing plans, advertising campaigns.

We can define operations and process management with:

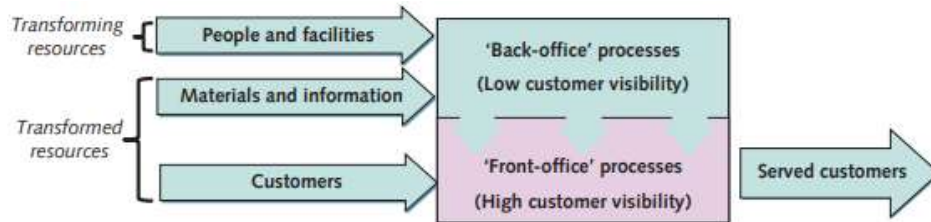
- Input – transformation – output model
- Network connection

take in a set of input resources ^{transformed into output} ^{FRONT-OFFICE} ^{BACK-OFFICE}
 do the transforming.

ALL PROCESSES

INPUT – TRANSFORMATION – OUTPUT MODEL

All **processes** and **operations** transform inputs into outputs. The transformation process model is used to describe the nature of processes and operations. **Processes** and **operations** take in a set of **input resources**, some of which are transformed into output and some do the transforming.



INPUT RESOURCES

- **TRANSFORMED**
The resources that are **changed** in some way **within a process**. They are **materials**, **information** or **customers**.
- **TRANSFORMING**
All processes have transforming resources such as **facilities** (equipment, technology, buildings) and **people** (who operate, maintain, plan and manage the operation).
The exact nature of both facilities and people will differ between processes.

TRANSFORMATION

- **FRONT OFFICE**
The front office parts of an operations are those **processes that interact with transform customers**. Customer have **high visibility** of the operation's processes.
- **BACK OFFICE**
The back-office operations are the processes that have **little or no direct contact with customers**, but perform the activities that support the front office in some way. Low customer visibility because customers receive their outputs without seeing inside the operations.

OUTPUT RESOURCES ^{For whole org., customers need to be served are external (provide revenues)}
^{For individual processes within an org., internal customers.}

All processes and operations produce products or service and **those outputs are intended to serve the requirements of customers**. For the operations function of a business as a whole, the customers who need to be served are likely to be **external customers**; for the **individual processes within an operation**, customers are usually (although not always) **internal customers**.

External customers provide an organization directly with its reason for being. They are the ones who **provide the revenue**. Serving **internal customers** is, in the longer term, just as important. **Poor service to internal customers** will eventually **result in poor service to external customers**, or **excessive costs to the organization**, or both.



SIPOC ANALYSIS

The input-transformation-output model can be the basis of a useful first step in understanding and improving processes. This is called **SIPOC = suppliers, inputs, process, outputs and customers**.

It's a method of formalizing **what a process requires** for its inputs, **where those inputs come from**, **what the process entails**, **what the process produces** as outputs and **for whom the outputs are intended**. The advantage of such an analysis is that it helps all those involved in the process to understand what it involves and where it fits within the business.

S suppliers	I inputs	P process	O output	C customers
Functional head Project manager Recruitment agency	Recruitment request Long-term resourcing plan Labour market reports	Agree essential and desirable characteristics ↓ Authorise budget and timescale ↓ Formally contract recruitment agency ↓ Agree shortlist of candidates ↓ Make offer and negotiate conditions ↓ Confirm start date and arrange induction	Qualified and experienced new staff member Completed staff records Payment authorisation	Functional head Project manager New recruit

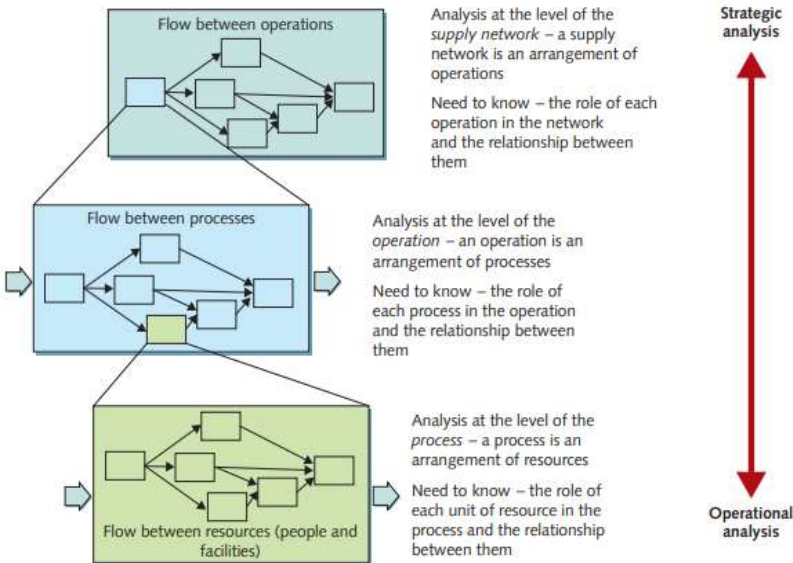
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# NETWORK CONNECTION – 3 LEVELS OF ANALYSIS

The concept of input – transformation – output process is the first idea that defines operations and process management. The second is that **all operations processes are connected in networks**.

The **network concept** can be applied at 3 levels of analysis:

1. The level of **operation** itself
2. The level of the **supply network**
3. The level of individual **processes**



Role of each process + relationship



## ▪ PP AT LEVEL OF OPERATION

Analysis at the level of the operation = arrangement of processes.

Need to know the role of each process in the operation and the relationship between them.

The operation is itself an input – transformation – output system, which transforms various inputs to produce a range of different products or services.

## PRODUCTS AND SERVICES

- PRODUCT = physical and tangible thing
- SERVICE = activity that usually involves interaction with a customer or something representing the customer. The resources that carry out the services may be tangible, but not the service they provide. It's more correct that the distinction between products and services is not confined to tangibility, but include other characteristics IHIP:

### IHIP CHARACTERISTICS

- I. Intangibility = services are not physical items
- H. Heterogeneity = services are difficult to standardize because each time a service is delivered it will be different because the needs and behaviors of customers will vary
- I. Inseparability = production and consumption are simultaneous
- P. Perishability = services cannot be stored because they have a very short shelf life

Relatively few operations produce either purely products or purely services. The output from most types of operation blends the characteristics of pure goods and pure services.



OPERATIONS = SERVICE PROVIDERS who produce physical products as a means of serving customers  
↳ Service-dominant logic

## SERVICES AND PRODUCTS ARE MERGING AND CHANGING

Increasingly the distinction between services and products is seen as not particularly useful. Some authorities see the **essential purpose of all businesses**, and therefore all operations, as being to serve customers. Therefore, they argue, **all operations are service providers** who **may produce physical products as a means of serving their customers**.

This idea, that all operations should be seen as offering value propositions through service, is called **service-dominant logic**.

→ TRANSPARENT INTERACTION

**CO CREATION** = CUSTOMERS INVOLVEMENT in stages of the creation process.

If all operations can be seen as producing services, and services act on customers or their surrogates, then **customers play a central part in how they derive value from an operation's outputs**. It's important the idea of customer involvement.

**Co-creation** is all types of customer involvement, engagement, participation or collaboration with the operation, in stages of the creation process.

This establishes the importance of a full two-way and **transparent interaction between an operation and its customers**.

changing the focus of what an op. produces from its products to those services that the products enable

**SERVITIZATION**

This means **any strategy that seeks to change the way in which product functionality is delivered to its markets**. It means **changing the focus of what an operation produces from its products to those services that the product enables**.

Advantages:

- It can generate extra revenues
- **Services** tend to provide a more stable source of revenue because they are resistant to the economic cycle that drive capital equipment purchases
- **Can improve relationship with customers**: offering services provides an opportunity to understand customer needs

Ex: Rolls-Royce, the aero-engine manufacturer, instead of selling its engines to its customers (mainly airlines), it offered them the option of **paying a price per engine flying or operating hour**, the price being **dependent on the amount of support needed** (for example the number of aircraft, amount of overhaul and repair) **and the performance guarantees required** (for example engine availability, turnaround time of repair, reliability). They originally called this service, **power by the hour**.

## ▪ PP AT LEVEL OF SUPPLY NETWORK

Analysis at the level of the supply network = **arrangement of operations**.

Need to **know** the **role of each operation** in the network and the **relationship** between them.

Any individual operation is part of a greater network of operations. The collection of interconnected operations is the supply network.

### SUPPLY NETWORK

- Complex: operations may have **many customers** and **suppliers** who themselves have large number of customers and suppliers. Also, the relationship between operations in the supply network can be subtle
- Boundaries of the supply network can be very wide
- Supply network are always changing

Thinking about operations management in a supply network context is important. Questions:

- Does my operation contribute to the supply network as a whole?
- Are we good customer to our suppliers?
- Are we good supplier to our customers?

## ▪ PP AT LEVEL OF INDIVIDUAL PROCESSES

Analysis at the level of the process = arrangement of resources.

Need to know the role of each unit of resource in the process and the relationship between them.

We can also view the operation as a network of individual processes that interact with each other, with each process being at the same time an internal supplier and an internal customer for other processes.

The internal customer concept provides a model to analyze the internal activities of an operation.

## LINE OF SIGHT WITHIN PROCESS NETWORKS

Thinking about operations as networks of processes also gives a further benefit.

If there is a clear line of sight forward through to the external customers who will eventually be affected by their performance, customers will have a better chance of seeing how they contribute to the final value.

Similarly, a clear line of sight backwards helps to understand what is required from internal (and eventually external) suppliers. Certainly, a failure to understand how internal process chains work will reduce the effectiveness of the whole operation.

## END TO END BUSINESS PROCESSES

Processes are defined by how the organization chooses to draw process boundaries. We can choose and define the boundaries between processes, the activities that they perform and the resources that they use.

Usually, processes are defined by the type of activity engaged in (example: assembly processes, product design processes).

One way of redefining the boundaries and responsibilities of processes is to consider the end-to-end set of activities that satisfy defined customer needs. This approach is closely identified with the business process engineering (or re-engineering) movement (examined in Chapter 12). It calls for a radical rethink of process design that will probably involve taking activities and resources out of different functions and placing them together to meet customer needs.

- Processes defined by the type of activities they engage in
- $\Rightarrow$   $\Rightarrow$  by their end-to-end set of activities that satisfy cust. needs

## 1.3 STRATEGIC IMPACT OF OPM

One of the biggest mistakes a business can make is to confuse operations with operational.

- OPERATIONAL = is the opposite of strategic: it means detailed, short-term, day-to-day
- OPERATIONS = set of resources that produce products and services

Operations can be treated both at an operational and a strategic level. Operations function is strategic because it represents the bulk of its assets and the majority of its people.

## OPERATIONS PERFORMANCE

Operations performance can be assessed at 3 different levels:

- Wider: how it impacts on long term societal issue
- Individual firm: how it support strategic aim
- Operational level: how well they serve their customer and improve efficiency

■ SOCIETAL LEVEL <sup>It is concerned about → Corp. + team societal issues</sup>  
→ Stakeholders → 3BL

At its broadest operations can be judge on how it impacts on long-term societal issue.

Operations decisions affect a whole variety of stakeholders. But although each of these groups will be interested in an operation's performance, they have very different views of which aspect of performance is important.

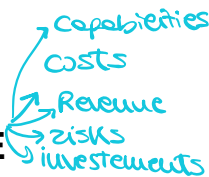
TRIPLE BOTTOM LINE (3BL) PEOPLE, PLANET, PROFIT

Operations should judge themselves on the triple bottom line principle of people, planet and profit. It's an idea that organizations should measure themselves not just on the traditional economic profit that they generate for their owners, but also on the impact their operations have on society (in the sense of communities and individually in terms of their employees) and the ecological impact on the environment.

A sustainable business is one that creates an acceptable profit for its owners but minimizes the damage to the environment and enhances the existence of the people with whom it has contact. In other words, it balances economic, environmental and societal interests.

A sustainable business is more likely to remain successful in the long term than one that focuses on economic goals alone.

## ▪ LEVEL OF ENTERPRISE



At the level of the individual firm or enterprise, it can be judged on how it supports (or not) their strategic aim. It is judged on how operations and process management impact on the business's costs, revenue, risk, investment and capabilities.

All operations should be expected to contribute to their enterprise by controlling costs, increasing revenue, reducing risk, making investment more effective and growing long term capabilities.

### • COST

Operations and processes management can reduce the costs of producing products and services by being efficient. The more productive the operation is at transforming inputs into outputs, the lower will be the cost of producing a unit of output.

### • REVENUE



It can increase revenue by increasing customer satisfaction through quality, service and innovation. Existing customers are more likely to be retained and new customers are more likely to be attracted to products and services if they are error-free and appropriately designed, if the operation is fast and responsive in meeting their needs and keeping its delivery promises, and if an operation can be flexible.

### • RISK

It can reduce the risk of operational failure, because well-designed and well-run operations should be less likely to fail and if fails, should be able to recover faster and with less disruption.

### • INVESTMENT

It can ensure effective investment (capital employed) to produce its products and services. Effective and efficient operations can reduce costs and increase revenues.

### • CAPABILITY

It can build capabilities that will form the basis for future innovation by building a solid base of operations skills and knowledge within the business. Every time an operation produces a product or a service it has the opportunity to accumulate knowledge about how that product or service is best produced. This accumulation of knowledge should be used as a basis for learning and improvement.

## ▪ LEVEL OF OPERATIONS' PROCESS

At its more operational level, the individual processes within an operation can be judged by how well they serve their (internal and external) customers and improve the efficiency with which they do it.

There are 5 aspects of operations and process performance, that will affect customer satisfaction and business competitiveness:

- **QUALITY** *doing things right*  
*error-free*  
*right services*  
Doing things right, providing error-free goods and services that are fit for their purpose
- **SPEED**  
Doing things fast, minimizing the time between a customer asking for goods and services and the customer receiving them in full
- **DEPENDABILITY**  
Doing things on-time, keeping the delivery promises that have been made to customers
- **FLEXIBILITY** *NEW PRODUCTS*  
*INDIVIDUAL TREATMENTS*  
*COPE WITH UNEXPECTED CIRCUMSTANCES.*  
Changing what you do or how you do it, the ability to vary or adapt the operation's activities to cope with unexpected circumstances or to give customers individual treatment, or to introduce new products or services.
- **COST**  
Doing things cheaply, producing goods and services at a cost that enables them to be priced appropriately for the market while still allowing a return to the organization

But do not think that these 'operational' aspects of performance have little strategic significance. On the contrary, they all contribute to the success of the organization as a whole.



## 1.4 REFLECT OPERATING CIRCUMSTANCES

Are processes managed to reflect their operating circumstances?

Because all processes differ in some way, all processes need to be managed differently. Processes differ in 4 characteristics: the 4Vs. The way in which processes need to be managed is influenced by volume, variety, variation in demand and visibility.

- **VOLUME**

HIGH volume: high degree of repeatability, and because tasks are repeated frequently it often makes sense for staff to specialize the tasks they perform. This allows the systemization of activities, with standard procedures. Also, because tasks are systemized and repeated, it is often worthwhile developing specialized technology that gives higher processing efficiencies.

By contrast, low-volume processes with less repetition cannot specialize to the same degree. Staff is likely to perform a wide range of tasks.

- **VARIETY**

HIGH variety of products and services: must engage in a wide range of different activities, changing relatively frequently between each activity. They must also contain a wide range of skills and technology. That also imply a relatively wide range of inputs to the process and the additional complexity of matching customer requirements to appropriate products or services. High variety processes are invariably more complex and costly than low-variety ones.

- **VARIATION IN DEMAND**

CONSTANT DEMAND: Processes are generally easier to manage when they only must cope with predictably constant demand, because resources can be geared to a level that is just capable of meeting demand, and all activities can be planned. Also, because processes with lower variation do not need any extra safety capacity and can be planned, they will generally have lower costs than those with higher variation.

By contrast, when demand is variable and/or unpredictable, resources will have to be adjusted over time. Worse still, when demand is unpredictable, extra resources will have to be designed into the process to provide a capacity cushion that can absorb unexpected demand.

- **VISIBILITY**

Process visibility indicates how much of the process is experienced directly by customers, or how much the process is exposed to its customers.

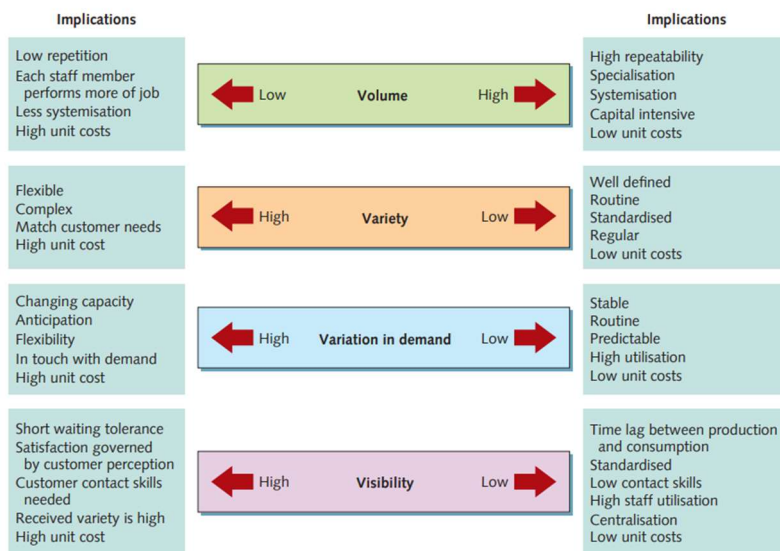
Generally, processes that act directly on customers (such as retail processes or healthcare processes) will have more of their activities visible to their customers than those who act on materials and information.

Also, because the customer interface needs managing, high-visibility processes tend to have higher costs than low-visibility processes.

## 4V PROCESSES

The 4 dimensions have implications for:

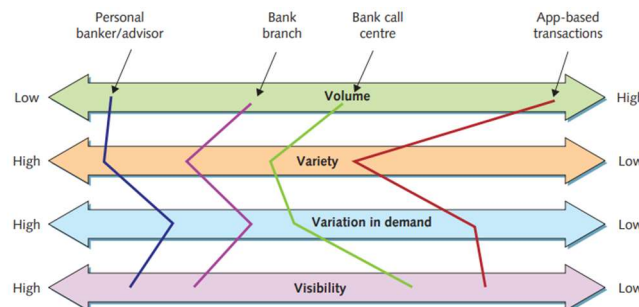
- **Processing costs**  
High volume, low variety, low variation and low visibility all keep processing costs down
- Types of activities that operations managers will have to focus on  
Volume and variety are particularly important in how processes are resourced and designed (ch4 and ch5), variation in demand calls for careful consideration of how capacity is to be managed (ch8), while visibility has a significant influence over how the quality of products and services is judged (ch13).



## 4V ANALYSIS

In almost any operation, processes can be identified that have different positions on the four dimensions, and which therefore have different objectives and will need managing in different ways.

Example: illustrates the different positions on the four Vs for some retail banking processes.



Different operation decisions serve different purposes. Their aim is to determine how well I can reach the overall purpose and contribute to the org. as a whole.

DIRECTING the overall strategy = understand processes, how they can be done in reality + how much of the value-adding processes should be outsourced

DESIGNING DELIVERY through the total S.N.

## 1.5 OPERATIONS & PROCESS DECISION-MAKING

DEVELOPING the capabilities to improve the total performance

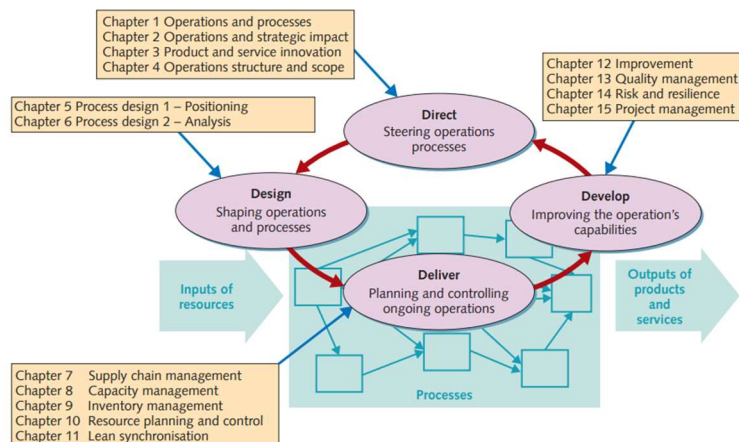
Is operations and process decision making appropriate?

The way operations managers approach decision-making is therefore of considerable importance in determining their effectiveness. But **different operations decisions serve different purposes.**

Managing operations and processes involves a whole range of separate decisions that will determine how well they achieve their overall purpose and contribute to the organization as a whole.

Operations management activities can be grouped into **4 broad categories:**

- **DIRECTING** the overall strategy of the operation  
It necessary to **understand operations and processes**, their **strategic purpose**, how it is translated into reality, how innovation is incorporated into products and services, and how much of the total value-adding process should be kept in-house and outsourced.
- **DESIGNING** the operation's processes  
Design is the activity of determining the physical form, shape and composition of operations and processes, together with the types of resources they contain.
- Planning and controlling process **DELIVERY**  
After being designed, the delivery of products and services from suppliers and through the total operation to customers must be planned and controlled.
- **DEVELOPING** process performance  
They have to develop the capabilities of their processes to improve process performance.



MODEL = representation of a decision. It structures and formalizes the info we possess in order to give a simplified and organized form of the real pattern.

QUALITATIVE  $\Rightarrow$  it does not provide an answer, but enhance understanding and stimulate discussion around the possible ways to organize the ops

## MODEL FOR DECISION-MAKING

QUANTITATIVE

Operations managers use models to help them to make decisions: models are at the core of operations decision-making.

- MODEL = an explicit statement of our image of reality. It is a representation of the aspects of a decision with which we are concerned. It structures and formalizes the information we possess about the decision, and in doing so presents reality in a simplified and organized form. A model therefore provides an abstraction of a more complex reality.

Model could be:

- QUALITATIVE  
They categorize or describe the relationships between aspects of decisions.  
It does not provide an 'answer', but it does enhance understanding and stimulate discussion around other possible ways to organize the operations.
- QUANTITATIVE  
They try to represent the underlying behaviors involved in a decision by using mathematical and/or statistical descriptions of relationships. They allocate numerical values to variables to produce a mathematical representation of reality.

## BEHAVIORAL OPERATIONS

Sometimes operations management are accused of ignoring the practical reality of how operations management decisions are made. They say that the models don't reflect how people really behave when making operations management decisions in practice.

This has led to the development of a new branch of operations management, called behavioral operations management.

BOM explores the interaction of human behaviors and operational systems and processes. It challenges the idea that managers are rational when making decisions that impact operations performance.

### BEHAVIORAL ASSUMPTION

- People are not a major factor in operations decisions
- People are deterministic and predictable
- People make decisions independently of each other
- People do not learn from experience
- People are not part of the product or service
- People are emotionless
- Work is perfectly observable and can be understood

PATTERN OF DECISIONS

## 2. OPERATION STRATEGIC IMPACT

### 2.1 OPERATIONS AND STRATEGY

Does the operation have a strategy?

- **OPERATIONS STRATEGY** = <sup>intends</sup> pattern of decisions and actions that shape the long-term vision, objectives and capabilities of the operations and its contribution to the overall strategy of the business

Strategy is the way in which operations resources are developed over the long term to create sustainable competitive advantage for the business. Increasingly, many businesses are seeing their operations strategy as one of the best ways to differentiate themselves from competitors.

Strategies are always difficult to identify because they have no presence in themselves but are identified by the pattern of decisions that they generate. Nevertheless, one can identify what an operations strategy should do.

### JUDGE AN OPERATION STRATEGY

It's difficult to judge an operation strategy because you can't see or touch strategy and also it may be years before an operation strategy decision can be judged to be a success or not.

### STAKEHOLDERS

- **STAKEHOLDERS** = people and groups who have a legitimate interest in the operation's strategy. Some are internal (employees), others are external (customers, society, shareholders).

All strategy should take significant stakeholders into account. It's a responsibility of the operations function to understand the objectives of its stakeholders and set its objectives accordingly.

But stakeholder relationship is not just one way: it's useful to consider what and individual organization wants of the stakeholder groups themselves.

The dilemma with using this range of stakeholders for judge the performance is that organizations have to cope with the conflicting pressures of maximizing profitability on one hand with the expectation that they will manage in the interests of all society in general.

### CSR ⇒ CORPORATE SOCIAL RESPONSIBILITY

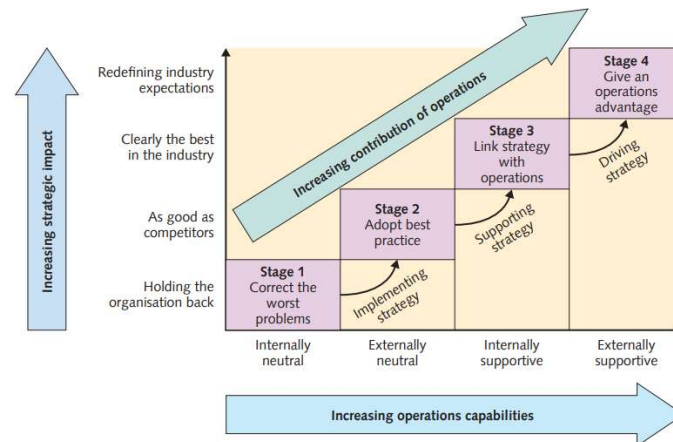
Corporate social responsibility means listening and responding to the needs of a company's stakeholders, including the requirements of sustainable development, and building good relationships with employees, suppliers, and wider society. The triple bottom line approach integrates economic, environmental, and social impacts.

HAYES-WHEELWRIGHT STAGE MODEL → STAGE 2: EXTERNAL NEUTRALITY → comparing themselves with others  
 & adopt the best practice from others  
 STAGE 3: INTERNALLY SUPPORTIVE →  
 STAGE 4:

## OS ARTICULATE A VISION FOR OPERATIONS CONTRIBUTION

Operations strategy should articulate a vision for the operations function's contribution to overall strategy. The vision is a clear statement of how operations intend to contribute value for the business.

A common approach to summarize operations contribution is the **Hayes and Wheelwright Four-Stage Model**, that traces the progression of the operations function from what is the largely negative role of Stage 1 operations to it becoming the central element of competitive strategy.



### STAGE 1: INTERNAL NEUTRALITY

This is the very poorest level of contribution by the operations function. The other functions regard it as holding them back from competing effectively. The operations function has very positives to contributes towards competitive success. Its goal is to be ignored.

Vision: be internally neutral, by avoiding the bigger mistakes.

### STAGE 2: EXTERNAL NEUTRALITY

Operations function start **comparing itself with similar companies** in the outside market.

Vision: become up to speed or **externally neutral with similar businesses** in its industry by adopting best practice ideas and norms of performance from others.

### STAGE 3: INTERNALLY SUPPORTIVE

They may not be better than competitors on every aspect of performance, but they are broadly up.

Vision: **be clearly and unambiguously the very best in the market**. They try to achieve this by **gaining a clear view of the company's competitive and strategic goals and developing appropriate operations resources to excel in the areas in which the company needs to compete effectively**.

### STAGE 4: EXTERNALLY SUPPORTIVE

Operations **look to the long term**. It **develops the operations-based capabilities** that will be required to **compete in future market conditions**. Operations are creative, innovative and capable of adaptation as markets change. They are trying to be one step ahead of competitors.

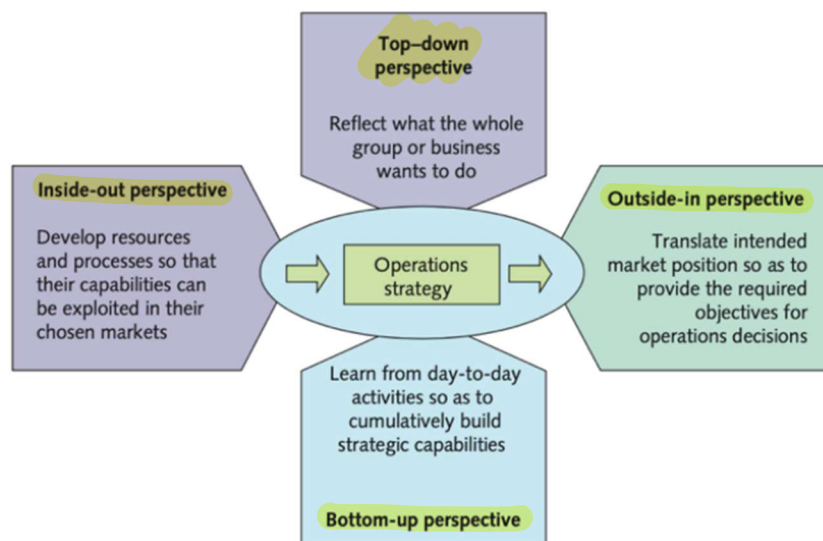
Vision: provide the foundation for competitive success.

## 4 PERSPECTIVES ON OPERATIONS STRATEGY

So as an enterprise moves through each of these stages, what should an operations strategy do?

1. Operations strategy should reflect what the whole business wants to do
2. Operations strategy should translate the enterprise's intended market position to provide the required objectives for operations decisions
3. Operations strategy should learn from day to day activities to build strategic capabilities
4. Operations strategy should develop its resources and processes so that their capabilities can be exploited in their chosen markets

Together the 4 perspectives provide some idea of the pressures that go to form the content of operations strategy.



→ direction gives the overall strategy; business then follows the direction  
CORPORATE ⇒ position in its global, economic, political and social environment. (Where, what)

TOP-DOWN: ↳ covers of strategy ↳ BUSINESS ⇒ set out mission and objectives (what)  
 ↳ FUNCTIONAL ⇒ (how) organize themselves  
 CORRESPONDENCE / COHERENCE

## 2.2 TOP DOWN (BUSINESS STRATEGY)

Operations strategy should reflect the business strategy in a top-down manner. The company's strategy gives a **direction** that must be **followed by business areas** and then must be followed in **defining functional strategies**.

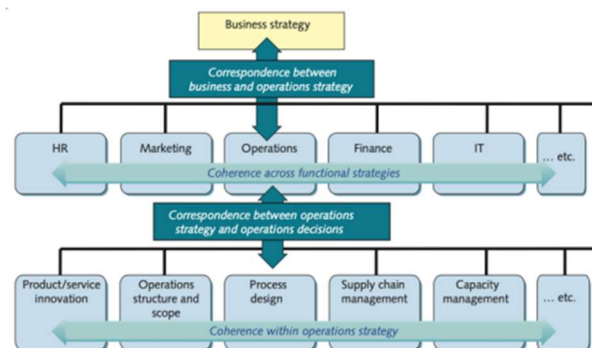
### LEVELS OF STRATEGY

- **CORPORATE**  
Position the corporation in its global, economic, political and social environment. Decisions about what types of business to be involved in, what parts of the world to operate in
- **BUSINESS**  
Set out each business's individual mission and objectives
- **FUNCTIONAL**  
Consider **what part each function should play in contributing to the strategic objectives** of the business. All functions need to consider how best they should organize themselves to support the business's objectives

Business Strategy  
 ↳ Operations Strategy  
 ↳ Individual decisions  
 connection between ↳  
 ↳  
**CORRESPONDENCE AND COHERENCE**  
 ↳ elements pull in the same direction.

A business strategy consists of aggregated and approximated objectives, because it gives an overall direction. Any top-down perspective on operations strategy should attempt to achieve both correspondence and coherence.

- **CORRESPONDENCE** = it is consistent with business strategy. There should be a **clear and connection between a business's strategy** and its **operations strategy**, and between an **operations strategy** and the **individual decisions** taken within the operations function
- **COHERENCE** = its **various elements all pull in the same direction**, means that all decisions should complement and reinforce each other in the promotion of the objectives



# BUSINESS AND OPERATING MODEL

One way of achieving correspondence and coherence is to create a business model that integrates with an operating model.

## BUSINESS MODEL *→ plan to generate revenue and make profit.*

It's the **plan** that is **implemented by a company** to **generate revenue and make a profit**. It includes **what a company does** and **how it makes money from doing**. It is a description of the value a company offers to one or several segments of customers and of the architecture of the firm and its network of partners for creating, marketing, and delivering this value and relationship capital, to generate profitable and sustainable revenue streams.

Includes:

- What is offered to the market
- Target customer segments
- Relationship established with customers
- Revenue streams generated
- Cost structure

## OPERATING MODEL *→ High-level design of the org. Defines <sup>STRUCTURE</sup> <sup>STYLE</sup> → meet business objectives.*

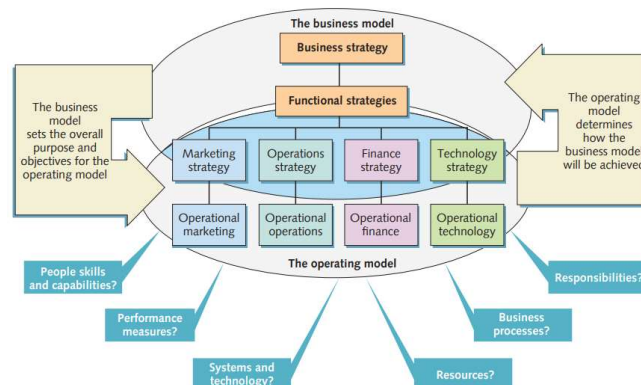
It's a **high-level design of the organization** that defines the **structure** and **style** which enables it to **meet its business objectives**. An operating model should provide a clear description of what the organization does and how it does it.

It should provide a way to examine the business in terms of the key relationships between business functions, processes and structures that are required for the organization to fulfil its mission.

Includes:

- *↑ importance of performance objectives*  
**Key performance indicators**: indication of the relative importance of performance objectives
- The **structure of the organization**
- Systems and **technologies**
- **Processes, responsibilities** and **interactions**
- **Key knowledge and competence**

An operating model focuses more on how an overall business strategy is to be achieved.



Reflects market position. Operations provides the ability to perform in a manner that is appropriate for the intended pos.  
~ TRANSLATING REQUIREMENTS INTO PERFORMANCE OBJECTIVES.

## 2.3 OUTSIDE IN (MARKET REQUIREMENTS)

Operations strategy should reflect the market position of the business. And because companies compete in different ways (cost leadership, differentiation), the operations function should respond by providing the ability to perform in a manner that is appropriate for the intended market position.

The outside-in perspective involves translating market requirements into performance objectives (quality, speed, dependability, flexibility and cost), the exact definition of which will be different for different operations.

These may be refined by distinguishing between those performance objectives that are order-winners and those that are qualifiers.

### DEFINE PERFORMANCE OBJECTIVES

Operations add value for customers and contribute to competitiveness by being able to satisfy the requirements of their customers.

So we have to translate market requirements into performance objectives:

- **QUALITY:** producing error-free goods and services that are fit for their purpose
- **SPEED:** minimizing the time between a customer asking for goods and services and the customer receiving them in full
- **DEPENDABILITY:** keeping the delivery promises that have been made to customers
- **FLEXIBILITY:** the ability to vary or adapt the operation's activities to cope with unexpected circumstances, to give customers individual treatment or to introduce new products/services
- **COST:** producing goods and services at a cost that enables them to be priced appropriately for the market, while still allowing a return to the organization

**RELATIVE PRIORITY** → 5 performance objectives depends on how the business competes in its market.

The interpretation of the 5 performance objectives will differ between different operations.

The relative importance of the five performance objectives depends on how the business competes in its market. This is because business compete in different ways, want different things from their operations functions and so will apply different priorities to its performance objectives.

Example business that compete on low prices should emphasize cost and efficiency, instead firms that compete on a high degree of customization should emphasize flexibility.

# ORDER WINNERS AND QUALIFIERS

Different customer needs imply different priorities of performance objectives. So it's necessary to determine the order winners and qualifiers for each group of customers.

A way for determine the relative importance of competitive factors is to distinguish between:

- ORDER WINNERS

Those things that directly and significantly contribute to winning business. They are **regarded by customers as key reasons for purchasing the product or service.**

Raising performance in an order-winner will either result in more business.

- QUALIFIERS

May not be the major competitive determinants of success but are important in another way. They are **those aspects of competitiveness where the operation's performance must be above a particular level just to be considered by the customer:** qualifiers are those things that are generally expected by customers.

Any further improvement above the qualifying level is unlikely to gain competitive benefit.



Life cycle:   
 INTRODUCTION   
 GROWTH   
 MATURITY   
 DECLINE

## PRODUCT/SERVICE LIFE CYCLE

Product/service life cycles will vary, but generally they are shown as the sales volume passing through four stages: introduction, growth, maturity and decline.

The important implication of this for operations management is that products and services will require different operations strategies in each stage of their life cycle.

INTRODUCTION → first appearance to the market.  
 slow growth in sales.  
 No/few competitors  
 Frequent design changes  
 ORDER-WINNERS: performance, novelty  
 ORDER-QUALIFIERS: Quality range

GROWTH →

|                                         | Introduction                                            | Growth                                    | Maturity                           | Decline                                    |
|-----------------------------------------|---------------------------------------------------------|-------------------------------------------|------------------------------------|--------------------------------------------|
| Sales volume                            | Product/service first introduced to market              | Product/service gains market acceptance   | Market needs start to be fulfilled | Market needs largely met                   |
| Volume                                  | Slow growth in sales                                    | Rapid growth in sales volumes             | Sales slow down and level off      | Sales decline                              |
| Customers                               | Innovators                                              | Early adopters                            | Bulk of market                     | Laggards                                   |
| Competitors                             | Few/none                                                | Increasing numbers                        | Stable number                      | Declining numbers                          |
| Variety of product/service design       | Possible high customisation or frequent design changes  | Increasingly standardised                 | Emerging dominant types            | Possible move to commodity standardisation |
| Likely order-winners                    | Product/service characteristics, performance or novelty | Availability of quality products/services | Low price Dependable supply        | Low price                                  |
| Likely qualifiers                       | Quality Range                                           | Price Range                               | Range Quality                      | Dependable supply                          |
| Dominant process performance objectives | Flexibility Quality                                     | Speed Dependability Quality               | Cost Dependability                 | Cost                                       |

- INTRODUCTION

When a product or service is first introduced, it is likely to be offering something new in terms of its design or performance. Given the market uncertainty, the operations management of the company needs to develop the flexibility to cope with these changes and the quality to maintain product/service performance

- GROWTH

In the growing market, standardized designs emerge that allow the operation to supply the rapidly growing market. Keeping up with demand through rapid and dependable response and maintaining quality levels will help to keep market share as competition starts to increase

- MATURITY

Demand starts to level off as the market becomes dominated by a few larger companies. Competition will probably emphasize price or value for money, so operations will be expected to get the costs down in order to maintain profits or to allow price-cutting, or both.

- DECLINE

After time, sales will decline. To the companies left there might be a residual market

## **2.4 BOTTOM UP (OPERATIONAL EXPERIENCE)**

Operations strategy should always reflect operational reality, taking into account the circumstances, experiences and capabilities of the various businesses that form the group.

So, business when reviewing their strategies, should consult the individual functions within the business about their constraints and capabilities and also incorporate the ideas that come from each function's day-to-day experience. In fact, many strategic ideas emerge over time from operational experience rather than originating exclusively at a senior level.

Sometimes companies move in a particular strategic direction because the ongoing experience of providing products and services to customers at an operational level convinces them that it is the right thing to do.

The bottom-up perspective should involve:

- Capturing the learning that comes from routine operations activities
- Transforming that learning into strategically valuable knowledge

Exploited capabilities in the chosen market developing resources and processes.  
↳ Competitors find difficult to imitate or match our capabilities.

↑

## 2.5 INSIDE OUT (DEVELOP THE CAPABILITY)

Operations strategy should develop its resources and processes so their capabilities can be exploited in the chosen market.

In the longer term, operations strategy must build the capabilities within its resources and processes that will allow it to provide something to the market that its competitors find difficult to imitate or match.

Operations strategy should do more than simply meet the short term needs of the market: it has to develop processes and resources within operations to provide the business with a set of capabilities (know-how). If the capabilities are refined and integrated, they can form the basis of the business's ability to offer unique and difficult to imitate products and services to its customers.

Difficult to imitate resources can be classified as strategic with the VRIO analysis.

Long-term objective = build operation-based capabilities. Take into account also intangible resources.

↑

## BUILD OPERATIONS CAPABILITIES

The long-term objective of operations strategy is to build operation-based capabilities.

Building resources capabilities means understanding the existing resources and processes within the operation, starting with the questions what do we have and what can we do?

We have to take into account also the intangible resources, because they are important, they have value and both tangible and intangible resources and processes shape is capabilities.

### INTANGIBLE RESOURCES

- Relationship with suppliers
- Knowledge and experience in handling its process technologies
- Way its staff can work together in new product development
- Way it integrates all its processes into a mutually supporting whole

Business doesn't have the resources to pursue every single  
→ action that might improve its ops performances.  
→ How the operation best achieve its performance objectives.

## IDENTIFY BROAD DECISION

Operations strategy should identify the broad decisions that will help the operation build its capabilities.

Because business don't have the resources to pursue every single action that might improve their operations performance, it's necessary to indicate broadly how the operation might best achieve its performance objectives.

Example: a business might specify that it will attempt to reduce its costs by aggressive outsourcing of its non-core business processes.

Business with an ABOVE-AVERAGE strategic performance are likely to have gained their sustainable competitive advantage.

RBV



## RESOURCE BASED VIEW (RBV)

Difficult to imitate (resources)

SCARCE  
IMPERFECTLY MOBILE  
IMPERFECTLY IMITABLE AND SUSTAINABLE

The idea that building operations capabilities is a particularly important objective of operations strategy is closely linked with the **resource-based view** (RBV) of the firm.

This holds that businesses with an **above-average strategic performance** are likely to have gained their **sustainable competitive advantage** because of their core competences (or capabilities). This means that **the way an organization acquires or develops its operations resources will, over the long term, have a significant impact on its strategic success.**

RBV sees firms being able to **protect their competitive advantage through barriers to imitation**. Difficult to imitate resources can be classified as strategic if they are:

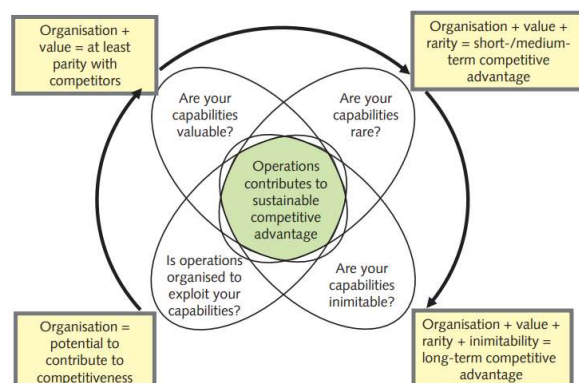
- SCARCE
- IMPERFECTLY MOBILE: resources that are difficult to move out of a firm (interconnected)
- IMPERFECTLY IMITABLE AND SUSTITUTABLE: The more the resources are connected with process knowledge embedded deep within the firm, the more difficult they are for competitors to understand and to copy

way of applying RBV  
VRIO

The most common way of applying RBV is the VRIO framework. Strategic resources have to be:

- **VALUABLE**: it's possible to **identify specific and definable competitive value from the resource**
- **RARE**: if you have or have access to resources that your competitors don't
- **INIMITABLE**: if you have resources that competitors can't imitate, purchase or find a suitable alternative to, at a realistic cost or in a realistic time frame.
- **ORGANIZED**: if you have within the business the **system, culture, capacity, motivation to exploit any capabilities embedded in your resources and processes**

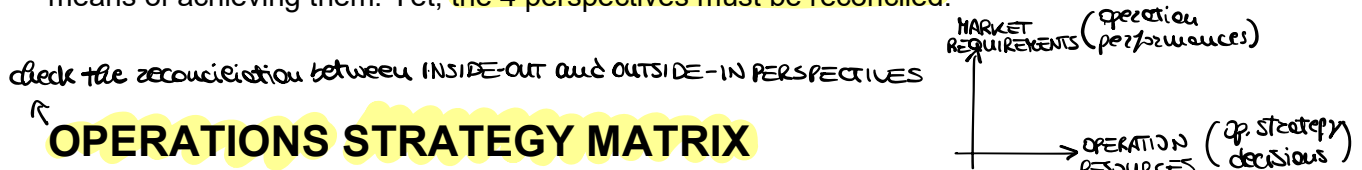
All these factors are time dependent. The organization is a necessary prerequisite because without the ability to exploit strategic resources, they are of little use.



## 2.6 FOUR PERSPECTIVES RECONCILED

None of the four perspectives individually can give a full picture of any operations strategy. But together they do provide a good idea of how operations are contributing strategically.

An operations strategy should articulate the relationship between operations objectives and the means of achieving them. Yet, **the 4 perspectives must be reconciled**.



The operations strategy matrix is one method of checking the **reconciliation** between the **inside-out** and **outside-in perspectives**. It brings together **market requirements** and **operations resources** to form the two dimensions of a matrix. It describes operations strategy as the intersection of a company's performance objectives and the strategic decisions that it makes.

Reconciliation between:

- MARKET REQUIREMENTS**  
 What is **required from the operations function** → performance objectives.  
**Outside-in perspective**
- STRATEGIC DECISIONS**  
**How the operation tries to achieve this** through the set of **choices made** and the **capabilities** that have been **developed in each decision area**.  
 What are the decisions that support operations capabilities → design, delivery, development  
**Inside-out perspective**

The operations strategy matrix is a good model for testing whether market requirements and the operations capability perspectives fit together.

The disadvantage is that it gives little sense of dynamics of reconciliation and how the balance between market requirements and operations capability can change over time.

| How operations resources are used |                               |                                        |                                |                                   |
|-----------------------------------|-------------------------------|----------------------------------------|--------------------------------|-----------------------------------|
| Operations performance objectives | ★★★<br>Quality of service     | Track and trace technology             |                                | Knowledge management system       |
|                                   | ★<br>Range of services        | Sorting systems<br>Variety of vehicles |                                |                                   |
|                                   | ★★★<br>Delivery dependability | Location of depots                     | Demand-capacity management     | Vehicle maintenance programme     |
|                                   | ★★<br>Innovation              |                                        |                                | Customer service centre feedback  |
|                                   | ★★<br>Cost                    | Process standardisation                | Route scheduling system        | Location                          |
|                                   |                               | Strategic 'design' decisions           | Strategic 'delivery' decisions | Strategic 'development' decisions |
| Operations strategy decisions     |                               |                                        |                                |                                   |

How market requirements are met

# LINE OF FIT MODEL <sup>↑ requirements of the market (op. strategy objectives) (outside-in)</sup> → capabilities (inside-out)

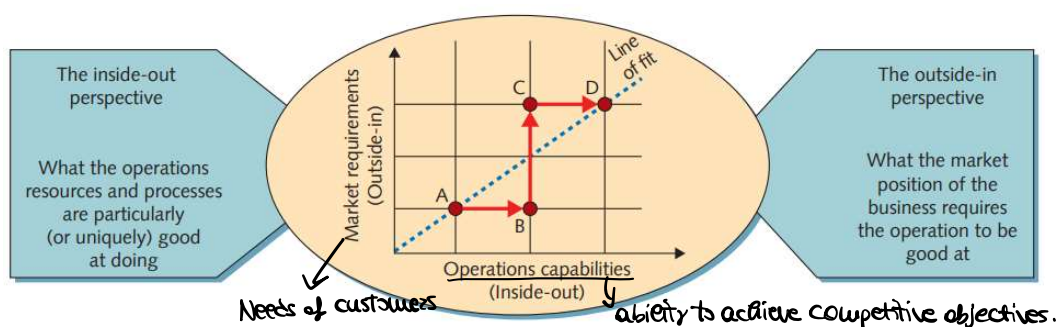
The line of fit model can be used to map the balance between requirements of the market and the capabilities of the operation over time.

Ideally, there should be a reasonable degree of alignment, or fit, between the two:

- MARKET REQUIREMENTS
  - Reflect the intrinsic needs of customers
  - Outside-in
  - Movement along the vertical dimension indicates a broadly enhanced level of market performance
  - Example: strength of the brand, reputation of degree of differentiation
- RESOURCE AND PROCESS CAPABILITIES
  - Inside-out
  - Movement along the horizontal dimension indicates a broadly enhanced level of operations capability
  - Example: performance of the operations in terms of its ability to achieve competitive objectives, the efficiency with which it uses its resources

If the market requirements and operations capability of an operation are aligned, they would be positioned diagrammatically on the line of fit.

FIT is to achieve an approximate balance between market requirements and operations capability. So when fit is achieved, firms' customers do not need, or expect, levels of operations capability that cannot be supplied. Nor does the operation have strengths that are either inappropriate for market needs or remain unexploited in the market.



## DEVIATES FROM LINE OF FIT

A company that has position A has achieved fit in a low level: the market doesn't want much from the business, which is just as well because its operation is not capable of achieving much. Over time its ambition is to move to position D, where it has also achieved fit, but at a much higher level.

However, like most strategic improvement, the company cannot always guarantee to keep on the line of fit as it moves from A to D over time. It has to move to B (operation is failing to exploit its operations capabilities) and then to C (risk of damage its reputation by failing its market promises).

Starting point: OPERATIONS STRATEGY

improve business's performance relative to its competitors in the LONG TERM.

## 2.7 IMPROVEMENT PATH

trade-off between performance objectives

An operations strategy is the starting point for operations improvement. The purpose of operations strategy is to improve the business's performance relative to its competitors in the long term.

It therefore must provide an indication of how this improvement is to take place. This is best addressed by considering the tradeoffs between performance objectives in terms of the efficient frontier model.

## TRADE-OFFS BETWEEN PERFORMANCE OBJECTIVES

An operations strategy should address the relative priority of operations performance objectives.

In the short term, operations can't achieve outstanding performance in all its operations objectives simultaneously. In the long term, a key objective of operations strategy is to improve all aspects of operations performance.

improve some objective, reduce others

How to manage trade off? There are 2 approaches:  $\nearrow$  improve without reductions.

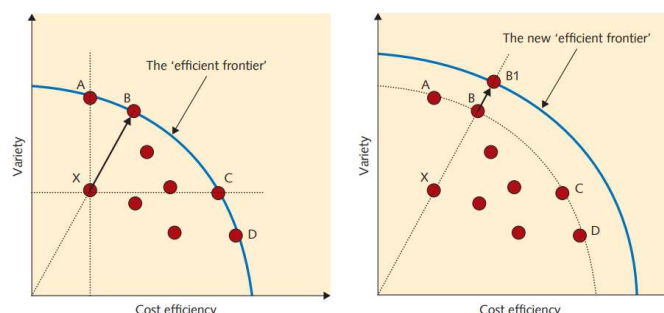
- Repositioning performance objectives by trading off improvements in some objectives for a reduction in performance in other. Taken to its extreme, this trade-off principle implies that improvement in one performance objective can only be gained at the expense of another.
- Overcoming trade off so that improvements in more aspects of performance can be achieved without any reduction in the performance of others

## EFFICIENT FRONTIER $\nearrow$ VARIETY $\nearrow$ COST EFFICIENCY

Presumably all the operations would ideally like to be able to offer very high variety, while still having very high levels of cost efficiency. However, the increased complexity that a high variety of product or service offerings brings will generally reduce the operation's ability to operate efficiently.

Operation X, however, has an inferior performance because operation A is able to offer higher variety at the same level of cost efficiency; and operation C offers the same variety but with better cost efficiency.

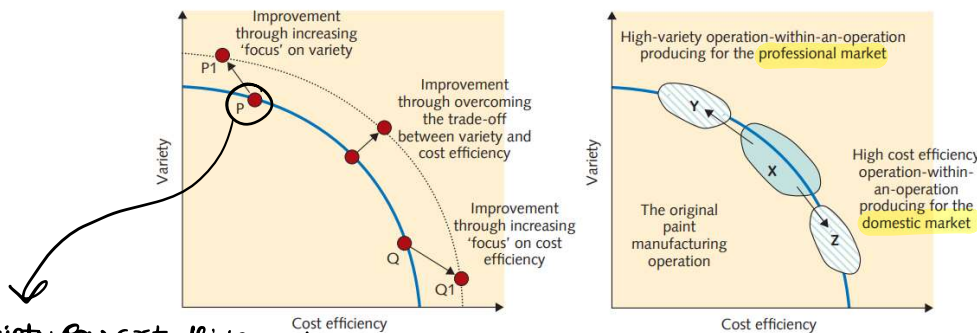
The convex line on which effective operations (A,B,C,D) lie is known as efficient frontier.



## IMPROVE BY USING TRADE OFF

Improving the effectiveness of an operation by **pushing out the efficient frontier** requires different approaches, depending on the original position of the operation on the frontier.

Example: P is designed to offer a high level of variety of products or services at the expense of low-cost efficiency (high technology, skilled employees). If he want to have greater variety of products (focus on variety), he have to sacrifice cost efficiency.



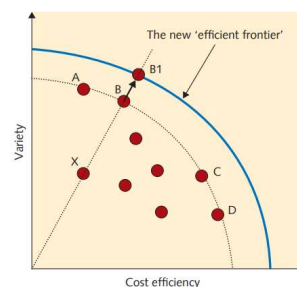
High variety, low cost efficiency, I can try pushing up the variety at the expense of cost efficiency or find some method to increase both the aspects, improving the efficient frontier

## IMPROVE BY OVERCOMING TRADE OFF

Those with a position on the efficient frontier will generally also want to improve their operation's effectiveness by overcoming the trade-off that is implicit in the efficient frontier curve.

Even trade-offs that seem to be inevitable can be reduced to some extent.

Example: supermarket manager has to decide **how many checkout positions to open at any time**. If **too many checkouts are opened** then there will be **times** when the **checkout staff don't have any customers to serve and will be idle**. The customers, however, will have **excellent service** in terms of little or no waiting time. Conversely, if **too few checkouts are opened**, the staff will be **working all the time** but **customers will have to wait in long queues**. This is a direct trade-off between staff utilization (cost) and customer waiting time (speed of service). For resolve, the manager might allocate a number of core staff to operate the checkouts but also **arrange other staff with other jobs to be on-call when demand suddenly increase**. By devising a **flexible system of staff allocation**, the manager can both improve customer service and keep staff utilization high.



# 4. OPERATION SCOPE & STRUCTURE

## 4.1 OPERATIONS' PLACE IN SUPPLY NETWORK

Does the operation understand its place in its supply network?

Both the structure and the scope of an operation's supply network are decisions that shape how the operation interacts with its markets, its suppliers, in fact with the world in general. All operations are part of a larger and interconnected network of operations (supply network).

- **SUPPLY NETWORK** = interconnection of organizations that relate to each other through upstream and downstream linkages between the different processes and activities that produce value in the form of products and services to the ultimate consumer. It includes the chain of suppliers, the chain of customers and sometimes other operations.

To understand the place and role of the supply network, an operation must have an idea of how it fits into the scope and the structure of its supply network.

⇒ How much an op. should own <sup>VERTICAL INTEGRATION</sup> + slope of their S.N. <sup>outsourcing</sup> <sup>location of each op.</sup> → how big the parts of the network that owns should be

- **SCOPE**  
At a strategic level, operations managers are involved in deciding how much of the network an operation should own. It defines what is going to do itself and what it will buy in from suppliers.
- **STRUCTURE**  
The managers are also concerned with the shape and form of their network. It involves deciding (or at least influencing) its overall shape, the location of each operation and how big the parts of the network that it owns should be.

## TWO-WAY FLOW THROUGH THE NETWORK

Flow is a two-way process with items flowing one way and information flowing the other.

- **Physical flow**  
Materials, parts, information, ideas and sometimes people all flow through the network of customer-supplier relationships formed by all these operations
- **Informational flow**  
Each customer-supplier linkage will feed back orders and information

Understanding COMPETITIVENESS: Look beyond first-tier customers/suppliers to understand why they act as they do. Don't put too much faith in first-tier sup/cust.

Look at the whole S.N. <sup>Identify links in networks:</sup>   
 • Not everyone has the same degree of importance, some op. contribute more to the performance objectives that add value to the final customers   
 • It helps understanding RELATIONSHIPS, LONG-TERM ISSUES etc.

## IMPORTANT TO CONSIDER THE WHOLE SUPPLY NETWORK

Supply network issues and the operations' positions in them, have a significant effect on their strategic performance. In addition, there are other reason why it is important to stand back and look at the whole (or a large part) of a supply network rather than an individual operation.

- It helps an **UNDERSTANDING OF COMPETITIVENESS**

Immediate customers and immediate suppliers are the main concern for companies. Yet sometimes they need to look beyond these immediate contacts to understand why customers and suppliers act as they do. Simply relying only on one's immediate network is arguably putting too much faith in someone else's judgement of things that are central.

- It helps identify **SIGNIFICANT LINKS IN NETWORK**

A supply network perspective helps to make sense of competitive, relationship and longer-term operations issues. Not everyone in a supply network has the same degree of influence over the performance of the network as a whole. Some operations contribute more to the performance objectives that are valued by end customers. So, an analysis of networks needs an understanding of the operations that contribute most to end-customer service.

- It helps **FOCUS ON LONG TERM ISSUES**

A long-term supply network view would involve examining technology and market changes constantly to see how each operation in the supply network might be affected.

Ex: street music store

## SCOPE AND STRUCTURE

RELATED

SCOPE: <sup>vertical integration</sup> degree of outsourcing   
 STRUCTURE: <sup>how the S.N. should be configured</sup> long-term capacity decision (physical capacity)   
 location decision

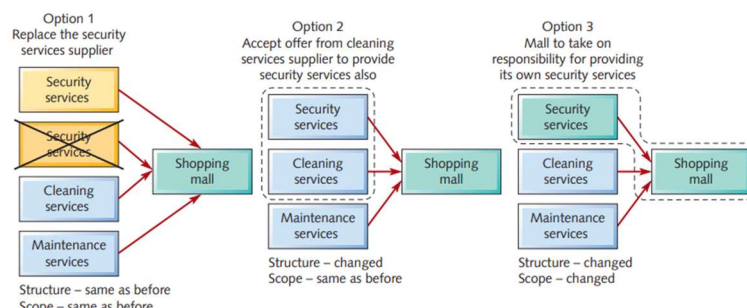
The scope and structure of an operation's supply network are strongly related. Both scope and structure decisions comprise a number of other constituent decisions

The **SCOPE** of an operation's activities within the network is determined by two decisions:

1. The extent and nature of the operation's vertical integration
2. The nature and degree of outsourcing it engages in

The **STRUCTURE** of an operation's supply network is determined by three sets of decisions:

1. How the network should be configured
2. The long-term capacity decision: what physical capacity each part of the network should have
3. The location decision: where each part of the network should be located



## 4.2 VERTICAL INTEGRATION

The scope of an operation's supply network determines to what extent an operation does things itself and the extent to which it will rely on other operations to do things for it. Vertical integration is the extent to which an organization owns the network of which it is a part.

Generally, an organization's vertical integration strategy can be defined in terms of:

- **DIRECTION OF INTEGRATION**

Does it expand by buying one of its suppliers (backward or upstream vertical integration) or should it expand by buying one of its customers (forward or downstream vertical integration)?

- Backward vertical integration: by allowing an organization to take control of its suppliers, is often used either to gain cost advantages or to prevent competitors gaining control of important suppliers
- Forward vertical integration: takes an organization closer to its markets and allows more freedom to make contact directly with its customers

- **EXTENT OF PROCESS SPAN OF INTEGRATION**

Some organizations deliberately choose not to integrate far from their original part

- **BALANCE AMONG THE VERTICALLY INTEGRATED STAGES**

It refers to the amount of capacity at each stage in the network that is devoted to supplying the next stage. So, a totally balanced network relationship is one where one stage produces only for the next stage in the network and totally satisfies its requirements.

## ADVANTAGE OF VERTICAL INTEGRATION

1. **SECURE DEPENDABLE ACCESS TO SUPPLY**

It can give a more secure supply or bring a business closer to its customers. Downstream vertical integration can give a firm greater control over its market positioning.

Example: in Apple both its hardware and at least some of its software are designed in-house

2. **REDUCE COST**

Vertical integration also reduces the transaction costs of dealing with suppliers and customers. If transaction costs can be lowered to the point where the purchase price plus transaction costs is less than the internal cost, there is little justification for the vertical integration of the activity

3. **HELP TO IMPROVE QUALITY**

Used to secure specialist or technological advantage by preventing product and service knowledge getting into the hands of competitors

4. **HELP IN UNDERSTANDING OTHER ACTIVITIES**

Some companies choose to own some parts of the supply network other than what they regard as core

# DISADVANTAGE OF VERTICAL INTEGRATION

## 1. CREATE INTERNAL MONOPOLY

Operations will only change when they see a pressing need to do so. Internal supply is less subject to the normal competitive forces that keep operations motivated to improve. If an external supplier serves its customers well, it will make higher profits; if not, it will suffer. Such incentives and sanctions do not apply to the same extent if the supplying operation is part of the same company.

## 2. CAN'T EXPLOIT ECONOMIES OF SCALE

Specialist suppliers who can serve more than one customer are likely to have volumes larger than any of their customers could achieve by doing things for themselves. This allows specialist suppliers to reap some of the cost benefits of economies of scale, which can be passed on in terms of lower prices to their customers.

## 3. LOSS OF FLEXIBILITY

Heavily vertically integrated companies do most things themselves. This means that a high proportion of their costs will be fixed costs. They have invested heavily in the capacity that allows them to do most things in-house. A high level of fixed costs relative to variable costs means that any reduction in the total volume of activity can easily move the economics of the operation close to, or below, its break-even point.

## 4. IT CUTS OFF FROM INNOVATION

Vertical integration means investing in the processes and technologies necessary to produce products and services in-house. But, as soon as that investment is made, the company has an inherent interest in maintaining it. Abandoning such investments can be both economically and emotionally difficult. The temptation is always to wait until any new technology is clearly established before admitting that one's own is obsolete. This may lead to a tendency to lag in the adoption of new technologies and ideas.

## 5. LOSS OF FOCUS

Vertical integration, by definition, means doing more things, which can distract from the (few) particularly important things (core activities).

## 4.3 IN-HOUSE OR OUTSOURCE

How do operations decide what to do in house and what do outsource?

Theoretically, the vertical integration decision and the outsourcing decision are almost the same thing. The difference between them is one of scale.

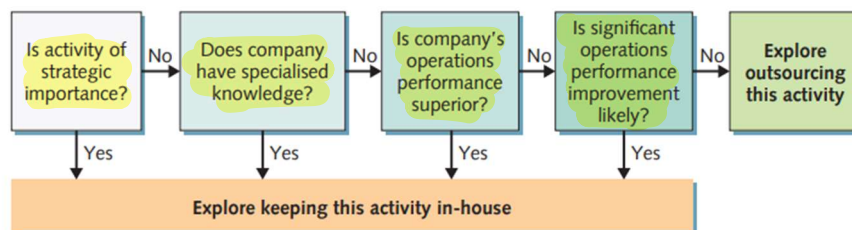
- Vertical integration is a term that is usually applied to whole operations  
Example: selling the part of your business that services your products to a specialist servicing company that can do the job better
- Outsourcing applies to smaller sets of activities that before have been performed in-house.  
Example: deciding to ask a specialist laboratory to perform some quality tests that your own quality control department used to do

## OUTSOURCING DECISION

The question is “Is it in-house or outsourced supply in a particular set of circumstances that gives the company the appropriate performance objectives that it requires to compete more effectively in its markets?”.

Assessing the advisability of outsourcing should also include consideration of:

- how it impacts on relevant performance objectives
- the strategic importance of the activity
- the relative operations performance
- the risks of supply disruption.



| Performance objective | 'Do it yourself' in-house supply                                                                                                                                                                                                                                                                    | 'Buy it in' outsourced supply                                                                                                                                                                                                                                                                          |
|-----------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Quality               | The origins of any quality problems are usually easier to trace in-house and improvement can be more immediate, but there can be some risk of complacency.                                                                                                                                          | Supplier may have specialised knowledge and more experience, also may be motivated through market pressures, but communication more difficult.                                                                                                                                                         |
| Speed                 | Can mean synchronised schedules, which speeds throughput of materials and information, but if the operation has external customers, internal customers may be low priority.                                                                                                                         | Speed of response can be built into the supply contract where commercial pressures will encourage good performance, but there may be significant transport/delivery delays.                                                                                                                            |
| Dependability         | Easier communications can help dependability but, if the operation also has external customers, internal customers may receive low priority.                                                                                                                                                        | Late delivery penalties in the supply contract can encourage good delivery performance, but organisational barriers may inhibit communication.                                                                                                                                                         |
| Flexibility           | Closeness to the real needs of a business can alert the in-house operation to required changes, but the ability to respond may be limited by the scale and scope of internal operations.                                                                                                            | Outsourced suppliers may be larger with wider capabilities than in-house suppliers and have more ability to respond to changes, but may have to balance conflicting needs of different customers.                                                                                                      |
| Cost                  | In-house operations do not have to make the margin required by outside suppliers, so the business can capture the profits that would otherwise be given to the supplier, but relatively low volumes may mean that it is difficult to gain economies of scale or the benefits of process innovation. | Probably the main reason why outsourcing is so popular, as outsourced companies can achieve economies of scale and they are motivated to reduce their own costs because it directly impacts on their profits, but costs of communication and coordination with supplier need to be taken into account. |

- OUTSOURCING = deciding to buy-in products rather than perform the activities in-house.
- OFFSHORING = obtaining products and services from operations that are based outside one's own country. Offshoring to a lower-cost region of the world is usually done to reduce an operation's overall costs

## GLOBALIZATION

Businesses across the world have been forging ever – closer relationships. Especially in the last few decades, the use of geographically dispersed suppliers to outsource at least some activities has become routine. This has been driven partly by:

- labor cost differentials
- cheap and efficient telecommunication between businesses
- trade deals
- reducing transport costs

This is globalization, where products, raw materials, money, technology and ideas move (relatively) smoothly across national boundaries.

## RISK AND RESHORING

Reducing reliance on complicated international supply chains can:

- save transport and inventory costs
- less polluting
- less prone to reputational risk if far – off suppliers misbehave
- increase supply flexibility

Developments in technology could reinforce this so-called reshoring process. Automation may encourage a trend towards 'radical insourcing', where more economically developed countries no longer need to outsource production to countries where wages are low.

## 4.4 CONFIGURATION OF SUPPLY NETWORK

What configuration should a supply network have?

- CONFIGURING = Configuring a supply network means determining its overall pattern. In other words, what should be the pattern, shape or arrangement of the various operations that make up the supply network?

Even when an operation doesn't directly own other operations in its network, it may still wish to change the shape of the network by reconfiguring it, in order to change the nature of the relationship between its various part.

TOO MANY SUPPLIERS = <sup>COSTS</sup>  $\rightarrow$  prevent from developing a close relationship with a supplier.

### RECONFIGURATION OF SUPPLY NETWORK

- REDUCTION OF SUPPLIERS

The complexity of dealing with many hundreds of suppliers may both be expensive and prevent the operation from developing a close relationship with a supplier

- DISINTERMEDIATION

Disintermediation means cutting out the middle men. It happen when companies within a network, bypass customers or suppliers to make contact directly with customers' customers or suppliers' suppliers.

### CO-OPETITION APPROACH

One approach to thinking about supply networks sees any business as being surrounded by 4 types of players: suppliers, customers, competitors and complementors.

- Complementors = enable one's products or services to be valued more by customers because they can also have the complementor's products or services, as opposed to when they have yours alone.
- Competitors = they make customers value your product or service less when they can have the competitor's product or service, rather than yours alone.

All the players in the network, whether they are customers, suppliers, competitors or complementors, can be both friends and enemies at different times.

### BUSINESS ECOSYSTEM

An idea that is closely related to co-opetition is the business ecosystem, defined as an economic community supported by a foundation of interacting organizations and individuals. The economic community produces goods and services of value to customers, who are themselves members of the ecosystem. The member organisms also include suppliers, lead producers, competitors, and other stakeholders. Over time, they coevolve their capabilities and roles, and tend to align themselves with the directions set by one or more central companies.

# DESCRIPTION OF SUPPLY NETWORK: DYADS AND TRIADS

Any supply network is complex. There are many operations, all interacting in different ways to produce end products and services.

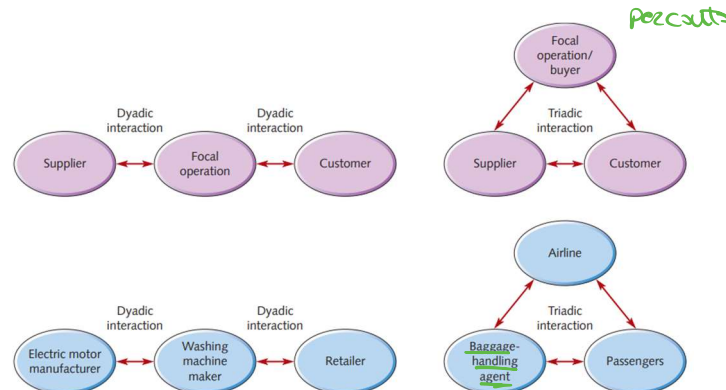
## DYADIC

Because of this, and to understand them better, supply network academics and professionals often choose to focus on the individual interaction between two specific operations in the network.

This is called a dyadic interaction, or dyadic relationship, and the two operations are referred to as a dyad. All relationships in a network are based on the simple dyad.

**TRIADIC** ⇒ relevant in service supply networks because they are outsourcing the delivery of some aspects of their service providers, who deal directly with customers on behalf of the focal operation.

More recently, the idea of a dyad has evolved into a triad: no matter how complex a network, it can be broken down into a collection of triadic interactions. The idea of triads is especially **relevant in service supply networks**. Operations are increasingly outsourcing the delivery of some aspects of their service to specialist providers, who deal directly with customers on behalf of the focal operation.



NON-HS CAPTS

↑  
**ADVANTAGE** → dependence of the organizations on their suppliers' performance

Thinking about supply networks as a **collection of triads** rather than dyads is strategically important:

- It emphasizes the dependence that organizations place on their suppliers' performance when they outsource service delivery.
- *Il controllo che ha il compratore sul servizio di consegna al cliente è diminuito*  
The control that the buyer of the service has over service delivery to its customer is diminished in a triadic relationship. In a conventional supply chain, with a series of dyadic relationships, there is the opportunity to intervene before the customer receives the product or service. However, products or services in triadic relationships bypass the buying organization and go directly from provider to customer.
- The direct link between service provider and customer can result in power gradually transferring over time from the buying organization to the supplier that provides the service.
- It becomes increasingly difficult for the buying organization to understand what is happening between the supplier and customer at a day-to-day level. It may not even be in the supplier's interests to be totally honest in giving performance feedback to the buyer.

# STRUCTURAL COMPLEXITY IN SN

Some supply networks are relatively straightforward, both to describe and to manage; other are more complex. Put many triadic relationships together into an interrelated network and things become even more complex. This is especially true when customers have some degree of choice over how and when they interact with a network.

## OMNICHANNEL RETAIL NETWORKS

Single-channel, multi-channel, cross-channel and omnichannel retail models represent increasing options for customers but increasing operations complexity.

### 1. SINGLE CHANNEL

The original relationship between a retailer and its customers was straightforward: the retailer expected customers to collect goods from its stores, or delivered them later if requested by the customer

### 2. MULTI CHANNEL

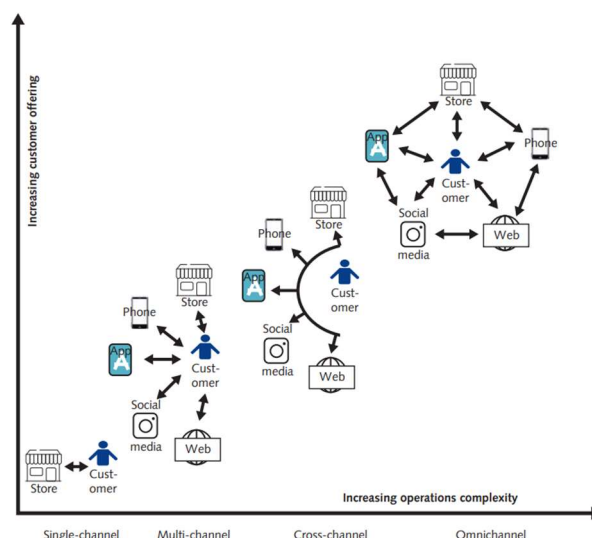
As more methods of contacting retailers became available, such as mobile phones, apps and social media, most retailers struggled to manage and fully integrate the many alternative channels of communication. In fact, they often treated them specifically as independent entities so that they could align each channel with specific targeted customer segments

### 3. CROSS CHANNEL

Later came the first attempts to integrate high-street stores with (initially) web and other channels in order to enhance their cross-functionality that, in turn, allowed a better shopping experience for customers

### 4. OMNICHANNEL

The omnichannel model is seen by many retail analysts as being one of the most important retail developments since the advent of online services. It seeks to provide a seamless all-inclusive customer experience by fully integrating all possible channels, so customers can use whichever is the most convenient for them at whatever stage of the transaction



How much capacity? Depends on its view of current and future DEMAND (PROBLEM: demand changing)

→ CHOOSE the optimum capacity for each site / balance the various capacities in the network / timing the changes in the capacity

## 4.5 CAPACITY OF OPERATIONS PLAN

How much capacity should operations plan to have? The set of structure decisions concern the size of capacity of each part of supply network. How big in terms of capacity the organization want to be?

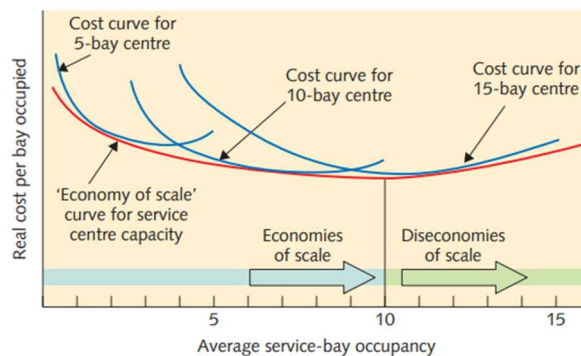
The amount of capacity an organization will have depends on its view of current and future demand. It is when its view of future demand is different from current demand that this issue becomes important. When an organization has to cope with changing demand, a number of capacity decisions need to be taken. These include choosing the optimum capacity for each site, balancing the various capacity levels of the operation in the network, and timing the changes in the capacity of each part of the network. Important influences on these decisions include the concepts of economy and diseconomy of scale.

### OPTIMUM CAPACITY LEVEL

All types of operations exhibit ECONOMIES OF SCALE effects where operating costs reduce as the scale of capacity increases. Economies of scale don't go on forever.

Above a certain size, the lowest cost point on curves may increase. This occurs because of diseconomies of scale. DISECONOMIES OF SCALE increase operating costs above a certain level of capacity, resulting in a minimum cost-level of capacity.

Example: low occupancy, will result in a high cost per customer served because the fixed costs of the operation are being shared between few customers. As demand, and therefore service-bay occupancy, increases, the cost per customer will reduce. However, operating at very high levels of capacity utilization (occupancy levels close to capacity) can mean longer customer waiting times and reduced customer service. This means that the average cost start to increase after a point that will often be lower than the theoretical capacity of the operation.



Excrete near to hot spots. I have a better social networks knowledge.

SMALL-SCALE OPERATIONS:

- Fast response to customers needs
- Taking advantage of the potential for HR development ⇒ high degree of local autonomy
- New techs explored

## ADVANTAGE OF BEING SMALL

Although large-scale-capacity operations will usually have a cost advantage over smaller units, there are also potentially significant advantages that can be exploited by small-scale operations.

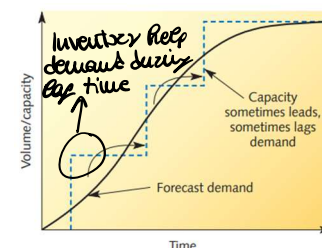
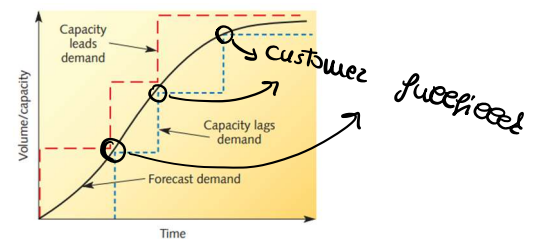
- Locating near to hot spots that can tap into local knowledge networks. Larger companies centralize their research and development efforts.
- Responding rapidly to regional customer needs and trends by basing more and smaller units of capacity close to local markets
- Taking advantage of the potential for human resource development, by allowing staff a greater degree of local autonomy.
- Exploring radically new technologies by acting as a smaller more entrepreneurial rival. Larger, more centralized development activities are more bureaucratic than smaller scale.

## TIMING OF CAPACITY CHANGE

Changing the capacity of any operation in a supply network is not just a matter of deciding on its optimum capacity: it needs to decide when to bring on-stream new capacity. 3 strategies:

1. **Capacity LEAD DEMAND** *demand forecast*  
Capacity is introduced to generally **lead demand**, timing the introduction of capacity in such a way that **there is always sufficient capacity to meet forecast demand**.  
→ there is always enough capacity to meet forecast demand
2. **Capacity LAG DEMAND**  
Capacity is introduced to generally **lag demand**, timing the introduction of capacity so that **demand is always equal to or greater than capacity**.  
→ demand > capacity
3. **SMOOTHING WITH INVENTORY**  
Capacity is introduced to **sometimes lead and sometimes lag demand**, but **inventory** built up during the lead times is used to help meet demand during the lag times.

| Advantages                                                                                                  | Disadvantages                                                                                                                                                                |
|-------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Capacity-leading strategies</b>                                                                          |                                                                                                                                                                              |
| Always sufficient capacity to meet demand, therefore revenue is maximised and customers are satisfied       | Utilisation of the plants is always relatively low, therefore costs will be high                                                                                             |
| Most of the time there is a 'capacity cushion' that can absorb extra demand if forecasts are pessimistic    | Risks of even greater (or even permanent) over-capacity if demand does not reach forecast levels                                                                             |
| Any critical start-up problems with new operations are less likely to affect supply                         | Capital spending on capacity will be early                                                                                                                                   |
| <b>Capacity-lagging strategies</b>                                                                          |                                                                                                                                                                              |
| Always sufficient demand to keep the operation working at full capacity, therefore unit costs are minimised | Insufficient capacity to meet demand fully, therefore reduced revenue and dissatisfied customers                                                                             |
| Over-capacity problems are minimised if forecasts prove optimistic                                          | No ability to exploit short-term increases in demand                                                                                                                         |
| Capital spending on the operation is delayed                                                                | Under-supply position even worse if there are start-up problems with the new operations                                                                                      |
| <b>Smoothing with inventory strategies</b>                                                                  |                                                                                                                                                                              |
| All demand is satisfied, therefore customers are satisfied and revenue is maximised                         | The cost of inventories in terms of working capital requirements can be high; this is especially serious at a time when the company requires funds for its capital expansion |
| Utilisation of capacity is high and therefore costs are low                                                 | Risks of product deterioration and obsolescence                                                                                                                              |



Costs implications of adding increments of capacity. Using inventories to overcome DEMAND-CAPACITY imbalance tends to increase WORKING CAPITAL REQUIREMENTS. → FIXED COSTS  
INCREASING CAPACITY = increasingly costs before any further activity can be undertaken in the operation.  
↑ Rises

## BREAK-EVEN ANALYSIS OF CAPACITY EXPANSION

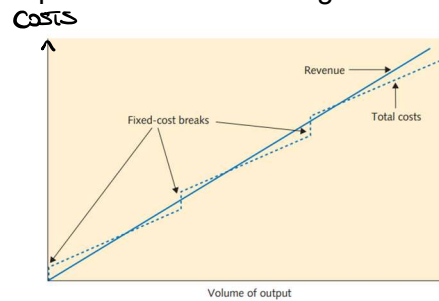
An alternative view of capacity expansion can be gained by examining the cost implications of adding increments of capacity on a break-even basis. Using inventories to overcome demand-capacity imbalance tends to increase working capital requirements.

Equilibrium

Incurring fixed costs repeatedly can raise total costs above revenue over some ranges of output. Increasing capacity can move an operation from profitability to loss. Each additional unit of capacity results in a fixed-cost break that is a further lump of expenditure that will have to be incurred before any further activity can be undertaken in the operation.

The operation is unlikely to be profitable at very low levels of output. Eventually, assuming that prices are greater than marginal costs, revenue will exceed total costs.

However, the level of profitability at the point where the output level is equal to the capacity of the operation may not be sufficient to absorb all the extra fixed costs of a further increment in capacity. This could make the operation unprofitable in some stages of its expansion.



## **4.6 LOCATION OF OPERATIONS**

Where should operations be located?

The location of each operation in a supply network is a key element in defining its structure, and also will have an impact on how the network operates in practice. Poor location of any operation in a supply network can have a significant impact, both on profits, but also those of others in the network.

Location decisions have effect on:

- Operation's costs
- Its ability to serve customers and so its revenue

In addition, location decisions, once taken, are difficult to undo. The costs of moving an operation can be hugely expensive and the risks of inconveniencing customers very high.

### **RELOCATION**

When operations change their location, their assumption is that the potential benefits of a new location will outweigh any cost and disruption involved in changing location. When operations do move, it is usually because of changes in demand and/or changes in supply. The factors that determine a location are such things as labor, land and utility costs, the image of the location, its convenience for customers and the suitability of the site itself.

### **WHY RELOCATE?**

- **CHANGES IN DEMAND**  
If customer demand shifts it may prompt a change in location. Changes in the volume of demand can also prompt relocation: expand or chose a larger site or open a new site.  
Example: suppliers that follow manufacturer
- **CHANGES IN SUPPLY**  
If there's a change in the cost, or availability, of the supply of inputs to the operation.  
Example: mining company need to relocate as the minerals it's extracting become depleted

An operation should only change its location if the benefits of moving outweigh the costs of operating in the new location plus the cost of the move itself.

# EVALUATE POTENTIAL CHANGES IN LOCATION

Evaluating possible locations is a complex task due to the number of location options, the criteria against which they could be evaluated and the rarity of a single location that clearly dominates.

The decision often involves high levels of uncertainty, so it is useful to be systematic in terms of identifying alternative options and evaluating each option against a set of rational criteria.

## ▪ IDENTIFY ALTERNATIVE LOCATION OPTIONS

The first relocation option to consider is not to relocate. Sometimes relocation is inevitable. It's worth evaluating the "do nothing" option, to provide a base case against which to compare other options.

The process of identifying location options usually involves selecting a limited number of sites that represent different attributes. The options may be chosen to reflect a range of both these factors.

## ▪ SET LOCATION EVALUATION CRITERIA

We can evaluate different location with these 5 broad categories:

### 1. CAPITAL REQUIREMENT

The capital or leasing cost of a site is a significant factor. This will probably be a function of the location of the site and its characteristics (such as the shape of the site, site access)

### 2. MARKET FACTORS

Location can affect how the market in general, or as individual customers, perceives an operation. (ex. hospital in countryside inconvenient for customers).

### 3. COST FACTORS

Two major categories of cost are affected by location:

- Cost of producing products or services: labor costs can vary between different areas in any country. There are other cost factors such as community factors, that derive from the social, political and economic environment of the site (local tax rates, capital movement restrictions, political stability, language, local amenities)
- Cost of transporting inputs from their source to the location of the operation and the cost of transporting products and services from the location to customers.

### 4. FUTURE FLEXIBILITY

Operations rarely change their location, so any new location must be capable of being acceptable, both under current circumstances and under possible future circumstances. The problem is that no one knows exactly what the future holds. Especially in uncertain environments, any evaluation of alternative locations should include some kind of scenario planning that considers the robustness of each in a range of possible futures.

### 5. RISK FACTORS

It's important evaluating the risk factors associated with possible locations. Long-term risks could include damaging changes in input factors such as exchange rates or labor costs but can also include more fundamental security risks to staff or property.

Conceiving the nature of the resources that make up the process and how they are arranged

## 5. PROCESS DESIGN: POSITIONING

Process design is concerned with conceiving the nature of the resources that make up the process and with their detailed workings. The first task of process design is to **conceive the nature of the resources that make up the process and how they are arranged**. Without appropriate resources it is difficult for the process to operate effectively. Only after this stage should the second task (of conceiving the detailed workings of the process) be attempted.

- 1) Consider the overall shape and nature of the resources that make-up the processes
- 2) Analyse the DETAILS of the activities in order to ensure that it fulfils its objectives effectively.

### 5.1 POSITION OF PROCESS RESOURCES

- POSITIONING the resources in processes is the first step in making sure that they are well designed.
- DESIGN = is to conceive the **looks, arrangement and workings of something before it is constructed**. Design is a conceptual exercise that deliver a solution that will work in practice.

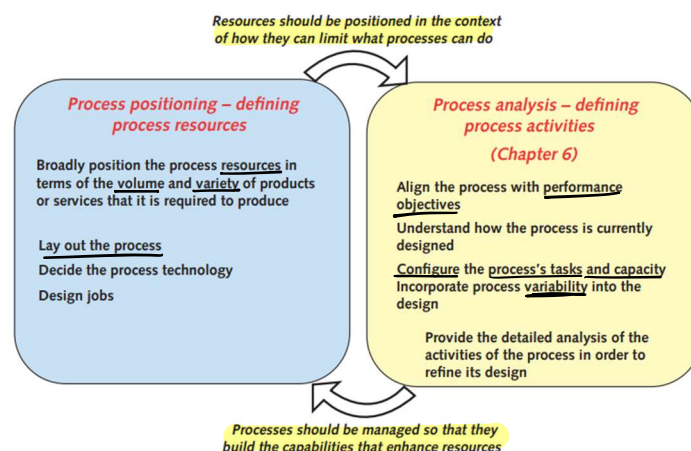
Designing processes:

- Consider the **overall shape and nature of the resources** that make up the process
- Analyze the **details of the activities** within the process in order to ensure that it fulfils its objectives effectively

### PROCESS POSITIONING AND PROCESS ANALYSIS

The positioning of resources and the detailed analysis of processes are not independent activities: they are a way of thinking about process design activity.

Defining process resources (process positioning) and defining process activities (process analysis) are strongly related and ideally should be considered alongside one another.



Process positioning ⇒ BROADLY position of the resources in terms of VOLUME-VARIETY of products/services that it is required to produce  
→ LAYOUT OF THE PROCESS  
→ DECIDE PROCESS TECHNOLOGY  
→ JOB DESIGN

Process analysis ⇒ ALIGN the process with performance objectives

Configure the process's tasks and capacity  
incorporate VARIABILITY

Refine design.

Volume-variety influence  
LAYOUT  
FLOW  
TECHNOLOGY  
JOB DESIGN

## 5.2 VOLUME-VARIETY REQUIREMENTS

Do processes match volume – variety requirements?

The volume–variety position of a process influences almost every aspect of its design. Processes with different volume–variety positions will be arranged in different ways, have different flow characteristics and have different technology and jobs. So, a first step in process design is to understand how volume and variety shape process characteristics, and to check whether processes have been configured in a manner that is appropriate for their volume–variety position.

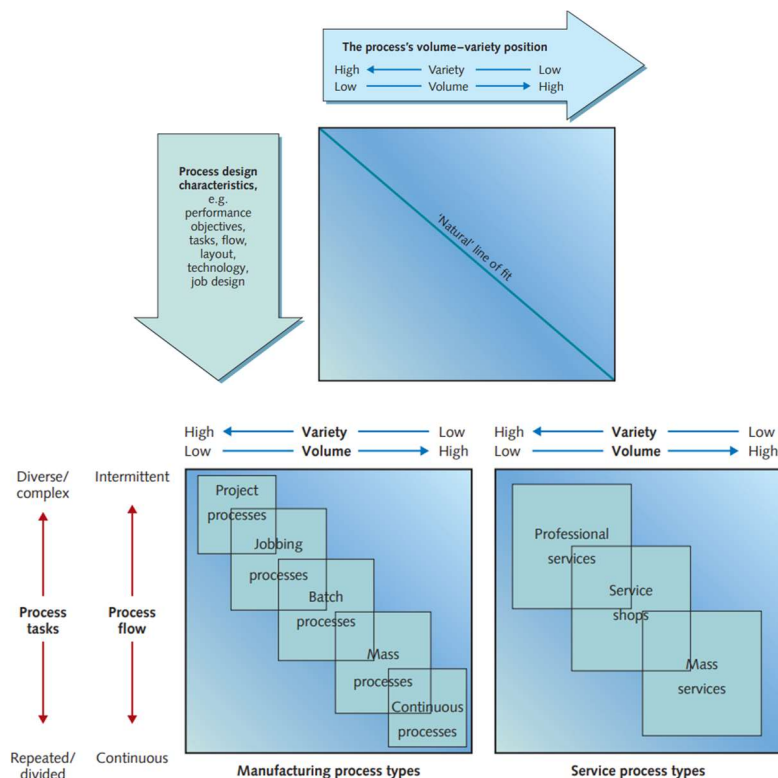
## PRODUCT-PROCESS MATRIX <sup>VOLUME-VARIETY PROCESS ALLOCATION (both for products/services)</sup>

The most common method of illustrating the relationship between a process' volume-variety position and its design characteristics is the product-process matrix. This can be used for any type of process, whether producing products or services.

For any process, the tasks that it undertakes, the flow of items, the layouts of its resources, the technology it uses and the design of jobs, are strongly influenced by its volume-variety position.

This means that most processes should lie close to the diagonal of the matrix that represents the fit between the process and its volume–variety position. This is called the natural diagonal.

So, the overall design of the process must fit its volume-variety position.



# PROCESS TYPES

Processes that inhabit different points on the diagonal of the product–process matrix are referred to as process types. Each process type implies differences in the set of tasks performed by the process and in the way materials information or customers flow through the process.

## ▪ PROJECT PROCESS

- Low volume and high variety, usually highly customized products
- The activities involved may be ill-defined and uncertain and the process map is complex. Often the timescale of making the product is relatively long, as are the intervals between the completion of each product. Each project has resources devoted exclusively to it.
- Example: shipbuilding, movie production company

## ▪ JOBBING PROCESS

- Low volume and high variety, but each product has to share operation's resources with many other.
- The process will work on a series of products but each will differ in its exact needs. Jobbing processes produce more and usually smaller items than project processes but, like project processes, the degree of repetition is low
- Example: made-to-measure tailor

## ▪ BATCH PROCESS

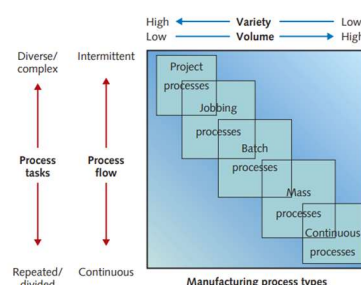
- Middle volume and middle variety
- Usually produce more than a product at a time so each part of the operation has periods when it is repeating itself while the batch is processed. The size of the batch may vary: the larger the batch the more repetitive the process.
- Example: clothing production

## ▪ MASS PROCESS

- High volume and low variety
- Its effective variety is low because the different variants do not affect the basic process of production. The activities are essentially repetitive and largely predictable and so process map will be straightforward sequences of activities
- Example: beer bottling plants

## ▪ CONTINUOUS PROCESS

- Higher volume and lower variety than mass production (production in an endless flow)
- Relatively inflexible, capital intensive technologies with highly predictable flow
- Example: electricity utilities



Working from low volume and high variety towards high volume and low variety, the **service** types:

- **PROFESSIONAL SERVICES**

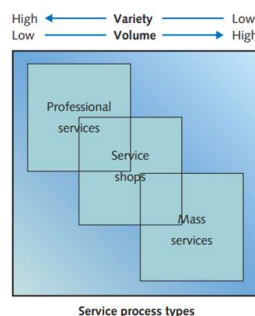
- Low volume and high variety
- Customers may spend a considerable time in the service process, because such services provide high levels of customization, so contact staff are given considerable discretion. They tend to be people-based with emphasis on the process
- Example: doctor's surgery

- **SERVICE SHOP**

- Medium volume and medium variety.
- The levels of customer contact, customization, volume of customers and staff discretion position them between the extremes of professional and mass services. Service is provided via a mix of front and back-office activities. The front-office staff have some technical training and can advise customers during the process of selling the product.  
Essentially, the customer is buying a fairly standardized product but will be influenced by the process of the sale, which is customized to the individual customer's needs.
- Example: hotels, travel agent

- **MASS SERVICE**

- High volume and low variety
- Many customer transactions and little customization. Such services are based and product oriented, with most value added in the back office, sometimes with comparatively little judgement needed by front-office staff, who may have a closely defined job and follow set procedures
- Example: supermarkets



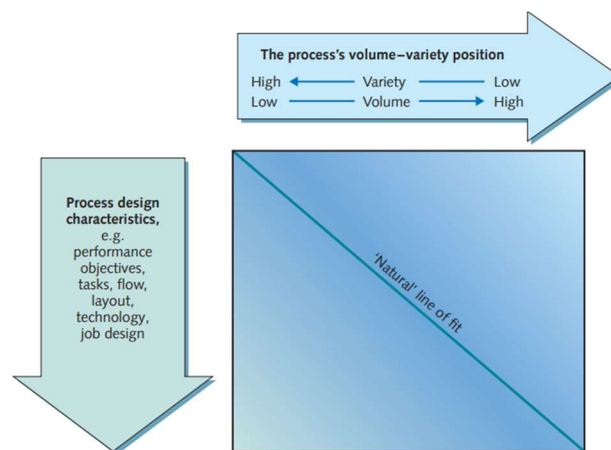
## MOVING OFF THE NATURAL DIAGONAL

A process lying on the natural diagonal of the matrix will have lower operating costs than one with the same volume-variety position that lies off the diagonal.

DIAGONAL represents the most appropriate process design for any volume-variety position.

Moving off the natural diagonal of the product–process matrix will incur excess costs:

- On right of the natural diagonal: would normally be associated with lower volumes and higher variety. This means that they are likely to be more flexible than seems to be warranted by their actual volume-variety position. They aren't taking advantage of their ability to standardize their activities. Because of this, their costs are likely to be higher
- On left of the diagonal: have adopted a position that would normally be used for higher volume and lower variety processes. Processes will therefore be over-standardized and probably too inflexible for their volume-variety position. This **lack of flexibility** can also **lead to high costs** because the **process will not be able to change from one activity to another as readily as a more flexible process**.

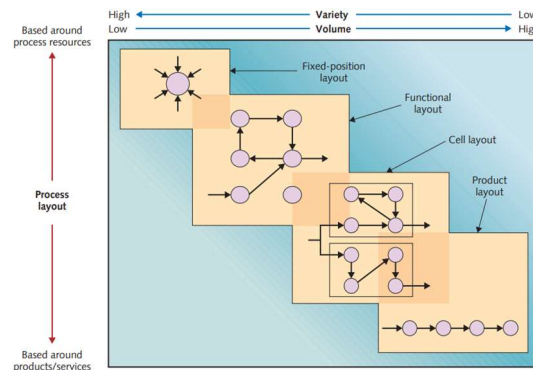


If movement down the natural diagonal of the product–process matrix changes the nature of a process, then the key elements of its design will also change. The key elements are technology and people, process resources, that are arranged in a layout.

## 5.3 PROCESS LAYOUTS

The objective of the layout decision is to minimize movements. The layout of a process is determined by its volume and variety characteristics.

Resources in low volume, high variety processes should be arranged to cope with irregular flow, while resources in high volume, low variety should be arranged to cope with smooth, regular flow.



### ▪ FIXED POSITION LAYOUT

- Low volume and high variety
- The transformed resources don't move between the transforming resources. The recipient of the processing is stationary and the equipment, machinery, plant and people who do the processing move as necessary. This could be because the product is too large or too delicate to move
- Example: open heart surgery

### ▪ FUNCTIONAL

- Similar activities or resources (or those with similar needs) are located together, because it's convenient to group them together or their utilization can be improved that way. Usually, this makes the flow pattern in the operation complex
- Example: hospitals department

### ▪ CELL

- Materials, information or customers entering the operation are pre-selected to move to one part of the operation (or cell) in which all the transforming resources are located. Internally, the cell itself may be arranged in any appropriate manner. After being processed in the cell, the transformed resources may go on to another cell. In effect, cell layout is an attempt to bring some order to the complexity of flow that characterizes functional layout
- Example: lunch products area in a supermarket.

### ▪ PRODUCT

- High volume and low variety (standardized requirements of products or services)
- Each product, information or customer follows a prearranged route in which the sequence of required activities corresponds to the sequence in which the processes have been located. The transformed resources flow along a line. The flow is clear, predictable and therefore relatively easy to control
- Example: car assembly

# LAYOUT SELECTION

Getting the process layout right is important, because of the cost, difficulty and disruption of making any layout change.

It isn't an activity many businesses would want to repeat very often. Also, an inappropriate layout could mean that extra cost is incurred every time an item is processed.

An effective layout gives clarity and transparency to the flow of items through a process. There is no better way of emphasizing that everyone's activities are really part of an overall process than by making the flow between activities evident to all participants in the process.

Layout is linked to the process types and so layout is governed by volume and variety.

|                            |                        |                                            |                                                           | Advantages                                                                                                                                                                                                                   | Disadvantages                                                                                                                                                                                |
|----------------------------|------------------------|--------------------------------------------|-----------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                            |                        |                                            |                                                           | Fixed-position <ul style="list-style-type: none"><li>Very high mix and product flexibility</li><li>Product or customer not moved or disturbed</li><li>High variety of tasks for staff</li></ul>                              | <ul style="list-style-type: none"><li>High unit costs</li><li>Scheduling of space and activities can be difficult</li><li>Can mean much movement of plant and staff</li></ul>                |
| Manufacturing process type | Potential layout types |                                            | Service process type                                      | Functional <ul style="list-style-type: none"><li>High mix and product flexibility</li><li>Relatively robust in the case of disruptions</li><li>Relatively easy supervision of equipment or plant</li></ul>                   | <ul style="list-style-type: none"><li>Low facilities utilisation</li><li>Can have very high work-in-progress or customer queueing</li><li>Complex flow can be difficult to control</li></ul> |
|                            | Project                | Fixed-position layout<br>Functional layout | Fixed-position layout<br>Functional layout<br>Cell layout | Professional service                                                                                                                                                                                                         |                                                                                                                                                                                              |
|                            | Jobbing                | Functional layout<br>Cell layout           | Functional layout<br>Cell layout                          | Service shop                                                                                                                                                                                                                 |                                                                                                                                                                                              |
|                            | Batch                  | Functional layout<br>Cell layout           |                                                           |                                                                                                                                                                                                                              |                                                                                                                                                                                              |
|                            | Mass                   | Cell layout<br>Product layout              | Cell layout<br>Product layout                             | Mass service                                                                                                                                                                                                                 |                                                                                                                                                                                              |
|                            | Continuous             | Product layout                             |                                                           |                                                                                                                                                                                                                              |                                                                                                                                                                                              |
|                            |                        |                                            |                                                           | Cell <ul style="list-style-type: none"><li>Can give a good compromise between cost and flexibility for relatively high-variety operations</li><li>Fast throughput</li><li>Group work can result in good motivation</li></ul> | <ul style="list-style-type: none"><li>Can be costly to rearrange existing layout</li><li>Can need more plant and equipment</li><li>Can give lower plant utilisation</li></ul>                |
|                            |                        |                                            |                                                           | Product <ul style="list-style-type: none"><li>Low unit costs for high volume</li><li>Gives opportunities for specialisation of equipment</li><li>Materials or customer movement are convenient</li></ul>                     | <ul style="list-style-type: none"><li>Can have low mix flexibility</li><li>Not very robust if there is disruption</li><li>Work can be very repetitive</li></ul>                              |

## LAYOUT AND SERVICESCAPES

Layout should include consideration of the look and feel of the operation to customers and/or staff.

**Servicescape** is used to describe the look and feel of the environment within an operation or process.

Ambient conditions, space factors, signs and symbols in a service operation will create an environmental experience for both employees and customers, and this environmental experience should support the service concept.

The individual factors that influence this experience will then lead to certain responses, in both employees and customers. These responses can be put into 3 main categories:

- **Cognitive**: what people think
- **Emotional**: what they feel
- **Physiological**: their body's experience

However, a servicescape will contain not only objective, measurable and controllable stimuli, but also subjective, immeasurable and uncontrollable stimuli, which will influence customer behavior.

In high-visibility processes the general look and feel of the process could be as or more important than more objective criteria such as cost or distance moved.

## 5.4 PROCESS TECHNOLOGIES

Are process technologies appropriate?

- **PROCESS TECHNOLOGIES** = are the machine's equipment and devices that help processes to transform materials, information and customers. They are different to the product technology that is embedded within the product or service itself.

### PROCESS TECHNOLOGIES AND VOLUME-VARIETY

Also process technology should reflect volume and variety: different process technologies will be appropriate for different parts of the volume-variety continuum.

- **DEGREE OF AUTOMATION** *Low volume, high variety → NOT AUTOMATED PROCESS, general purpose, small-scale and flexible-tech*  
*High volume, low variety → AUTOMATED, LARGE-SCALE TECH, CAPITAL INTENSIVE.*  
All technology needs human intervention.
  - Low volume and high variety: requires relatively unautomated, general-purpose, small-scale and flexible technologies
  - High volume and low variety: require automated, dedicated and large-scale technologies that are sometimes relatively inflexible
- **SCALE/SCALABILITY**
  - **Larger scale technology**: can process items cheaper than small-scale technologies, but usually need high volume and can cope only with low variety
  - **Smaller scale technology**: are often the nimbleness and flexibility that are suited to high-variety, lower-volume processing and can be more robust
  - Example: suppose the choice is between 3 small machines and 1 larger one. In the first case, if one machine breaks down only a third of the capacity is lost, but in the second, capacity is reduced to zero

Scalability is the ability to shift to a different level of useful capacity quickly and cost effectively.

- **COUPLING/CONNECTIVITY**  
Coupling means the linking together of separate activities within a single piece of process technology to form an interconnected processing system.
  - **Tight coupling**: gives fast process throughput. The flow is simple and predictable, making it easier to keep track of parts when they pass through fewer stages, or information when it is automatically distributed to all parts of an information network.
  - **Closely coupled technology**: expensive (each connection require capital costs) and vulnerable (a failure in one part of an interconnected system can affect the whole system)
  - Coupling is more suited to low variety and high volume. Higher-variety processing generally requires a more open and unconstrained level of coupling because different products and services will require a wider range of processing activities.

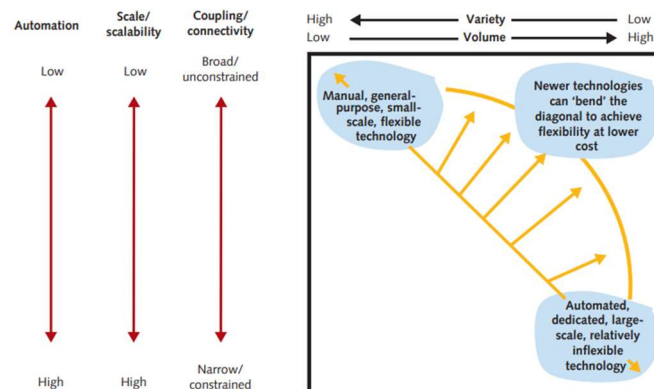
# NEW TECHNOLOGY CHANGING DIAGONAL

Different process technologies are important for different volume-variety combinations, but some newer technologies can achieve both flexibility and low cost.

Some developments in technology have overcome the implied trade-off between flexibility and cost. Specifically, digitization and the vastly increased computing power embedded in many new technologies has made it easier to achieve lower costs while not having to sacrifice the ability to provide variety. Not only does this allow a wide variety of garments to be produced at relatively low cost, it reduces the waste of material.

*fixes*

The effect of this is to 'bend' what has hitherto been accepted as the natural diagonal.



## 5.5 JOB DESIGN

How people do their tasks within a process  
 Governs people's expectations and perceptions of their contribution  
 Define activities in relation to their work colleagues  
 Channels the flow of communication between different parts of operations.

Are job design appropriate?

- **JOB DESIGN** = is about how people carry out their tasks within a process. It's important because it governs people's expectations and perceptions of their contribution to the organization. It also defines their activities in relation to their work colleagues and it channels the flow of communication between different parts of operation. Also it helps to develop the culture of the organization: it shares values and beliefs.

Job must be design to fit the nature of the process.

Some aspects of job design are common to all processes, irrespective of volume-variety position:

- **Safety**: the primary objective of job design is to ensure the safety of everyone affected by the process
- **Ethical issue**: no one should be asked to perform any task that is illegal or conflicts with held ethical beliefs
- **Work-life balance**: all jobs should be structured to promote an appropriate balance between time spent at work and time away from work

### JOB DESIGN AND VOLUME-VARIETY

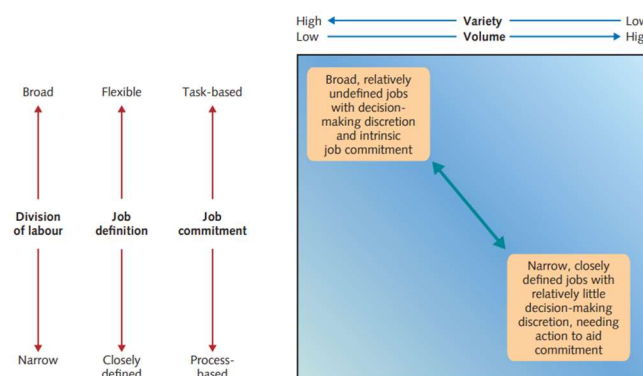
The nature and challenges of job design are governed by volume-variety characteristics of a process.

However, other aspects of job design are influenced by volume and variety:

- The extent of division of labor
- The degree to which jobs are defined
- The way in which job commitment is encouraged

Job designs in:

- **High volume and low variety** processes: are relatively closely defined with little decision-making discretion and needing action to encourage commitment. Require jobs that are relatively narrow in scope
- **Low volume and high variety** processes: require broad, undefined jobs with decision-making discretion. Such jobs tend to have intrinsic job commitment



# DIVISION OF LABOUR

How many of the tasks within any process are allocated to an individual? Separating tasks into smaller parts between individuals is called the division of labor.

## ADVANTAGE

- Promote **faster learning**  
It is obviously easier to learn how to do a relatively short and simple task than a long and complex one, so new members of staff can be quickly trained and assigned to their tasks
- **Automation becomes easier**  
Substituting technology for labor is considerably easier for short and simple tasks
- **Non-productive work is reduced**  
In complex tasks the proportion of time between individual value-adding elements can be very high

## DRAWBACKS

- **Monotonous**  
Repeating the same task may lead to an increased likelihood of absenteeism, staff turnover and error rates
- Cause **physical injury**  
The continued repetition of a very narrow range of movements can, in extreme cases, lead to physical injury. The over-use of some parts of the body (especially the arms, hands and wrists) can result in pain and a reduction in physical capability
- **Low flexibility**  
Dividing up a task into many small parts often gives the job design a rigidity that is difficult to adapt under changing circumstances
- **Poor robustness**  
Highly divided jobs imply items **passing between several stages**. If one of these stages is not working correctly (faulty equipment), the whole operation is affected. **If each person is performing the whole of the job, any problems will affect only that one person's output**

## DEGREE OF JOB

To what degree should jobs be defined?

- Job in high variety processes: difficult to define in anything but the most general terms. Such jobs may require **tacit knowledge gained over time and through experience**, and often require individuals to exercise significant discretion in what they do and how they do it
- Job in high volume and low variety processes: is likely to be defined more closely, with the exact nature of each activity defined and individual **staff trained to follow a job step by step**

# JOB COMMITMENT

How should job commitment be encouraged?

- **JOB ENLARGEMENT**

Involves allocating a larger number of tasks to individuals, usually by combining tasks that are broadly of the same type as those in the original job. This may provide a more complete and therefore slightly more meaningful job and less repetitive

Example: each operator would have twice the number of tasks to perform

- **JOB ENRICHMENT**

Increases the number of tasks in a job, but also implies allocating tasks that involve more decision-making, greater autonomy and greater control over the job. This could include the maintenance and adjustment of any process technology used, the planning and control of activities within the job and the monitoring of quality levels. The effect is both to reduce repetition in the job and to increase personal development opportunities

- **JOB ROTATION**

Moving individuals periodically between different sets of tasks to provide some variety in their activities. It can increase skill flexibility and make a small contribution to reducing monotony. However, it is not always viewed as beneficial either by management (can disrupt the smooth flow of work) or by the people performing the jobs (can interfere with their rhythm of work)

- **EMPOWERMENT**

Enhancing individuals' ability, and sometimes authority, to change how they do their jobs. Some technologically constrained processes, such as those in chemical plants, may limit the extent that staff can dilute their highly standardized task methods without consultation.

- **TEAM WORKING**

Team-based work organization is where staff, often with overlapping skills, collectively perform a defined task and have some discretion over how they perform the task

- **WORK LIFE BALANCE**

Reaching a healthy work-life balance is important because employee retention is improved, an operation's reputation in the jobs market is enhanced (helping to attract more able staff), staff physical or mental health problems are reduced and staff effectiveness is increased.

# 6. PROCESS DESIGN: ANALYSIS

To design is to conceive the looks, arrangement and workings of something before it is constructed. Process design should be **treated at two levels**: the **broad, aggregated level** and the **more detailed**.

Chapter 5 set the broad parameters for process design: it showed how volume and variety shape the resources of the process in terms of its appropriate layout, process technology and the design of its jobs. Chapter 6 takes a more detailed view of the activities undertaken by the process.

## 6.1 DETAILED PROCESS DESIGN

How individual processes are designed is important to any operation because they're the building blocks of an operation. They link together to form internal process chains through which flow material, information or people. However, no chain is stronger than its weakest link, so the output from an operation will be limited by the effectiveness of its weakest process.

Process design is so important because it involves **calculating the details of the process**: its **objectives, sequences of activities**, allocation of tasks and **capacity** and its ability to incorporate the effects of **variability**. It is the complementary activity to the broad positioning of processes.

Processes should always be designed to reflect customer and/or market requirements.

*Processes are designed in the context of the other processes that they are connected with.  
OPERATIONS ARE MADE UP OF NETWORKS AND THEY'RE A PART OF NETWORKS.*

### **PROCESS NETWORKS**

**Processes** should be **designed** in the **context of the other processes** that they are **connected with**. For making all networks work efficiently is important to understand that **any operation** is both **made up of networks** (of processes) and **is a part of networks** (of other operations). Through this network there are many process chains: that is, a thread of processes within the network.

Thinking about processes as part of a network has a number of advantages:

- **Understanding how and where a process fits into the internal network helps to establish appropriate objectives for the process**
- Possibility to make sure that **everyone** in a process has a **clear line of sight through to end customers**, so that the **people working in each process** have a **better chance of seeing how** they contribute to satisfying the operation's customers
- A clear line of sight backwards to the operation's suppliers makes the role and importance of suppliers easier to understand
- **Not understanding how process chains interact can reduce the effectiveness of the whole operation**, and **increase the risk of disruption spreading**. Process chains can become channels for disruption when things go wrong

# IDEA OF PROCESS DESIGN

The very idea of process design can be controversial. The whole idea of detailed process design can polarize opinion among operations managers. Two very opposing views are often expressed:

- **EVERYONE AND EVERYTHING IS A PROCESS**

If all work is a process, one should make an effort to apply a standardized set of disciplines and rules that can make what we do both more effective and more efficient.

The objective is to find the best sequence of activities that describe the way things should be done, then preserve it in a formal set of rules that will ensure the same result 'every time, all the time. The advantage of formal processes is the discipline it imposes.

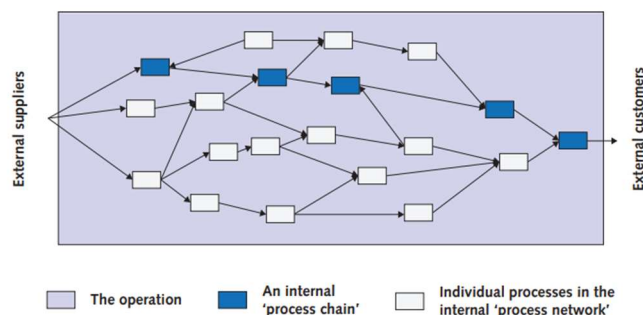
- **UNDERSTANDING, IMAGINATION AND <sup>intuitive</sup> INSIGHT**

Effective management must always rely on understanding, imagination and insight. You can't reduce everything to a standardized set of steps.

People are the most important element in any processes. Effective management must always rely on the quality of people and how they are led with inspiration and imagination. Constraining all activities into a standardization kills the essential humanity of working life

Both of these views are extreme, but they represent the two ends of a spectrum of attitudes commonly expressed.

A process is a set of resources and activities that produces value and, everyone is part of a process. Not all processes are the same; some need to be highly specified, but not all. Process design is important because it determines the actual performance of an operation's processes and the performance of process networks makes up the performance of the total opera.



## 6.2 PROCESS PERFORMANCE OBJECTIVES

The major objective of any process in the business is to support the business' overall objectives. Process design must therefore reflect the relative priority of the normal performance objectives of quality, speed, dependability, flexibility and cost. In addition to the standard performance objectives, process designs are increasingly being judged on their environmental impact.

The whole point of process design is to make sure that the performance of the process is appropriate for whatever it's trying to achieve.

| Strategic performance objective | Typical process design objectives                                                                                                                                                                                                                                                                                                        | Some benefits of good process design                                                                                                                                                                                                                                                                 |
|---------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Quality                         | <ul style="list-style-type: none"><li>• Provide appropriate resources, capable of achieving the specification of product or services</li><li>• Error-free processing</li></ul>                                                                                                                                                           | <ul style="list-style-type: none"><li>• Products and service produced 'on-specification'</li><li>• Less recycling and wasted effort within the process</li></ul>                                                                                                                                     |
| Speed                           | <ul style="list-style-type: none"><li>• Minimum throughput time</li><li>• Output rate appropriate for demand</li></ul>                                                                                                                                                                                                                   | <ul style="list-style-type: none"><li>• Short customer waiting time</li><li>• Low in-process inventory</li></ul>                                                                                                                                                                                     |
| Dependability                   | <ul style="list-style-type: none"><li>• Provide dependable process resources</li><li>• Reliable process output timing and volume</li></ul>                                                                                                                                                                                               | <ul style="list-style-type: none"><li>• On-time deliveries of products and services</li><li>• Less disruption, confusion and rescheduling within the process</li></ul>                                                                                                                               |
| Flexibility                     | <ul style="list-style-type: none"><li>• Provide resources with an appropriate range of capabilities</li><li>• Change easily between processing states (what, how, or how much is being processed)</li></ul>                                                                                                                              | <ul style="list-style-type: none"><li>• Ability to process a wide range of products and services</li><li>• Low cost/fast product and service change</li><li>• Low cost/fast volume and timing changes</li><li>• Ability to cope with unexpected events (e.g. supply or processing failure)</li></ul> |
| Cost                            | <ul style="list-style-type: none"><li>• Appropriate capacity to meet demand</li><li>• Eliminate process waste in terms of:<ul style="list-style-type: none"><li>- excess capacity</li><li>- excess process capability</li><li>- in-process delays</li><li>- in-process errors</li><li>- inappropriate process inputs</li></ul></li></ul> | <ul style="list-style-type: none"><li>• Low processing costs</li><li>• Low resource costs (capital costs)</li><li>• Low delay/inventory costs (working capital costs)</li></ul>                                                                                                                      |

## PROCESS FLOW OBJECTIVES

All the strategic performance objectives translate directly to process design, as shown in the table. But, because processes will be managed at a very operational level, process design also needs to consider a more micro and detailed set of objectives, concerned with flow through the process.

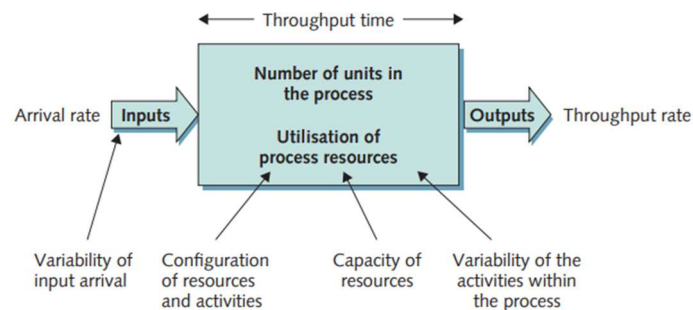
### FLOW OBJECTIVES

- Throughput rate (flow rate)  
It's the rate at which units emerge from the process  
Example: number of units passing through the process per unit of time
- Throughput time  
The average elapsed time taken for inputs to move through the process and become outputs
- Number of units in the process  
Average over a period of time. Also called the work in process or in-process inventory
- Utilization of process resources  
The proportion of available time that the resources in the process are performing useful work

## DESIGN FACTORS

The design factors that will influence the flow objectives are:

- Variability of input arrival to the process
- Configuration of the resources and activities within the process
- Capacity of the resources at each point in the process
- Variability of the activities within the process



## 6.3 CURRENT PROCESS DESIGN

How are processes currently designed?

Existing processes aren't always sufficiently well defined or described and this may lead to confusion and inhibition of improvement. So, it's important to have some recorded visual descriptor of a process that can be agreed by all those who are involved in it.

Process mapping is needed to expose the reality of process behavior. In fact much process design is a redesign and so a useful starting point is to fully understand how the current process operates. The most effective way of doing this is to map the process.

## PROCESS MAPPING

Process mapping involves describing processes in terms of how the activities within the process relate to each other.

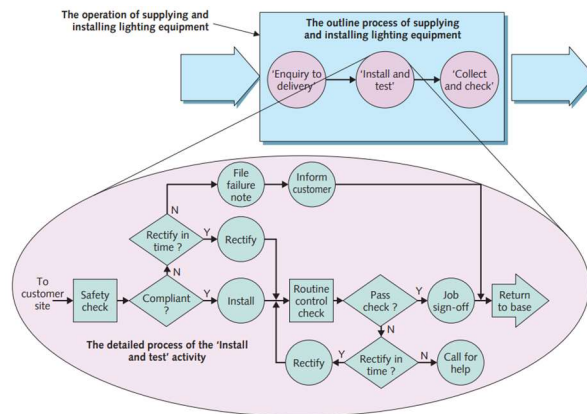
Process mapping can be done at different levels using slightly different mapping techniques. Sometimes it is useful to define the degree of visibility for different parts of the process, indicating how much of the process is transparent to customers. All the techniques have 2 main features:

- They identify the different types of activity that take place during the process
- They show the flow of materials or people or information through the process (or the sequence of activities that materials, people or information are subjected to)

## LEVELS OF PROCESS MAPPING

For a large process, drawing process maps at this level of detail can be complex. So processes are often mapped at a **more aggregated level**, before more detailed maps are drawn.

- At the highest level: the process can be drawn simply as an input– transformation–output process with materials and customers as its **input resources** and lighting services as outputs. No details of how inputs are transformed into outputs are included
- At a **more detailed level**: an outline process map (or chart) **identifies the sequence of activities** in a general way. All the activities are shown in a detailed process map
- An even more micro set of process activities could be mapped within each of the detailed process activities.



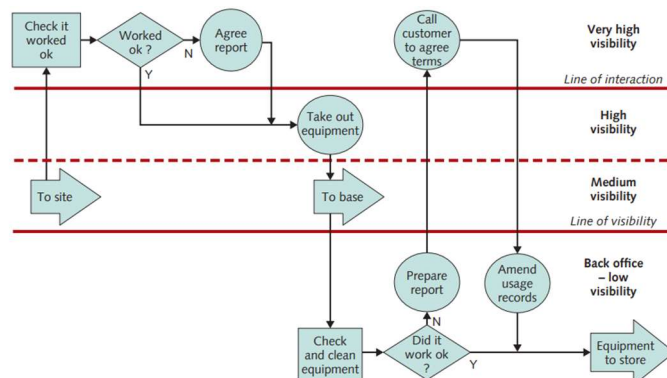
## PROCESS VISIBILITY

Sometimes it's **useful to map such processes in a way that makes the degree of visibility of each part of the process obvious**. This allows those parts of the process with high visibility to be designed so that they **enhance the customer's perception of the process**.

The process is mapped to show the visibility of each activity to the customer. The boundary between those activities that customers can and can't see is the **line of visibility**.

At the very **highest level of visibility**, above the line of interaction, are those activities that involve direct interaction between the lighting company's staff and the customer.

Other activities take place at the customer's site or in the presence of the customer but involve less or no direct interaction. **Other activities have some degree of visibility** because they take place away from the company's base and are visible to potential customers, but not to immediate customer.



## VISIBILITY

Process with a high level of customer visibility cannot be designed in the same way as processes that deal with inanimate materials or information. Processing people is different.

When a customer experiences a process it results in them feeling emotions, not all of which are necessarily rational. Any high customer-contact product (or, more likely, service) always creates an **experience for the customer**. Moreover, customer experience will affect customer satisfaction, and therefore has the potential to produce customer loyalty, influence expectations and create emotional bonds with customers.

This is why many service organizations perceive how customers experience their processes (the so-called customer journey) to be at the core of their process design.

## DESIGNING THE CUSTOMER EXPERIENCE

The design of processes that deal with customers should consider the emotions engendered at each stage of the process. Because the experience is a personal interpretation of a service, it exists only in the customer's mind and it's personal, individual. **The concept of a servicescape** is strongly related to consideration of **engaging customers** so that they connect with the process in a personal way.

One of the most **common methods of designing such processes** is to **consider TOUCHPOINTS**. These have been described as **everything the consumer uses to verify their service's effectiveness**.

They are the points of contact between a process and customers, and there might be many different touchpoints during the customer journey. It is the **accumulation of all the experiences from every touchpoint interaction** that shapes customers' judgement of the process.

The features of a process at the touchpoints are called **CLUES**: these are the messages that customers receive or experience as they progress through the process.

The emotions that result from these cues contain the messages that the customer will receive and therefore influence how a customer will judge the process.

Various tools and techniques have been developed to help design and improve the customer's journey and influence their experience. Most try to capture the series of touchpoints that customers encounter during a service.

Example: mapping the route customers take through websites, or clinical pathways in hospitals.

Another approach is to ask staff, managers or independent advisers to act as surrogate customers and conduct a walk-through audit. Extending this idea, emotion mapping captures the emotions felt by customers as they move through the process.

## **6.4 CONFIGURATION OF PROCESS**

Process map show how the activities of any process are currently arranged and help to suggest how they can be reconfigured.

Getting to grips with process capacity means understanding the task precedence, examine how alternative process design options can incorporate series and parallel, calculate cycle time and process capacity and establish the relationship between throughput, cycle time and work-in-process.

### **TASK PRECEDENCE**

Process design must respect task precedence. It defines what activities must occur before others, because of the nature of the task itself.

At its simplest level, task precedence is defined by:

- The individual activities that comprise the total process task
- The relationship between these individual activities

Task precedence is described by using a precedence diagram, which includes:

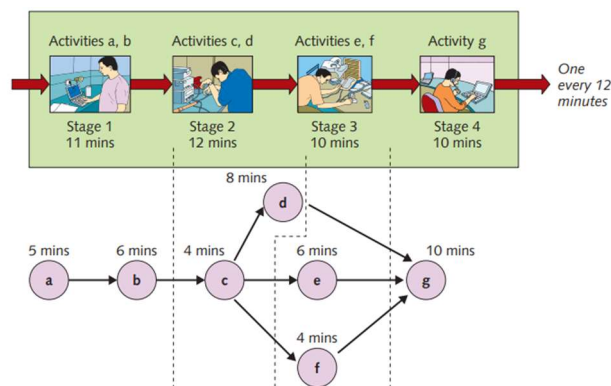
- The time necessary to perform the total task (work content)
- The time necessary to perform each of the individual activities within the task

# SERIES AND PARALLEL CONFIGURATION

The general configuration of a process involves deciding the extent to which activities are arranged sequentially and the extent to which they are arranged in parallel. Most processes will adopt a combination of series and parallel configurations.

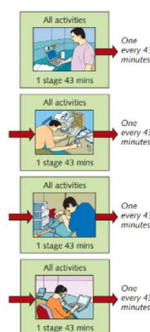
## ADVANTAGE OF SERIES

- A more controlled flow through the process that is relatively easy to manage
- Simple material handling: especially if a product being manufactured is heavy or large
- Lower capitals requirements
- More efficient operation. If each stage is performing only a small part of the total job, the person at the stage may have a higher proportion of direct productive work as opposed to the non-productive parts of the job, such as picking up tools and materials.



## ADVANTAGE OF PARALLEL

- Higher mix flexibility  
If the process needs to produce several types of products or services, each stage could specialize in different types
- Higher volume flexibility  
As volume varies, stages can simply be closed down or started up as require
- Higher robustness  
If one stage breaks down or ceases operation in some way, the other parallel stages are unaffected; a long–thin arrangement would cease operating completely
- Less monotonous work



## CYCLE TIME

The cycle time is the **time between completed units emerging from the process**.

- Indicates the output that must be achieved by a process
- It's a vital factor in process design
- Has a significant influence on most of the other detailed design decisions
- It can be used to represent both the demand placed on a process and the process's capacity
- **Sets the pace of the process** and process analysis derives from an understanding of the required process cycle time

It is calculated by considering the likely demand for the products or services over a period and the amount of production time available in that period.

$$\text{Cycle time} = \frac{\text{Available time}}{\text{Number to be processed}}$$

$$\text{Take time} = \frac{\text{Available time}}{\text{N}^\circ \text{ of products asked by customers}}$$

In ideal case: CT = TT

## PROCESS CAPACITY

After decided the cycle time as the **output** that must be achieved by a process, the next decision must be how capacity is needed by the process in order to meet the cycle time.

In order to calculate process capacity, we need to know the **work content of the process task**. The **larger the work content of the process task and the smaller the required cycle time**, the more capacity will be necessary if the process is to meet the demand placed upon it.

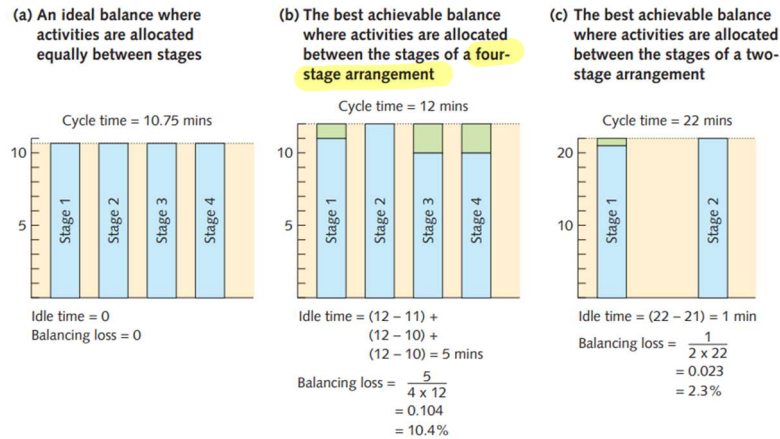
## PROCESS BALANCING

Balancing a process involves attempting to **allocate activities to each stage as equally as possible**.

Because of cycle time, the more equally work is allocated, the less time will be wasted at the other stages in the process. In practice it's impossible to achieve perfect balance.

- **The effectiveness of the balancing activity** is measured by balancing loss = is the **time wasted through the unequal allocation of activities** as a percentage of the total time invested in processing.
- The BOTTLENECK is the stage of a process that has the longest allocation of work and this **stage will dictate the output of the whole process**

Balancing loss is that proportion of the time invested in processing the product or service that is not used productively. Allocating work equally to each stage in a process (balancing) smooths flow and avoids bottlenecks.



## THROUGHPUT RATE

The cycle time of a process is a function of its capacity: for a given amount of work content in the process task, the **greater the capacity of the process the smaller its cycle time**.

In fact, the capacity of a process is often measured in terms of its cycle time, or more commonly the reciprocal of cycle time that is called throughput rate.

$$\text{Throughput rate} = \frac{1}{\text{Cycle time}}$$

However, a **high level of capacity** (short cycle time and fast throughput rate) doesn't necessarily mean that material, information or customers can move quickly through the process.

This will **depend on how many other units are contained within the process**. If there is a large number of units within the process they may have to wait in **work-in-process** inventories for part of the time they are within the process (throughput time).

## LITTLE'S LAW

The Little's Law is the **mathematical relationship that relates cycle time, work-in-process and throughput time**. It is simple but very useful, and it works for any stable process.

$$\text{Throughput time} = (\text{work in process}) \times (\text{Cycle time})$$

$$\text{Work in process} = (\text{Throughput time}) \times \frac{1}{\text{Cycle time}}$$

$$\text{Work in process} = (\text{Throughput time}) \times (\text{Throughput rate})$$

## **6.5 PROCESS VARIABILITY**

Processes have to cope with variability, both in terms of time and the tasks that are performed within the process. This variability can have very significant effects on process behavior, usually to reduce process efficiency.

Queuing theory can be used to understand this effect. It's important to understand the relationship between **process utilization** and the **number of units waiting to be processed** (or throughput time).

### **SOURCES OF VARIABILITY IN PROCESSES**

There are many reasons why variability occurs in processes:

- **The late** (or early) **arrival of material**, information or customers at a stage within the process
- The temporary malfunction or **breakdown** of process technology within a stage
- The necessity for **recycling misprocessed materials**, information or customers to an earlier stage in the process
- The **misrouting of material**, information or customers that then needs to be redirected
- Each **product** or service being **processed** might be **different**
- **Products or services**, although essentially the same, might require slightly **different treatment**.  
Example: in the computer test and repair process, the time of some activities will vary depending on the results of the diagnostic check
- With any human activity there are slight variations in the physical coordination and effort on the part of the person performing the task that result in variation in activity times, even of routine activities

All these sources of variation within a process will interact with each other, but result in 2 types of variability:

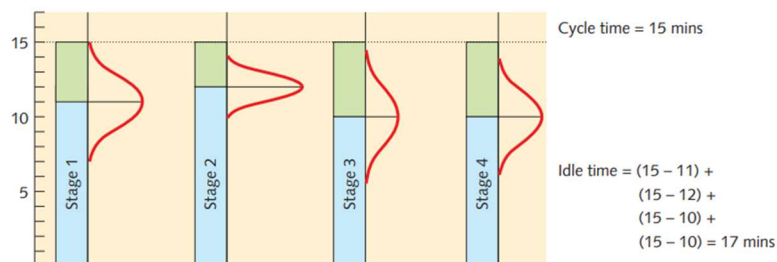
- Variability in **demand** for processing at an individual stage within the process, usually expressed in terms of variation in the inter-arrival times of units to be processed
- Variation in the time taken to perform the activities at each stage.

## ACTIVITY TIME VARIABILITY

The effects of variability within a process will depend on whether the movements of units between stages, and hence the inter-arrival times of units at stages, are synchronized or not.

Variability in a process acts to reduce its efficiency.

- When movement is synchronized, the inter arrival time of units at each stage is fixed at the cycle time
- Without synchronization, the inter-arrival time at each stage will itself be variable

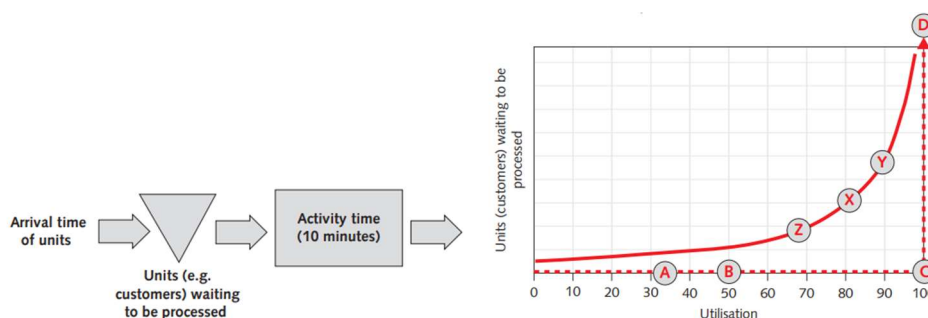


Example: the graph shows the average activity time at each stage of the process, but also the variability around the average time. Suppose that it was decided to synchronize the flow between the 4 stages by using a simple traffic light system that ensured all movement between the stages happened simultaneously. The interval between each synchronized movement would have to be set at an interval that would allow all stages to have finished their activities. This synchronized indexing time would have to be set at 15 and this becomes the effective cycle time of the process. Every stage will experience some degree of idle time, the average idle time at each station being the cycle time minus the average activity time at that station. This reduction in the efficiency of the process is only partly a result of its imbalance. The extra lost time is as a result of activity time variability.

## ARRIVAL TIME VARIABILITY

Process variability results in simultaneous waiting and resource underutilization.

Example: the process in the graph comprises one stage that performs exactly 10 minutes of work. Units arrive at the process at a constant and predictable rate. If the arrival rate is one unit every 30 minutes, then the process will be utilized for only 33.33% of the time, and the units will never have to wait to be processed (point A). If the arrival rate increases to one arrival every 10 minutes, the process is now fully utilized, but, because a unit arrives just as the previous one has finished being processed, no unit has to wait (point C). So, in a perfectly constant and predictable world, the relationship between process waiting time and utilization is a rectangular function.



1 unit every 30 min }  $\frac{10}{30} = 33,3\% \rightarrow$  No unit will wait  
10 mins of work

1 unit every 10 min  $\Rightarrow \frac{10}{15} = 100\% \rightarrow$  fully utilized

## ACTIVITY AND ARRIVAL TIME VARIABILITY

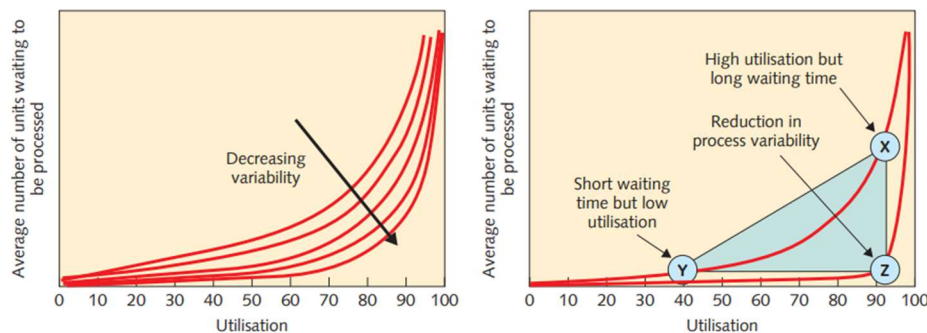
It's difficult to imagine any real process with absolutely zero variation. In practice there is likely to be a combination of activity and arrival time variation.

As a process moves closer to 100% utilization, the higher the average waiting time will become. Because of variability, the only way to guarantee very low waiting times for the units is to suffer low process utilization.

In fact, the greater the variability, the more the waiting time utilization deviates from the simple rectangular function of the no variability conditions. This phenomenon, called the OM triangle, has important implications for the design of processes.

In effect it presents 3 options to process designers wishing to improve the waiting time or utilization performance of their processes. Either:

- X. Accept long average waiting times and achieve high utilization
- Y. Accept low utilization and achieve short average waiting times
- Z. Reduce the variability in arrival times, activity times, or both, and achieve higher utilization and short waiting times



To analyze processes with both inter-arrival and activity time variability, queuing or waiting line analysis can be used. The only way to have both fast throughput time and high utilization simultaneously is to reduce variability in its processes, which may itself require strategic decisions such as limiting the degree of customization of products or services, or imposing stricter limits on how products or services can be delivered to customers, and so on.

It also demonstrates an important point concerned with the day-to-day management of process – the only way to absolutely guarantee 100% utilization of resources is to accept an infinite amount of work-in-process and/or waiting time.

Process design involves some choice between utilization, waiting time and variability reduction.



# 7. SUPPLY CHAIN MANAGEMENT

## 7.1 SUPPLY CHAIN MANAGEMENT

How are supply chains and networks managed? This involves determining **key performance objectives** for the supply network, deciding on supplier **relationships**, developing **sourcing strategies**, selecting appropriate **suppliers** and **negotiating**.

- **SUPPLY CHAIN** = is the **interconnected journey** that **raw materials, components and goods** take before their **assembly and sale to customer**.
- **SUPPLY NETWORK** = all **operations** that are **linked together** as **provide products and services** through **to end customers**. It's a set of supply chains.
- **SUPPLY CHAIN MANAGEMENT** = is the **management of the relationships** and flows **between the chain of operations and processes** that deliver value in the form of products and services to the ultimate consumer.

FLows of INFO and GOODS  
↑

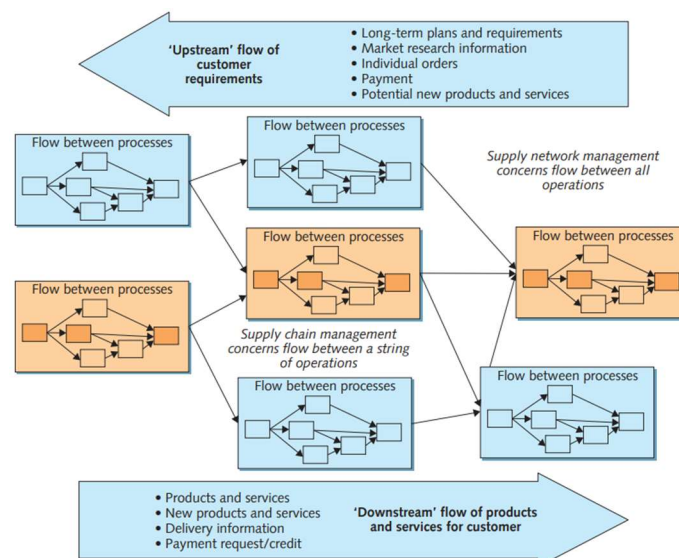
## MANAGING SUPPLY CHAIN AND SUPPLY NETWORK

Supply chain management is concerned with managing supply chains and supply networks is equally **concerned with flows of information** as with **flows of products or services**.

It's an essential activity because the overall success of any individual firm is fundamentally impacted by the effectiveness of its supply network.

Supply chain management applies equally to:

- **Physical flows**: movement, storage or sale the physical products
- **Non-physical flows**: financial services, insurance providers, consultants. This is because they all have to manage their supply chains (they have suppliers and customers, they purchase services)



In addition, most principles applied to external networks (organization-to-organization) can also be applied to internal networks (process-to-process).

Supply chain management concept apply to:

- Internal relationships between processes
- External relationships between operations

## SUPPLY CHAIN OBJECTIVES

The fundamental objective of SCM is to satisfy the needs of the end customer. So each operation or process in the network should contribute to deliver products or services to end customers that meet their expectations in terms of quality, speed, dependability, flexibility, cost and sustainability.

### ▪ QUALITY

The quality of a product or service when it reaches the customer is a function of the quality performance of every operation in the chain that supplied it. The implication of this is that errors in each stage of the chain can multiply in their effect on end-customer service

### ▪ SPEED

- How fast customers can be served (the elapsed time between a customer requesting a product or service and receiving it in full). However, fast customer response can be achieved simply by over-resourcing or overstocking within the supply chain
- The time taken for goods and services to move through the chain.  
Products that move quickly down a supply chain from raw material suppliers through to retailers will spend little time as inventory

### ▪ DEPENDABILITY

Guarantee on-time delivery by keeping excessive resources within the chain. Dependability of throughput time is a desirable aim because it reduces uncertainty within the chain.

### ▪ FLEXIBILITY

It's the chain's ability to cope with changes and disturbances.

The flexibility includes focusing on the end customer and ensuring fast throughput and responsiveness to customer needs and flexibility to cope with changes

### ▪ COST

In addition to the costs incurred within each operation to transform its inputs into outputs, the supply chain incurs additional costs that derive from each operation in a chain doing business with each other. These transaction costs include: the costs of finding appropriate suppliers, setting up supplier agreements, running ongoing supply, supplier training

### ▪ SUSTAINABILITY

Any organization that subscribes to sustainability objectives will want to make sure that it purchases its input products and services from suppliers that are similarly responsible. Example: buying from local suppliers, sourcing supplies from suppliers with ethical practices, choosing environmentally friendly products and services, using minimal packaging.

# EFFICIENT VS RESPONSIVE SUPPLY NETWORK

Different products or services often exhibit clear differences in how they compete in markets. Efficient (lean) supply networks are suitable for stable functional services and products, while responsive (agile) supply networks are more suitable for less predictable innovative services and products.

- FUNCTIONAL → EFFICIENT SN

Demand is stable, predictable and the profit margin is low

Efficient supply chain because:

- Keep inventory or service capacity low
- Focus on maximize utilization of all resources in the supply network to minimize cost

- INNOVATIVE → RESPONSIVE SN

Demand is more uncertain but the profit margin is high

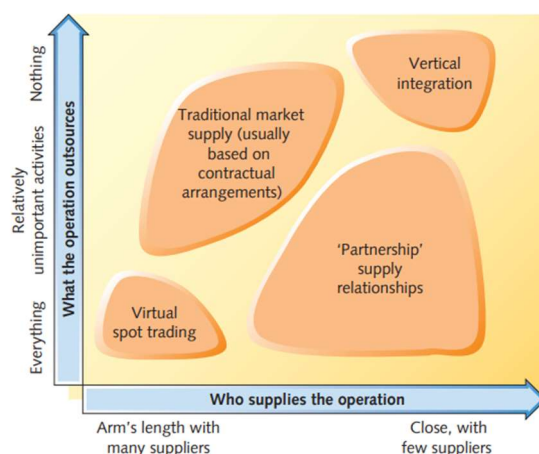
Responsive supply chain because:

- Inventory or service capacity in the network will be deployed as close as possible to the customer, so the network is still able to supply even when dramatic changes occur in customer demand
- Focus on high service levels and responsive supply to end customer

## 7.2 SUPPLY CHAIN RELATIONSHIP

What type of supply chain relationship should be adopted? The **relationship between operations** in a **supply chain** is the basis on which the exchange of products, services, information and money is conducted. So, **managing supply networks** is about **managing relationships**.

In considering the type of supply chain relationship, we can consider **what the operation outsources** (ch 4: scope and structure) and **who supplies the products or services**. In considering who supplies, managers must decide **how many suppliers** to use and **how close the relationship should be**.



## TRANSACTIONAL VS PARTNERSHIP RELATIONSHIP

### TRANSACTIONAL

Involve purchasing products and services in a pure market fashion, often **seeking the best supplier every time it is necessary** to make a purchase.

Can be **long or short term**, but there's **no guarantee beyond the immediate contract**.

They are appropriate when short term benefits are important.

### PARTNERSHIP

**Longer term** and involve a **commitment to work together over time to gain mutual advantage**.

These **relationships** emphasize **cooperation**, **frequent interaction**, information sharing, joint problem solving and even profit sharing. The concept of mutuality is important here.

Most businesses will have a portfolio of, possibly, widely differing relationships. All supply chain relationships can be describe by the balance between their contractual and partnership elements.

| Transactional relationship benefits                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Partnership relationship benefits                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <ul style="list-style-type: none"> <li>- Retains competition between alternative suppliers</li> <li>- If demand changes, customer can change the number and type of suppliers; a faster and cheaper alternative to redirecting internal activities</li> <li>- Wider variety of innovation sources (though access may be harder than in partnership relationships)</li> <li>- Useful in assessing a supplier before potential move to partnership model</li> <li>- Good for one-off or irregular purchases</li> </ul> | <ul style="list-style-type: none"> <li>- Higher levels of loyalty to customers than transactional relationships</li> <li>- Reduced time and effort of frequent re-contracting</li> <li>- Reduced transaction costs</li> <li>- Reduced cost of compliance monitoring</li> <li>- Fewer quality failures and unanticipated failures</li> <li>- Earlier failure identification</li> <li>- Emphasis on joint problem solving during failure episodes, rather than assigning blame</li> <li>- Generation of increased value by leveraging shared competences</li> </ul> |

## 7.3 SOURCING CONFIGURATION

Once managers have determined the type of supply chain relationship appropriate, they must consider preferences in relation to sourcing configuration. This involves deciding on the alternative sourcing methods (multiple, single, delegated, parallel sourcing) that should be used for different categories of products or services (non critical, bottleneck, leverage and strategic).

### SOURCING APPROACHES

- **MULTIPLE** <sup>performance objectives: PRICE, DEPENDABILITY</sup> <sup>1 PRODUCT, MANY SUPPLIERS</sup>  
<sup>COMPETITIVE MARKETS → switching costs are low</sup>

Involves obtaining a product or service component from more than one supplier.

It is commonly seen in competitive markets where switching costs are low and performance objectives are primarily focused on price and dependability.

Advantages:

- Can help maintain competition in the supply market
- Reduce supply risk
- Increase flexibility in the face of supplier failure or changes in customer demand
- Allow the buyer to tap into a more diverse range of knowledge and expertise
- Prevent supplier dependence, thus allowing for changes in purchase volumes without the risk of supplier bankruptcy

Disadvantages:

- Requires effort to ensure quality
- Hard to encourage supplier commitment with suppliers who are less willing to invest in new processes in the absence of longer-term contracts

- **SINGLE** <sup>→ MORE PRODUCT, 1 SUPPLIER</sup>

Involves buying all of one product or service component from a single supplier.

Often these components represent a high proportion of total spend and are of strategic importance. In other cases firms simply prefer the simplicity of single sourcing.

Advantages:

- Reduced transactional cost
- Longer-term outlook and focus on a wider range of performance objectives
- The mutual dependency effects encourage more commitment, effort, and higher-quality

Disadvantages:

- Increased risk of lock-in and upward price pressure
- Reduction in the firm's <sup>CONTRADICTORY</sup> bargaining power
- Relationships are vulnerable to disruption if a failure to supply occurs

- **DELEGATED** → 1 SUPPLIER THAT DELIVERS AN ENTIRE SUB-ASSEMBLY PART

One supplier is responsible for delivering an entire sub-assembly (or a package of services) and not a single part (or an individual service).

Advantages:

- Reduction of the number of first-tier suppliers
- Allow a focus on strategic partners

Disadvantages:

- Can alter the dynamics of the supply market
- Risk of creation of mega-suppliers with significant power in the network

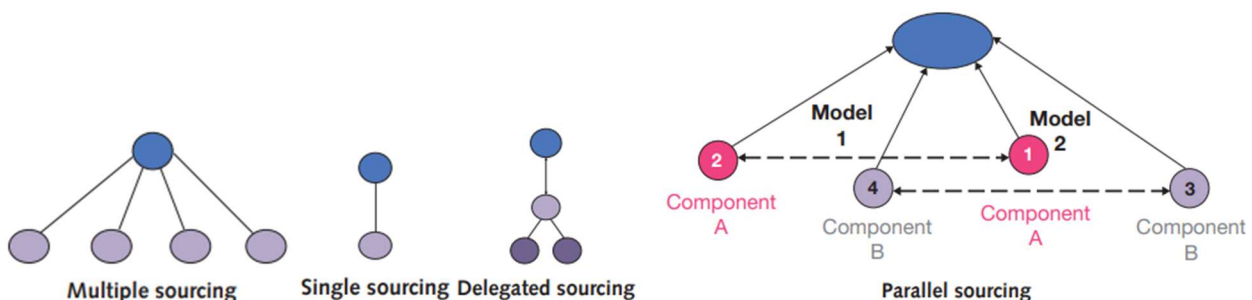
- **PARALLEL** → <sup>TRADE-OFF</sup> Single-source relationships for a single comp. for different service packages or prod. models.

Has the aim of providing the advantages of multiple sourcing and single sourcing simultaneously.

It involves having single-source relationships for a single component for different service packages or product models. If a supplier is unsatisfied, It's possible to switch to the alternative supplier who currently provides the same component but for a different service package or product model.

Advantages:

- It maintains competition
- Allows switching in circumstances of supplier failure, but it's complex to manage



The last years has seen a trend for organizations to reduce their supplier base. With creation of partnership relations, a firm can reduce running costs such as costs of ordering, expediting, supplier visits. But this increased supply risk because some suppliers are more able to act opportunistically.

depends on the risks in the supply market and what is critical for the production.

## SOURCING STRATEGY DECISION

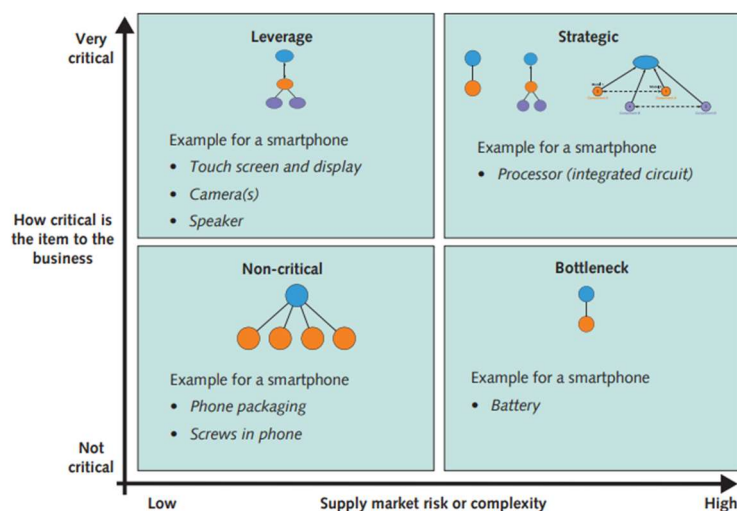
It's important to **decide** which **sourcing strategy** is most suitable under different circumstances. What is the risk in the supply market and what is the criticality of the product or service to the business?

Sourcing strategy is governed partly by:

- **RISK** in supply market  
number of alternative suppliers, how easy is to switch from one supplier to another, exit barriers, cost of bringing operations back in-house
- **CRITICALITY** of the item  $\rightarrow$  % of total purchase cost  
A product/service component's importance is consider in terms of % of total purchase cost or the impact on business growth

By looking at these 2 dimensions, it's possible to position components in one of the 4 quadrants:

- **NON CRITICAL**
  - Low risk of supply market and low complexity: large number of alternative suppliers
  - Non critical item: low proportion of the total cost of the product (consumption material)
- **BOTTLENECK**
  - High risk: limited supply alternatives and high switching costs
  - Non critical item (not strategic)
  - Single sourcing or reduce purchasing
- **LEVERAGE**
  - Low risk: easy to source because there are large numbers of suppliers
  - Critical item
  - Exploit bargaining power for enhance suppliers' competition and obtain advantages
- **STRATEGIC**
  - High risk: complex to acquire, few suppliers, high switching cost
  - Critical item (strategic)
  - Single sourcing and partnership for have close relationship with the client



## 7.4 SUPPLIER SELECTION AND NEGOTIATION

Supplier selection is vital for effective supply chain management. Supplier selection involves initial qualification, identifying key measurement criteria, collecting information on potential suppliers, making a final selection and negotiating with suppliers to agree the terms of the engagement.

### QUALIFICATION OF SUPPLIERS

Minimize possible supp.  
Pre-qualification → establishing a threshold against 3 capabilities → TECHNICAL, OPERATIONS, FINANCIAL

Initial qualification is aimed at reducing possible suppliers. Pre-qualification criteria are used for establishing a minimum <sup>specific</sup> threshold against 3 capabilities:

- **Technical capability**  
Does the supplier have service or product knowledge to supply to the required specification?
- **Operations capability**  
Does the supplier have the process knowledge to ensure consistent, responsive, dependable and reasonable-cost supply? *~ performance objectives*
- **Financial capability**  
Does the supplier have the financial strength to fund the business in short and long term?

### MEASUREMENT CRITERIA

Measurement criteria for supplier selection should align with the overall performance objectives of the supply network.

- Determination of the relative importance of key performance objectives (6)
- Determination of measurable criteria for key performance objectives

### INFORMATION ON SUPPLIERS PROSPECTIVE

It's necessary to:

- collect information via requests for quotes or requests for information
- visit the suppliers and test to assess supplier competence

The amount of time and effort invested in information search is influenced by the strategic importance of the purchase and the perceived capability of the supply base.

It's necessary to collect more information when the firm is buying a service or product of strategic importance and the supply market is more uncertain.

AHP ⇒ Analytic Hierarchy Process ⇒ provides quantifiable info for each key-selection criteria and a weighting of their relative importance allow for an "objective" assessment of different suppliers.

TCO ⇒ Total Cost of Ownership ⇒ provide detailed info of all possible costs associated with procurement to reach more rational decisions.

## SELECTION

Having arrived at a group of viable alternatives, rarely is one potential supplier so clearly superior that the decision is self-evident.

Usually alternative attributes need trading off, by using some kind of supplier scoring or assessment procedure to rate alternative suppliers. These decision-making models include:

- Weighted score method and the Analytic hierarchic process (AHP)  
Aim to provide quantifiable information for each key selection criterion and a weighting of their relative importance to allow for an objective assessment of different suppliers
- Total cost of ownership (TCO)  
Seeks to provide detailed information on all possible costs (rather than simply the price of the service or product) associated with procurement to reach more rational decisions

## NEGOTIATION

Once the preferred supplier has been identified by the scoring assessment methods, negotiation takes place to try to agree the key terms of engagement.

For those involved in this activity, it's important to understand the various tactics (including emotion, logic, threat, compromise) that may be employed during the negotiation process.

Negotiation could be:

- LOSE-LOSE  
When the negotiator advocates for one party to gain its most favorable outcome  
But emphasis on short term solutions, personalized conflicts, damage to the relationship between the buyer and supplier
- WIN-WIN  
Collaborative negotiation or mutual gains bargaining. Emphasis on meeting the needs of both parties, investment in developing personal relationships using creative brainstorming techniques and favoring long term objectives

## 7.5 SUPPLY CHAIN RUN EFFECTIVELY

Now we consider how such **relationships** can be **managed to maximize the ongoing performance of the supply network**. This involves examining information flows, perception gaps, the development of suppliers and the use of logistics (distribution).

### MANAGING INFORMATION FLOW

Another issue is how to ensure a **two-way flow of information between customer and supplier**. Managing supply chain relationships is not just choosing the right suppliers, but also **ensuring that suppliers are given the right information and encouragement to maintain smooth supply**.

If customer and supplier see themselves as partners, the free flow of information and a mutually supportive tolerance of occasional problems is the foundation of smooth supply.

### SERVICE-LEVEL AGREEMENT <sup>PARTS</sup> SLA

Service-level agreements are employed to **help formalize expectations between parties in a supply network**. Some organizations bring a degree of formality to supplier relationships by encouraging (or requiring) all suppliers to agree service-level agreements (SLAs).

SLAs are **mechanisms for governing the ongoing relationship between suppliers and customers** (the organization), covering:

- Response times
- Range of services
- Dependability of service supply

SLAs must be treated as working documents that establish the details of ongoing relationships in the light of experience.

### LOGISTICS AND INTERNET OF THINGS (IOT)

Internet-based communication has had a significant impact on logistics and distribution, in particular through the adoption of the Internet of Things (IoT).

IoT is a network of **physical objects** (such as products, equipment, trucks) **that have electronics, software and sensors implanted in them that can gather and exchange data**. Combined with global positioning systems (GPS) it permits instantaneous tracking of trucks, materials and people, allowing logistics companies, warehouses, suppliers and customers to share **knowledge of where products are in the network and where they are going next**.

This permits more effective coordination and creates the potential for cost savings.

# MINIMIZE PERCEPTION DIFFERENCES

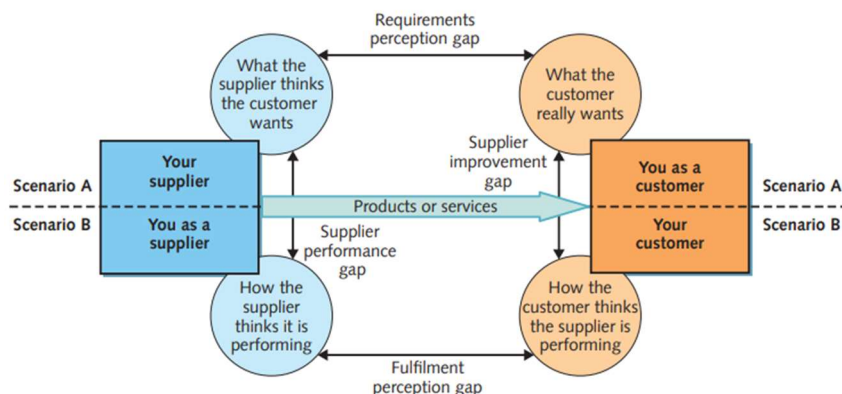
Another issue is how to minimize the common mismatches between how customer and supplier perceive both what is required and how the relationship is performing.

From a customer perspective:

- **REQUIREMENTS PERCEPTION GAP**  
First you (the customer) have an idea about what you really want from a supplier. But no SLA can capture everything about what is required. There may be a gap between how you as a customer describe what is required and how the supplier interprets it.
- **FULFILMENT PERCEPTION GAP**  
As a customer, you will have a view on how your supplier is performing in terms of fulfilling your requirements.
- **SUPPLIER IMPROVEMENT GAP**  
Gap between what the customer wants from its supplier and how it perceives the supplier's performance. This will influence the kind of supplier development goals set by the customer.
- **SUPPLIER PERFORMANCE GAP**  
Gap between your supplier's perceptions of your needs and its own assessment of performance. This will indicate how a supplier initially sees themselves improving their own performance.

From a supplier perspective, the same approach can be used to understand customer perceptions, both of their requirements and their view of your (the supplier's) performance.

Unsatisfactory supply chain relationships can be caused by requirement, fulfilment, supplier improvement and supplier performance perception gaps.



Supply chain relationships are multi-tiered configurations and so a formal contract between two organizations in a supply network has to be interpreted by managers in these organizations.

# DEVELOPING SUPPLIERS

Unless a relationship is purely market-based and transactional, it is in a customer's long-term interests to take some responsibility for developing supplier capabilities. Supplier development:

- enhances the service from the supplier
- lead to greater supplier loyalty and long-term commitment
- improve the effectiveness of the supply network as a whole

## PROCESS OF SUPPLIER DEVELOPMENT

### 1. Select product and supplier

The suppliers that can be developed are the ones that provide strategic products/services; the ones that have relatively weak current performance but with potential for improvement (inputs are particularly important); the ones where switching to another supplier is difficult.

### 2. Form a project team and gain buy-in

The project team should bring together key stakeholders from the buyer, supplier and other relevant parties. It's important the cooperation from key stakeholders.

### 3. Agree goals and measures

Supplier development requires clear scoping in terms of timelines, costs and key deliverables to ensure that both buyer and supplier agree on what constitutes success.

### 4. Implement, monitor and learn

It's important to monitor progress and intervene if the project is deviating from its performance targets. Celebrate success helps to reinforce the value of supplier development to all stakeholders. Supplier development doesn't stop when a project is complete: the objective is to develop suppliers' capabilities to enable them to continue improving.

# MANAGE LOGISTICS

The last issue is how to effectively manage logistics (or distribution). Logistics is the activity of moving products from suppliers to their customers.

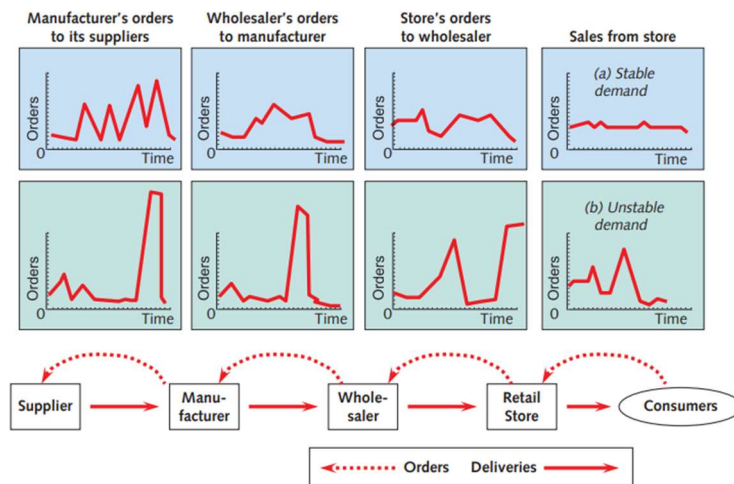
- 1<sup>st</sup> party logistics (1PL): logistics activity is an entirely internal process
- 2<sup>nd</sup> party logistics (2PL): outsource logistics services over a segment of a supply chain
- 3<sup>rd</sup> party logistics (3PL): a firm contracts a logistics company to work with other transport companies to manage their logistics operations more fully
- 4<sup>th</sup> party logistics (4PL): integrator that assembles the resources, capabilities and technology of its own organization and other organizations to design, build and run supply chain solutions
- 5<sup>th</sup> party logistics (5PL): defining themselves as broadening their scope to include e-business

## 7.6 SUPPLY CHAIN DYNAMIC UNDER CONTROL

There are dynamics that exist between firms in supply chains that cause errors, inaccuracies and volatility. This is known as **BULLWHIP EFFECT**.

Relatively **small changes at the demand end of the chain increasingly amplify into large disturbances as they move upstream**. The further back up the supply chain an operation is placed, the more drastic are the fluctuations.

There are other factors that make the fluctuations even more pronounced: **lack of forecasting**, errors in forecasting, quantity discounting (which encourages less frequent but larger orders), price fluctuations, time lags between the flows of information (order) and flow of materials (deliveries).



Supply chain dynamics have a number of harmful impacts on those operating in a supply network:

- Costs of outsized facilities and **excess inventories**
- For human resources, service capacity typically oscillates between **underutilization** and **overutilization**, and many firms are **forced to hire, fire and rehire employees**
- **Irregular patterns of work cause inefficiencies, delays and customer dissatisfaction**

## BLOCKCHAIN

Blockchain technology has significant potential for supply chain management:

- Supply networks are complex and involve many operations. However, for a distributed ledger network, **the larger the number of nodes, the more secure it is**, so the innate **complexity of supply networks actually strengthens the technology**
- **Large number of transactions necessary for the smooth running of most supply networks**. So blockchain solutions are useful if the supply network **crosses national boundaries**
- Blockchain technology is secure, with its **cryptographic techniques making it practically impossible for hackers to alter information**.

# CONTROL SUPPLY CHAIN DYNAMICS

An effective way to improve the performance of supply networks is to minimize the bullwhip effect. This usually means coordinating the activities of the operations in the chain in several ways.

- **ALIGN ALL THE CHANNELS**

Channel alignment of information and supply means the adjustment of scheduling, material movements, stock levels, pricing to bring all the operations into line with each other. Methods of planning and control decision-making are harmonized through the chain.

Allow an upstream supplier to manage the inventories of its downstream customer: vendor managed inventory (VMI)

- **INCREASE OPERATIONAL EFFICIENCY**

Operational efficiency is the efforts that each operation in the chain makes to reduce its own complexity, the cost of doing business with other operations in the chain, and the supply chain's throughput time. The effect of this is to simplify the whole chain.

A chain whose operations have high levels of operations performance is more predictable and has faster throughput, both of which help to minimize supply chain fluctuations.

- **IMPROVE FORECASTS**

Estimates of demand more accurate:

- reduce the inventory-holding requirements
- is required less excess capacity to buffer uncertainty
- achieve customer service-level targets

- **SHARE INFORMATION**

If information had been available and shared throughout the chain, it is unlikely that such wild fluctuations would have occurred.

An improvement is to make information on end-customer demand available to upstream operations with electronic point-of-sale (EPOS) systems.

# 8. CAPACITY MANAGEMENT

## 8.1 CAPACITY MANAGEMENT

- **CAPACITY MANAGEMENT** = activity of understanding the nature of demand and supply (capacity) and attempting to reduce mismatches between them.

In doing this, operations managers must reconcile 2 competing requirements: maintaining customer satisfaction by delivering products and services to customers reasonably quickly; and the need for operations to maintain efficiency by minimizing the costs of excess capacity.

- **DEMAND** = quantity of products or services that customers request from an operation or process at any point in time.
- **CAPACITY** = output that an operation or process can deliver over a defined unit of time.

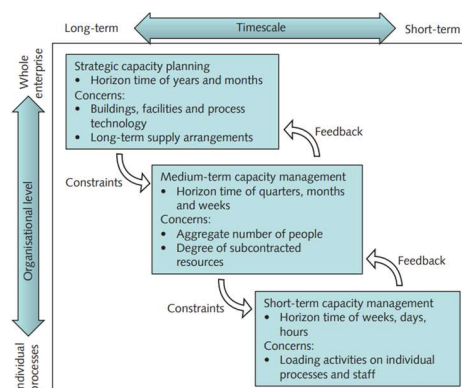
Capacity management is concerned with fluctuations in demand and supply, and so the process is a dynamic issue. A mismatch between demand and capacity can occur because demand fluctuates over time, capacity fluctuates over time, or both. It is worth noting that coping with mismatches between demand and capacity doesn't always mean that capacity should match demand.

### INTEGRATING ACROSS THE LEVELS

The central issue of capacity (deciding if and how to meet demand) can be considered at different organizational levels and over different time periods.

- In long term: total capacity, its distribution between sites and its location
- In medium term: extent to which capacity levels fluctuate, staffing levels and the degree of subcontracting
- In short term: which resources should be allocated to each set task and how tasks should be allocated to resources

Longer time frame capacity decisions impose constraints on shorter time frame decisions, while decisions made for shorter time frames provide important input into longer-term decisions.

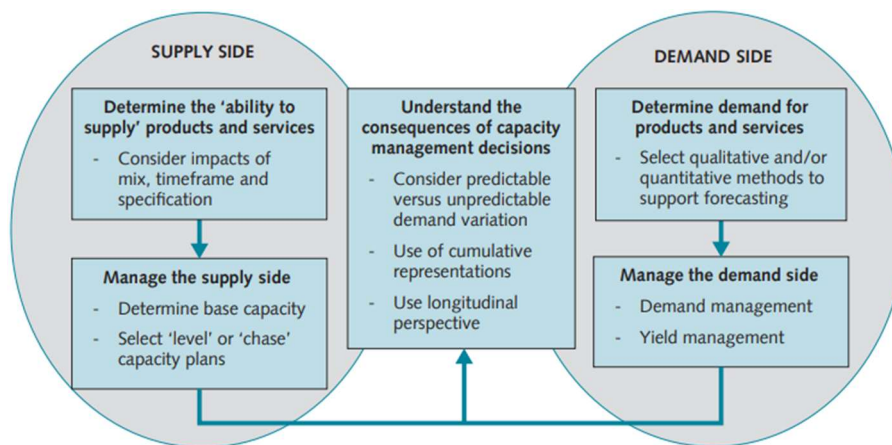


## FRAMEWORK FOR CAPACITY MANAGEMENT

Activities involved in capacity management:

- **DEMAND SIDE** *YIELD MANAGEMENT / DEMAND MANAGEMENT*
  - Measure demand for products and services
  - If and how to change demand patterns using demand or yield management
- **SUPPLY SIDE**
  - Measure the ability to supply products and services
  - Manage the supply side by determining the appropriate base level of capacity
  - Decide whether to keep this constant over time (level capacity plan) or make adjustments in line with changing demand patterns (chase capacity plan)
- **CONSEQUENCES**

Managers must understand the consequences of different capacity management decisions.



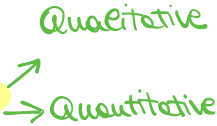
## 8.2 DEMAND SIDE

On the demand side of the capacity management framework, there are 2 tasks:

- understanding patterns of demand
- deciding if and how to influence these patterns: manage the demand side

Managers need to consider demand patterns over different time frames and various qualitative and quantitative methods are useful to support this analysis and in improving forecasts.

### DEMAND FORECASTING



There 1<sup>st</sup> step is to determine the demand for products and services: select from a range of methods to support more accurate forecasting. There are 2 approaches:

#### ▪ QUALITATIVE (Gale style)

Use perceptions, opinions

- Panel approach: a number of experts attempt to reach a consensus forecast
- Delphi method: experts make initial estimates and review an anonymous summary of estimates from other experts on the panel, after which their original estimates can be revised. It's a process that may be repeated several times
- Scenario planning: long-term forecasting, where experts develop and then discuss the operational implications for a range of future scenarios

#### ▪ QUANTITATIVE (Tommy style)

Predict demand mathematically

- Time series: use historical data to model future demand patterns
- Causal models: regression analysis or structured equation modelling to describe and evaluate the complex cause-effect relationship

Without some attempt to understand future demand fluctuations, it's not possible to plan effectively for future events, only to react to them. There are 3 ways to assess the usefulness of a demand forecast from an operations manager's perspective:

#### ▪ ACCURACY

Demand forecasts should be as accurate as possible.

- Firms must often decide output in advance, based on a forecast that might change before the demand occurs.
- Managers need to be aware that some forecast errors are particularly damaging. Example: if a process is operating at a level close to its max capacity, over-optimistic forecasts could result in unnecessary capital expenditure to increase capacity.

- **UNCERTAINTY** → give an indication of relative uncertainty → necessary to have reserve capacity  
Demand forecasts should give an indication of relative uncertainty.  
Operations managers need an understanding of when increased uncertainty makes it necessary to have reserve capacity.
- **USEFUL** → useful to capacity management  
Demand forecasts should be expressed in terms that are useful to capacity management.  
If forecasts are expressed only in monetary terms and give no indication of the demands that will be placed on an operation's capacity, they will need to be translated into realistic expectations of demand, expressed in the same units as the capacity.

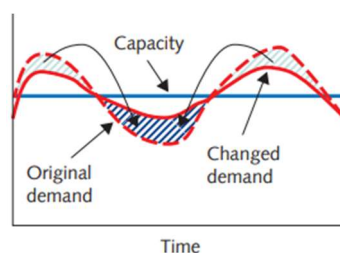
## CHANGE DEMAND PATTERNS

The second step on the demand side is to consider how to change demand patterns using demand management or yield management.

**DEMAND MANAGEMENT** → change demand to bring it closer to the available capacity. How?   
 CONSTRAINT  
 PRICE  
 SCHEDULING  
 SERVICE  
 ALTERNATIVE  
 PRODUCTS

Demand management involves changing the pattern of demand by stimulating off-peak demand or constraining peak demand. Demand management change demand to bring it closer to the available capacity.

- Constraining customer access: customers may only be allowed access to the operation's products or services at particular times
- Price differentials: adjusting price to influence demand by increasing prices during high demand periods and reducing prices during low demand periods
- Scheduling promotion: varying the degree of market stimulation through promotion and advertising in order to encourage demand during normally quieter periods
- Service differentials: adjusting service levels to reflect demand by allowing service to deteriorate in periods of high demand and increase in periods of low demand
- Creating alternative products and service: attempting to create alternative products or services to fill capacity in quiet periods



MARKET IS SEGMENTED

## YIELD MANAGEMENT MAXIMIZE REVENUES BY OPERATION WITH FIXED CAPACITIES WHEN

SERVICE CANNOT BE STORED  
SERVICE SOLD IN ADVANCE  
MARGINAL COST LOW.

Yield management is used to maximize revenues by operations that have relatively fixed capacities. They adopt a collection of methods to try to maximize the yield (profit) from their capacity.

It's useful when:

- the market can be fairly clearly segmented
- the service cannot be stored in any way
- the service is sold in advance
- the marginal cost of making a sale is relatively low

Example: overbooking capacity may be used to compensate for passengers who do not show up for the flight. However, if more passengers show up than they expect, the airline will have a number of upset passengers.

CAPACITY (SUPPLY): maximum level of value-added activity over a period of time that the process can achieve under normal operating conditions. Ability to supply products and services

CAPACITY / output of product that an op. can deliver over a unit of time

Plus a qu

## 8.3 SUPPLY SIDE: CURRENT CAPACITY

- CAPACITY = maximum level of value-added activity over a period of time that the process can achieve under normal operating conditions. It's the ability to supply products and services.

Every operation needs to know their capacity, because if they have too little, they cannot meet demand (bad for revenue) and if they have too much, they are paying for more capacity than they need (bad for costs).

Measuring capacity is a function of product/service mix, duration and product/service specification.

## CAPACITY DEPENDS ON

- **ACTIVITY MIX**

An operation's ability to supply is partly dependent on what it is being required to do: output depends on the mix of activities. As the mix changes, so does the effective capacity. Some of the problems caused by variation mix (output difficult to predict) can be partially overcome by using aggregated capacity measures: different products and services are bundled together in order to get a broad view of demand and capacity.

Example: a hospital may have a problem in measuring its capacity because the nature of its service may vary significantly.

- **DURATION**

Capacity depends on the duration over which output is required. Capacity is the output that an operation can deliver in a defined unit of time.

The level of activity and output that may be achievable over short periods of time is not the same as the capacity that is sustainable on a regular basis.

Example: staff need vacations, they can't work long hours continually, and so capacity is taken to be the level of activity (output) that can be sustained over a period of time

- **SPECIFICATION OF OUTPUT**

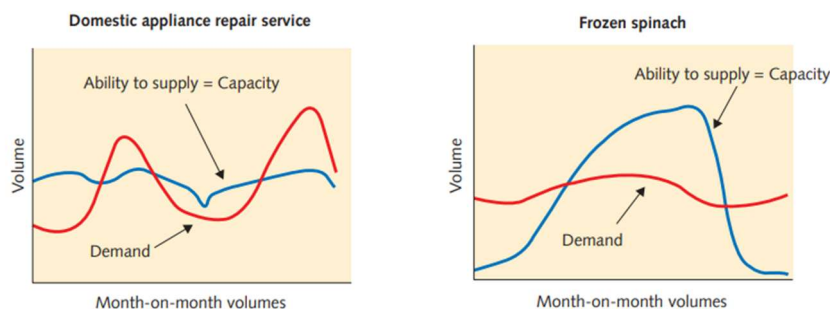
Some operations can increase their output by changing the specification of the product or service. The important task is to distinguish between the:

- must do elements of the service that should not be sacrificed
- nice to do parts of the service that can be omitted or delayed in order to increase capacity in the short term (busy period)

## CHANGES IN CAPACITY

While many operations are most concerned with dealing with changes in demand, some operations also have to cope with variation in capacity (if it's defined as the ability to supply).

Example: in a domestic appliance repair service, the capacity varies slightly because the field service operatives in the business prefer to take their vacations at particular times of the year. However, it's the demand side of the equation that exhibits more variability. The food manufacturer have the demand for this product relatively constant throughout the year but the capacity of the business varies significantly. During the growing and harvesting season capacity to supply is high.



# CAPACITY LEAKAGE

Even after allowing for all the difficulties inherent in measuring capacity, the **theoretical capacity** of a process is **rarely achieved in practice**. Different products or services may have different requirements, so people and machinery will have **delays** when **switching between tasks**. Maintenance will need to be performed on machines, while training will be required for employees.

This loss in capacity is sometimes called capacity leakage and one method of assessing this leakage is the OEE overall equipment effectiveness measure.

## OVERALL EQUIPMENT EFFECTIVENESS

$$OEE = a \times p \times q$$

### a) AVAILABILITY

Availability of the process can be lost through:

- Time losses: set up, changeover losses
- Breakdown failures (machine is being repaired, employees are being trained)

### p) PERFORMANCE

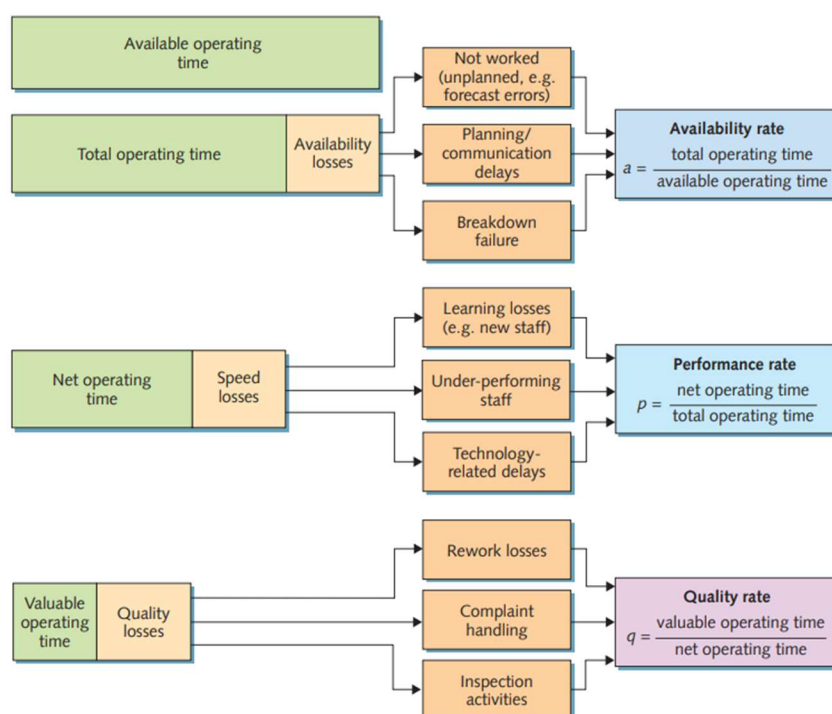
Some capacity is lost through speed losses when:

- Equipment is **idling** (temporarily waiting for work from another process)
- Equipment is **being run below its optimum work rate**
- Individuals are not working at an optimum rate

### q) QUALITY of product or service

Not everything processed by an operation will be error free

For processes to operate effectively, they need to achieve high levels of performance against all 3 dimensions: availability, performance (speed) and quality.



## 8.4 STRATEGIES FOR MANAGE SUPPLY SIDE

Having determined the capacity of the operation, managers must take 2 supply-side decision: determining the appropriate base level of capacity and then deciding to keep this constant over time or make adjustments in line with changing demand

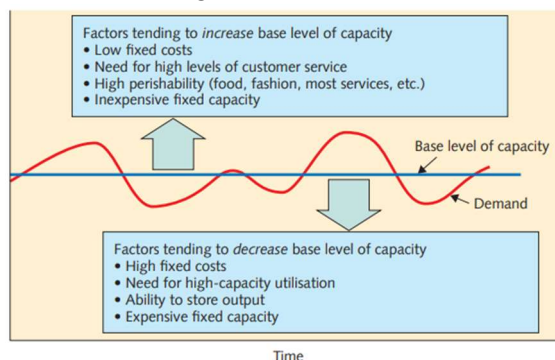
### SETTING BASE CAPACITY LEVEL

The starting point in managing supply side is to decide on a **base level of capacity** and then adjust is periodically up or down to reflect fluctuations in demand. The base capacity is influenced by:

- **OPERATION'S PERFORMANCE OBJECTIVES**

The base levels of capacity should reflect the relative **importance** of the **operation's performance objectives**.

Example: setting the base level of capacity high compared to average demand will result in relatively low levels of utilization and high costs.



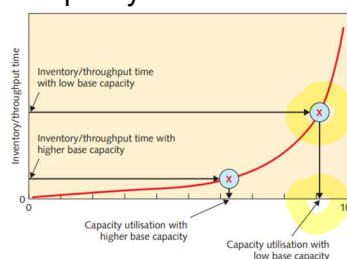
- **PERISHABILITY OF OUTPUTS**

When either **supply or demand is perishable**, base capacity will need to be set at a relatively **high level** because inputs or outputs from the operation cannot be stored for long periods.

Example: a factory that produces frozen fruit will need sufficient freezing and storage capacity to cope with the rate at which the fruit crop is being harvested during its harvesting season.

- **VARIABILITY IN DEMAND OR SUPPLY**

Variability, either in demand or capacity, will **reduce the ability of an operation to process its inputs**. The **greater the variability**, the more **extra capacity** will be **needed to compensate for the reduced utilization of available capacity**.



# DECISION FOR MANAGE SUPPLY SIDE

## ▪ LEVEL CAPACITY PLAN

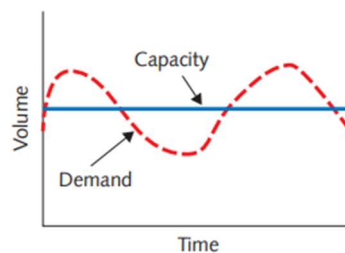
Keep the level of base capacity fixed throughout the planning period, regardless of the fluctuations in forecast demand. Although the demand fluctuates, the level of capacity remains constant.

It means that the same number of staff or machines operate the same processes and should therefore be capable of producing the same aggregate output in each period.

It requires that the operation absorb demand-capacity mismatches, usually through under or overutilization of its resources and the use of inventory.

When inventory is not possible (service operations), demand fluctuations are absorbed through underutilization of the operation's resources and/or failure to meet demand immediately. The more demand fluctuates, the higher the level of inventory or underutilization there is.

Setting capacity below the forecast peak demand level will reduce the degree of underutilization, but in the periods where demand is expected to exceed capacity, customer service will deteriorate.



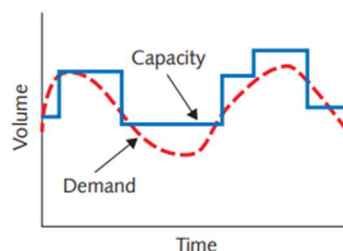
## ▪ CHASE (DEMAND) CAPACITY PLAN

It attempts to match demand patterns closely by varying levels of capacity.

Chase capacity plans are more difficult to achieve than level capacity plans, because different n° of staff, different working hours and different amounts of equipment may be necessary in each period.

Usually used by operations that are not able to store their output (manufacturers of perishable). Where output can be stored, the chase demand plan can be adopted in order to minimize or eliminate finished goods inventory, especially if the nature of future demand is relatively unpredictable.

It avoids the wasteful provision of excess staff that occurs with a level capacity plan, and yet should satisfy customer demand throughout the planned period.



There are other methods of adjusting capacity:

| <i>Method of adjusting capacity</i>                                                                                                                         | <i>Advantages</i>                                                                                                                                            | <i>Disadvantages</i>                                                                                                                                                                                                             |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Overtime</b> – staff working longer than their normal working times                                                                                      | Quickest and most convenient                                                                                                                                 | Extra payment normally necessary and agreement of staff to work; can reduce productivity over long periods                                                                                                                       |
| <b>Annualised hours</b> – staff contracted to work a set number of hours per year rather than a set number of hours per week                                | Without many of the costs associated with overtime the amount of staff time available to an organisation can be varied throughout the year to reflect demand | When very large and unexpected fluctuations in demand are possible, all the negotiated annual working time flexibility can be used up before the end of the year                                                                 |
| <b>Staff scheduling</b> – arranging working times (start and finish times) to vary the aggregate number of staff available for working at any time          | Staffing levels can be adjusted to meet demand without changing job responsibilities or hiring in new staff                                                  | Providing start and finish (shift) times that both satisfy staff's need for reasonable working times and shift patterns as well as providing appropriate capacity can be difficult                                               |
| <b>Varying the size of the workforce</b> – hiring extra staff during periods of high demand and laying them off as demand falls                             | Reduces basic labour costs quickly                                                                                                                           | Hiring costs and possible low productivity while new staff go through the learning curve; lay-offs may result in severance payments and possible loss of morale in the operation and loss of goodwill in the local labour market |
| <b>Using part-time staff</b> – recruiting staff who work for less than the normal working day (at the busiest periods)                                      | Good method of adjusting capacity to meet predictable short-term demand fluctuations                                                                         | Expensive if the fixed costs of employment for each employee (irrespective of how long he or she works) are high                                                                                                                 |
| <b>Skills flexibility</b> – designing flexibility in job design and job demarcation so that staff can transfer across from less-busy parts of the operation | Fast method of reacting to short-term demand fluctuations                                                                                                    | Investment in skills training needed and may cause some internal disruption                                                                                                                                                      |
| <b>Subcontracting/outsourcing</b> – buying, renting or sharing capacity or output from other operations                                                     | No disruption to the operation                                                                                                                               | Can be very expensive because of subcontractor's margin and subcontractor may not be as motivated to give the same service, or quality; also, a risk of leakage of knowledge                                                     |
| <b>Changing output rate</b> – expecting staff (and equipment) to work faster than normal                                                                    | No need to provide extra resources                                                                                                                           | Can only be used as a temporary measure, and even then can cause staff dissatisfaction, a reduction in the quality of work, or both                                                                                              |

## ZERO HOURS CONTRACTS

A zero-hours contract is one where an employer doesn't offer any guarantee of a specific number of hours of work for a worker. Neither is any person working under this type of contract under any obligation to accept those hours when they are offered. It can be seen as a method of varying effective capacity for chase demand approaches.

## 8.5 CONSEQUENCE OF CAPACITY MANAGEMENT

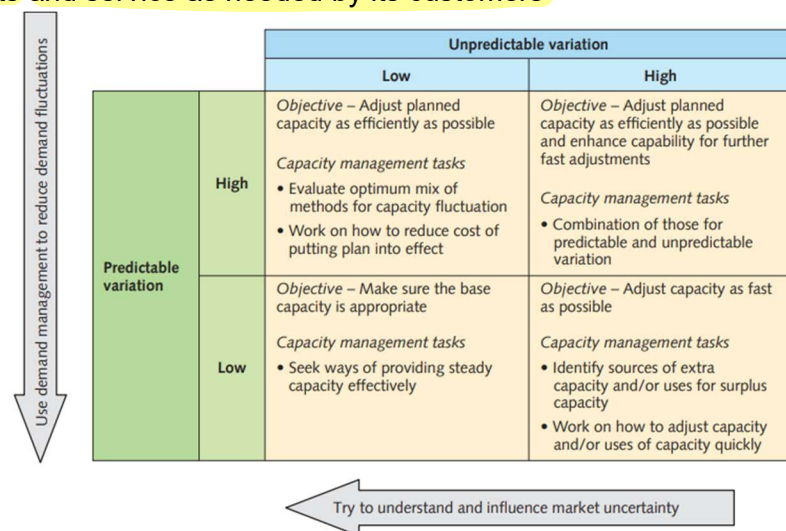
When making capacity management decisions, firms look to balance the need for customer service with the need to maintain operational efficiency. In practice most use a combination of demand-side and supply-side methods to achieve this balance.

There are 3 useful ways of assessing the mix of pure capacity management plans (demand management, level capacity or chase capacity) that may be most appropriate for an organization.

### • FACTORING IN PREDICTABLE VS UNPREDICTABLE DEMAND VARIATION

Managers should analyze the relative balance of predictable and unpredictable variation in demand, given its effect on objectives and tasks of capacity management.

- **Predictable variation in demand:** capacity adjustments may be needed but at least they can be planned in advance
- **Unpredictable variation in demand:** if an operation is to react to it at all, it must do so quickly; the change in capacity will have little effect on the operation's ability to deliver products and service as needed by its customers

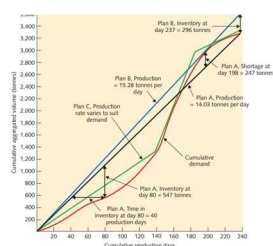


### • USING CUMULATIVE REPRESENTATIONS OF DEMAND AND CAPACITY

When an operation's output can be stored, a useful method for planning capacity are cumulative representations of demand and capacity.

For any capacity plan to meet demand as it occurs, its cumulative production line must always lie above its cumulative demand line. By cumulative representation we can:

- judge the adequacy of a plan
- gain an impression of the inventory implications by judging the area between the cumulative production and demand curves (amount of inventory carried over period)

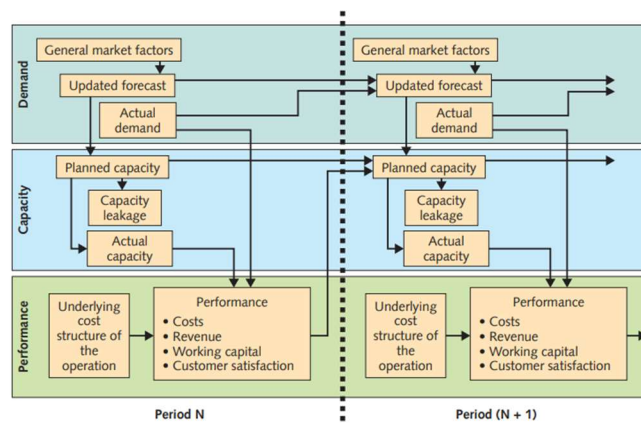


|                       | J   | F    | M    | A    | M    | J     | J     | A    | S     | O     | N    | D    |
|-----------------------|-----|------|------|------|------|-------|-------|------|-------|-------|------|------|
| Demand (tonnes/month) | 100 | 150  | 175  | 150  | 200  | 300   | 350   | 500  | 650   | 450   | 200  | 100  |
| Production days       | 20  | 18   | 21   | 21   | 22   | 22    | 21    | 10   | 21    | 22    | 21   | 18   |
| Demand (tonnes/day)   | 5   | 8.33 | 8.33 | 7.14 | 9.52 | 13.64 | 16.67 | 50   | 30.95 | 20.46 | 9.52 | 5.56 |
| Cumulative days       | 20  | 38   | 59   | 80   | 102  | 124   | 145   | 155  | 176   | 198   | 219  | 237  |
| Cumulative demand     | 100 | 250  | 425  | 575  | 775  | 1075  | 1425  | 1925 | 2575  | 3025  | 3225 | 3325 |

- **TAKING A LONGITUDINAL PERSPECTIVE**

Taking a longitudinal perspective to consider short and long-term outlooks helps align different time frames for capacity investments (or divestments).

Although planning capacity levels in advance, and even planning how to respond to unexpected changes in demand, is an important part of capacity management, it doesn't fully reflect the dynamic nature of the activity. Capacity management must react to actual demand and actual capacity as it occurs. Period by period, managers consider demand forecasts, understanding of current capacity and (if outputs can be stocked) how much inventory in the process have been carried forward from the previous period. Based on all this information, they make plans for the following period's capacity.



Finally, capacity management is a dynamic process with decisions reviewed period by period. It is essential that capacity decisions made in one period reflect the knowledge accumulated from experiences in previous periods. The success of capacity management is generally measured by some combination of costs, revenue, working capital and customer satisfaction.

|                              |                                   | Short-term outlook for volume                                                                                       |                                                                                                       |                                                                                                                          |
|------------------------------|-----------------------------------|---------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------|
|                              |                                   | Decreasing below current capacity                                                                                   | Level with current capacity                                                                           | Increasing above current capacity                                                                                        |
| Long-term outlook for volume | Decreasing below current capacity | Reduce capacity (semi) permanently: for example, reduce staffing levels; reduce supply agreements                   | Plan to reduce capacity (semi) permanently: for example, freeze recruitment; modify supply agreements | Increase capacity temporarily: for example, increase working hours and/or hire temporary staff; modify supply agreements |
|                              | Level with current capacity       | Reduce capacity temporarily: for example, reduce staff working hours; modify supply agreements                      | Maintain capacity at current level                                                                    | Increase capacity temporarily: for example, increase working hours and/or hire temporary staff; modify supply agreements |
|                              | Increasing above current capacity | Reduce capacity temporarily: for example, reduce staff working hours, but plan to recruit; modify supply agreements | Plan to increase capacity above current level; plan to increase supply agreements                     | Increase capacity (semi) permanently: for example, hire staff; increase supply agreements                                |

# 11. LEAN SYNCHRONIZATION

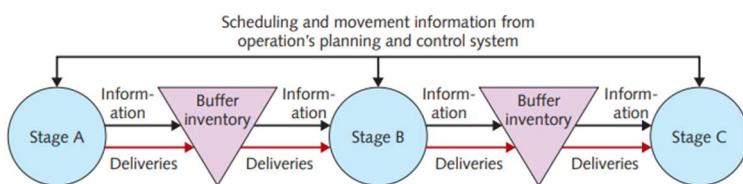
Lean synchronization uses lean (just-in-time) principles to supply perfect-quality products and services in synchronization with the demand for them, with zero waste and at low cost.

The focus of lean synchronization is to achieve a flow of products and services that delivers exactly what customers want, in exact quantities, exactly when needed, exactly where required and at the lowest possible cost. It is a concept that is almost synonymous with terms such as just-in-time (JIT) and lean operations principles.

## 11.1 BENEFITS OF LEAN

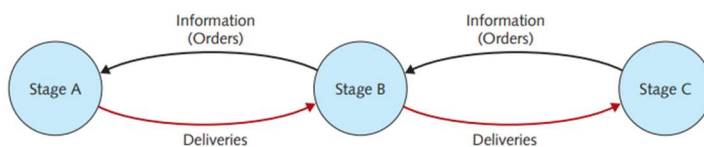
### TRADITIONAL

The traditional approach assumes that each stage in the process will place its output in an inventory or queue that buffers that stage from the next one downstream in the process. The next stage down will then take outputs from this buffer, process and pass them through to the next buffer inventory.



### LEAN SYNCHRONIZED FLOW

Products, customers or information are processed and then passed directly to the next stage just-in-time for them to be processed further. Now if A stops processing, stage B will notice immediately and also stage A's problem is quickly exposed to the whole process.



Benefits:

- Throughput time reduced
  - Negative effects of in-process inventory or queues are avoided.
- Inventories are seen as obscuring the problems that exist within processes and also inhibiting process improvement.

## RISK OF LOW INVENTORY

Reducing the inventories within processes, operations or supply networks bring risk because it increases **vulnerability** to large **disruptions** or changes in supply or demands.

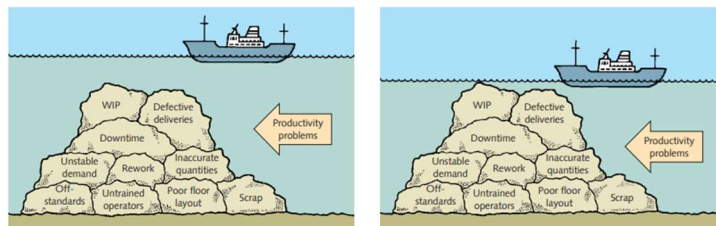
So in lean, where inventory reduction is a central tenet, is going to have to accept that its adoption inevitably incurs some degree of risk. The issue for many operations manager is whether that risk is worth it.

## RIVER AND ROCKS

The idea of obscuring the effects of inventory is often illustrated with the river and rocks analogy.

- Water = inventory in the operation
- Rocks = problems of the operation

Reducing the level of inventory or queues (water) allows operations management (the ship) to see the problems in the operation (the rocks) and work to reduce them.



# SYNCHRONIZATION, LEAN AND JIT

Lean synchronization aims to meet demand instantaneously, with perfect quality and no waste. It is a concept that is almost synonymous with terms such as:

- Just-in-time emphasizes the idea of producing items only when they are needed
- Lean stresses the elimination of waste
- Stockless production emphasizes the negative effects of in-process inventory or queues
- Toyota Production System encompasses all of these things

## 3 PERSPECTIVES ON LEAN

- Lean is a philosophy of how to run operations

It is a coherent set of principles that are founded on smoothing flow through processes by doing all the simple things well, gradually doing them better, meeting customer needs exactly

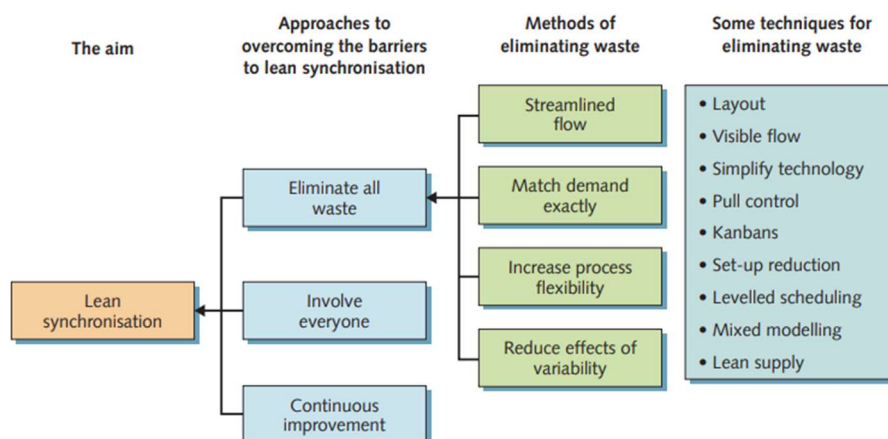
- Lean is a method of planning and controlling operation

How operations managers can manage this flow. Uncoordinated flow causes unpredictability, and unpredictability causes waste because people hold inventory, capacity or time to protect themselves against it

- Lean is a set of tools that improve operations performance

Collection of improvement tools and techniques that are the means for cutting out waste. Lean as a philosophy helped to shift the focus of operations management towards viewing improvement as its main purpose.

It is important to be clear on the distinction between the aim (lean synchronization), the approach to overcoming the barriers to achieving lean synchronization, the methods of eliminating waste, and the various techniques that can be used to help eliminate waste.



## **11.2 BARRIERS TO LEAN**

The aim of lean synchronization represents an ideal of **smooth**, uninterrupted flow without delay, waste or imperfection. The supply and demand between stages in each process, between processes in each operation and between operations in each supply network are all perfectly synchronized.

But there are barriers to achieving this ideal state:

- Failure to **eliminate waste** in all parts of the operation
- Failure to **involve all the people** within the operation in the shared task of smoothing flow and eliminating waste
- Failure to adopt continuous improvement principles

### **WASTE ELIMINATION BARRIER**

The most significant part of the lean philosophy is its focus on the elimination of all forms of waste. Waste can be defined as any activity that does not add value.

Identifying waste is the first step towards eliminating it.

Toyota has described 7 types of waste: waste from over-production, from waiting time, transportation waste, inventory waste, processing waste, waste of motion and waste from product defects.

- **IRREGULAR FLOW**

Perfect synchronization means smooth and even **flow through processes, operations and supply networks**. Barriers that prevent streamlined flow include:

- Waiting time: time when products, customers or information wait as queues
- Transport: **moving items or customers** around the operation does **not add value**. **Layout changes** that bring processes closer together, improvements in transport methods and workplace organization can all reduce transport waste.
- **Process inefficiencies**: some operations could be eliminated
- **Inventory**: all inventory should become a target for elimination
- Wasted motions: simplification of work is a source of reduction in the waste of motion

- **INEXACT SUPPLY**

Perfect synchronization means **supplying exactly what is wanted, exactly when it is needed**. Barriers to achieving an exact match between supply and demand include:

- Over production or under production
- Early or late delivery
- Inventory

- **INFLEXIBLE RESPONSE**

Customer needs can vary in terms of **what they want**, **how much they want** and **when they want it**. However, processes usually find it more convenient to change what they do relatively infrequently, because every change implies some kind of cost. Responding to customer demands exactly and instantaneously requires a high degree of **process flexibility**. Symptoms of inadequate flexibility include:

- **Large batches**
- **Delays between activities**: the longer the time (and cost) of changing over from one activity to another, the more difficult is to synchronize flow to match customer demand
- **More variation in activity mix than in customer demand**: if the mix of activities in different time periods varies more than customer demand varies, then some batching of activities must be taking place

- **VARIABILITY**

Synchronization implies **exact levels of quality**. If there is **variability in quality levels** then customers will **not** consider themselves as being **adequately supplied**. Symptoms include:

- Poor reliability of equipment: unreliable equipment usually indicates a lack of conformance in quality levels
- Defective products or services: waste caused by poor quality is significant

## **MUDA, MURA, MURI**

- MUDA = activities in a process that are wasteful because **they do not add value to the operation** or the **customer**. The main cause is likely to be **poorly communicated objectives** or the inefficient use of resources
- MURA = lack of consistency that results in periodic overloading of staff or equipment
- MURI = absurd or unreasonable. It is based on the idea that **unnecessary or unreasonable requirements put on a process** will **result in poor outcomes**. Appropriate skills, effective planning, accurate estimation of times and schedules will avoid this muri overloading waste

## **SHORT TERM PAIN FOR LONG TERM GAIN**

Focusing on lean synchronization can initially reduce resource utilization.

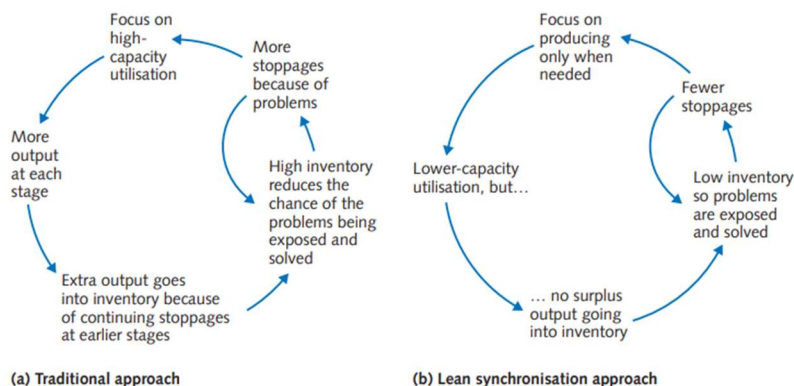
When stoppages occur in the traditional system, the buffers allow each stage to continue working and thus achieve high-capacity utilization. In a synchronized lean process, any stoppage will affect the rest of the system, causing stoppages throughout the operation.

# INVOLVEMENT BARRIER

An organizational culture that supports lean synchronization must place a very significant emphasis on involving everyone in the organization.

This approach to PEOPLE MANAGEMENT encourages (often requires) team-based problem solving, job enrichment, job rotation and multi-skilling. The intention is to encourage a high degree of personal responsibility, engagement and ownership of the job. Include:

- **Discipline**: work standards must be followed by everyone all the time
- **Flexibility**: expand responsibilities to the extent of people's capabilities
- **Equality**
- **Autonomy**: delegate responsibility to people involved in direct activities so that management's task becomes one of supporting processes
- **Development of personnel**: over time, the aim is to create more company members who can support the rigors of being competitive
- **Quality of working life**: security of employment, enjoyment
- **Creativity**: element of motivation, means as improving how the job is done and building the improvement into the process
- **Total people involvement**: staff take on more responsibility, continuous learning



## CONTINUOUS IMPROVEMENT

Lean synchronization objectives are expressed as ideals: to meet demand instantaneously, with perfect quality and no waste. While any operation's current performance may be far removed from such ideals, a fundamental lean belief is that it is possible to get closer to them over time.

Its aims are expressed in terms of ideals that individual organizations may never fully achieve, and so the emphasis must be on the way in which an organization moves closer to the ideal state: kaizen.



## **11.3 STREAMLINED FLOW**

The smooth flow of materials, information and people is central in lean synchronization.

Long process routes are wasteful, cause delay and inventory build-up. So the first contribution any operation can make to streamline flow is to **reconsider the basic layout of its processes**.

Reconfiguring the layout of a process to aid lean synchronization involves **moving it down the natural diagonal of process design** (chapter 5): moving **from functional layouts towards cell-based layouts**, or from **cell-based layouts towards line layouts**. Either way, it's necessary to move towards a layout that brings more systematization and control to the process flow.

### **EXAMINE THE SHAPE OF PROCESS FLOW**

Physically reconfiguring processes to **reduce distance travelled** and aid cooperation between staff can help to streamline flow.

Processes that have adopted the practice of curving line arrangements into U-shaped arrangements can have a number of advantages:

- Staff flexibility and balance: one person is enabled to do several activities
- Rework: it's easy to return faulty work to an earlier station
- Free flow: long straight lines interfere with cross travel in the rest of the operation
- Teamwork: the shape encourages a team feeling

### **ENSURE VISIBILITY**

Ensuring flow visibility helps to make improvement to flow easier. Sometimes this can involve **small-scale technologies** that can reduce **fluctuations in flow volume**.

Appropriate layout also includes the extent to which all **movement is transparent** to everyone within the process. High visibility of flow:

- Easier to **recognize potential improvements to flow**
- **Promotes quality** within the process because the more transparent the operation or process, the easier it is for all staff to share in its management and improvement
- Problems are more easily detectable
- Information becomes simple, fast and visual
- Performance measures displayed in the workplace (colored lights for indicate stoppages)
- Visual control systems (kanban and signal)

An important technique to ensure flow visibility is the use of VISUAL SIGNALS that can help everybody to see how work is progressing, indicate that a problem has occurred.

## USE SMALL – SCALE PROCESS TECHNOLOGY

Another tool to encourage smooth **streamlined flow** is using small-scale technologies. Advantages:

- Can process **different products** and services **simultaneously**
- The system is **more robust**: if one large machine breaks down, the whole system ceases to operate. If 1 of the 3 smaller machines breaks down, it is still operating at 2/3 effectiveness
- Easy to moved and so layout flexibility is enhanced

## EXAMINE ALL ELEMENTS OF THROUGHPUT TIME

The longer that items being processed are held in inventory, moved, checked that does not add value, the longer they take to progress through the process. So, looking what happens to items within a process is an excellent method of identifying sources of waste.

**VSM** = value stream mapping

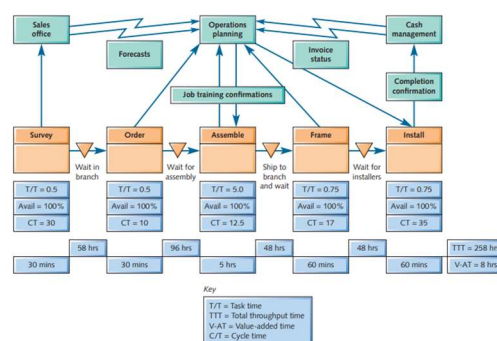
It's a simple but effective **approach** to **understanding the flow of material and information** as a product or service has value added as it progresses through a process, operation or supply chain. **It visually maps a product or service's production path** from start to finish. It's called value stream because it focuses on value-adding activities.

Similar to process mapping but:

- Higher level (5-10 activities)
- Wider scope (whole supply chain)
- Used to **identify where to focus future improvement activities**

A value stream perspective involves working on (and improving) the big picture, rather than just optimizing individual processes. It is a **4 step technique** that **identifies waste and suggests ways in which activities can be streamlined**:

1. Identifying the value stream (the process, operation or supply chain) to map
2. Physically mapping a process, **then above it mapping the information flow that enables the process to occur = current state map**
3. Problems are diagnosed and changes suggested = future state map
4. The changes are implemented.



## **11.4 SUPPLY AND DEMAND**

The aim of lean synchronization is to meet demand exactly (not too much or too little), and only when it is needed. Delivering only and exactly what and when is needed, smooths flow and exposes waste.

### **PULL**

Pull control principles are typically used to achieve the exact matching of supply and demand. Production is being triggered only by real customer demand.

This reduces clutter, speeds up throughput time and reduces confusion and inventories.

The essence of pull control is to let the downstream stage in a process, operation or supply network pull items through the system rather than have them pushed to them by the supplying stage.

Example: supermarkets usually replenish their shelves only when customers have taken sufficient products off the shelf. The movement of goods from the back office store to the shelf is triggered only by the empty shelf demand signal.

### **KANBAN**

The use of kanbans is one method of operationalizing pull control. Kanban is a card used by a customer stage to instruct its supplier stage to send more items.

The receipt of a kanban triggers the movement, production or supply of one unit or a standard container of units: kanbans are the only means by which movement, production or supply can be authorized.

Using kanbans, which are simple signalling devices, prevent the accumulation of excess inventory.

## **11.5 FLEXIBLE PROCESSES**

Responding exactly to demand only when it is needed often requires a degree of flexibility in processes, both to cope with unexpected demand and to allow processes to change between different activities without excessive delay.

This often means **reducing changeover times in technologies**. Flexible processes (often with flexible technologies) can enhance smooth and synchronized flow.

### **REDUCE SET UP TIMES**

For many technologies, increasing process flexibility means reducing setup times = **time taken to change over the process from one activity to the next**.

Setup reduction can be achieved by a variety of methods:

- Cutting out time taken to search for tools and equipment, the pre-preparation of tasks that delay changeovers and the constant practice of setup routines
- **Convert work** that was previously performed while the process was stopped (internal work) to work that is performed while the process is running (external work).

Example: in a manufacturing context, making this transition from internal to external work may include making equipment capable of performing all required tasks so that changeovers become a simple adjustment, and using simple devices to facilitate change of equipment

## 11.6 VARIABILITY MINIMIZED

Variability in processes (variability in product/service quality or quantity or timing) disrupts flow, in the means that acts against smooth flow and waste elimination. Variability includes quality variability and schedule variability.

### LEVEL SCHEDULES

Levelled scheduling (heijunka) means keeping the mix and the volume of flow between stages even over time.

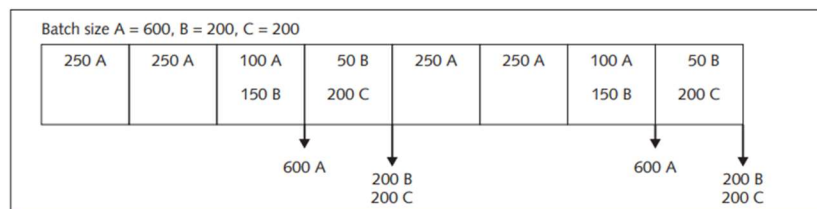
Example: instead of producing 500 parts in one batch, which would cover the needs for the next 3 months, levelled scheduling would require the process to make only one piece per hour regularly.

#### ▪ SCHEDULING IN LARGE BATCHES

If a mix of products or services was required in a time period, a batch size would be calculated for each product and the batches produced in some sequence.

Consequence:

- Relatively large amounts of inventory (or queues) accumulate within and between the units
- Most days are different from one another in terms of what they are expected to produce

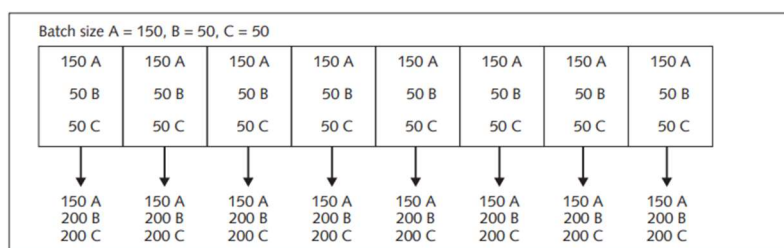


#### ▪ LEVELLED SCHEDULING

A batch of each product or service can now be completed in a single day, at the end of which the three batches are dispatched to their next stage.

Advantages:

- Reduce the overall level of work-in-progress in the operation
- Regularity and rhythm of the work in each unit: every day is the same
- Easy planning and control of each stage



## LEVEL DELIVERY SCHEDULES

A similar concept to levelled scheduling can be applied to many transportation processes.

### ▪ TRADITIONALLY

It may have dispatched a truck loaded with one particular product around all its stores so that each store received the appropriate amount of the product that would last it for one week.



### ▪ LEVELLED DELIVERY

Dispatch smaller quantities of all products in a single truck more frequently. Advantages:

- Each store would receive smaller deliveries more frequently
- Inventory levels would be lower
- The system could respond to trends in demand more readily because more deliveries means more opportunity to change the quantity delivered to a store.



## MIXED MODELLING

The principle of levelled scheduling can be taken further to give mixed modelling, that is a repeated mix of outputs. Doing the process as a sequence, it achieves relatively smooth flow.

Low Degree of levelling High

High Setup times Low

Low System flexibility High

|                        |                     |                       |
|------------------------|---------------------|-----------------------|
| Large batches, e.g.    | Small batches, e.g. | Mixed modelling, e.g. |
| 200 A<br>120 B<br>80 C | 5 A<br>3 B<br>2 C   | A A B A B C A B C A   |

## TPM

Total productive maintenance aims to eliminate the variability in operations processes caused by the effect of breakdowns. This is achieved by involving everyone in the search for maintenance improvements. Process owners are encouraged to assume ownership of their equipment and to undertake routine maintenance and simple repair tasks.

## **11.7 LEAN THROUGH SUPPLY NETWORK**

The same benefits and principles of lean synchronization that apply within operations can also apply across the whole supply networks.

Ensuring lean synchronization throughout an entire supply network is more difficult, because:

- The complexity of flow: interaction between whole operations is more complex than between individual stages within a process
- Supply networks are prone to the type of unexpected fluctuations that are easier to control within operations

Essentially, the principles and the techniques of lean synchronization are the same for a supply chain as they are for a process.

## **LEAN AND AGILE**

Lean means thin, having no superfluous fat. Lean attempts to eliminate waste and provide value to the customer throughout the entire supply chain. Synchronized, regular flow and low inventory.

### **AGILE**

Being responsive, quick moving, flexible, active and constantly ready to change. Responsiveness, flexibility and fast delivery.

Agility means rapid strategic and operational adaptation to large scale, unpredictable changes in the business environment. It focuses upon eliminating the barriers to quick response, be they organizational or technical. Agility is the capability of operating profitably in a competitive environment of continually changing customer opportunities.

It's more than simply organizational flexibility because it involves an organizational mastery of uncertainty and change, where people within the organization, their capacity to learn from change and their collective knowledge are regarded as the organization's greatest assets because they allow the operation to respond effectively to uncertainty and change. Continually inventing innovative business process solutions to new market demands becomes a key operations objective.

Agile philosophies require:

- Effective demand management to keep close to market needs
- Focus on customer relationship management
- Responsive supply coordination and visibility across the extended supply chain
- Continuous rescheduling and a quick response to changing demand
- Short planning cycles



# 12. IMPROVEMENT

## 12.1 IMPROVEMENT

Improvement means to **make something better**. It is increasingly seen as the **ultimate objective** for all operations and process management activity.

Improvement is the activity of closing the **gap between the current and the desired performance** of an operation or process.

It involves assessing the gaps between current and required performance, **balancing the use of continuous improvement** and **breakthrough improvement**, adopting appropriate improvement techniques and attempting to **ensure that the momentum of improvement does not fade over time**.

### WHY FOCUS ON IMPROVEMENT?

- There's a perception of increased competitive pressure: market are getting more competitive
- The nature of world trade is changing. Some **developing economies** are emerging as both **producers** and **consumers of products and services**. Politically inspired **trade conflicts**, alliances and separations have created operating uncertainty. All of which has increased the value of operational flexibility
- **New technology** has introduced opportunities to both **improve operations practice** and disrupt existing market
- Development of many new ideas and approaches to improving operations

### RED QUEEN EFFECT

A scientist established that no matter how long a family of animals had already existed, the probability that the family will become extinct is unaffected. In other words, the **struggle for survival never gets easier**, and so it can never relax.

In the business, improvements and innovations may be imitated or countered by competitors. The implication for operations improvement is clear: it's even more important, especially when competitors are actively improving their operations.

## 12.2 GAP CURRENT-REQUIRED PERFORMANCE

The gap between **how an operation or process is currently performing**, and how it wishes to perform, is the key driver of improvement. The wider the gap, the more importance to improvement.

This requires 2 sets of activities:

1. **Assess** the operation's and each process's **current performance**. This involves:
  - a. Decide what aspects of performance to measure
  - b. Which are the most important aspects of performance
  - c. What **detailed measures** should be used to assess each factors
2. Decide an appropriate level of target performance.

## ASSESSING CURRENT PERFORMANCE

Some kind of performance measurement is a prerequisite for judging whether an operation is good or bad (assessment of operations performance).

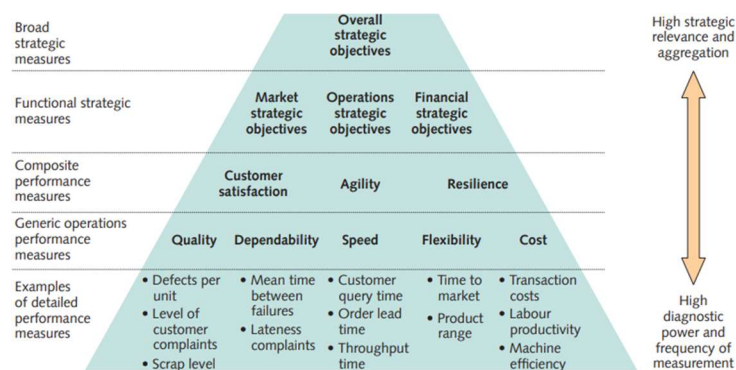
- PERFORMANCE MEASUREMENT = the attempt to **assess how well** (or not) **an operation is performing**
- PERFORMANCE MANAGEMENT = all the **actions taken to affect performance in some way**. It's what operations managers do to achieve the measures of performance that they are aiming for

### • WHICH FACTORS INCLUDE AS PERFORMANCE MEASUREMENT

A starting point for deciding which performance measures to adopt is to use the **5 generic performance objectives**: quality, speed, dependability, flexibility and cost.

These can be:

- **More detailed** measures: monitored more closely and more often. They provide a **limited view of an operation's performance**, but also a more descriptive and complete picture of what should be and what is happening within the operation
- More aggregated measures: **greater strategic relevance**, help to draw a picture of the overall performance (such as customer satisfaction, overall service level)



- **CHOOSE MOST IMPORTANT PERFORMANCE MEASUREMENT**

Choosing a useful performance measurement means achieve some balance between having a few key measures (simple but may not reflect the full range of organizational objectives) or having many detailed measures (complex and difficult to manage, but capable of conveying many nuances of performance).

A compromise is reached by a clear link between the operation's overall strategy, the most important performance indicators (KPIs) that reflect strategic objectives, and the bundle of detailed measures that are used to flesh out each key performance indicator.

- **WHICH DETAILED MEASURES TO USE**

The 5 performance objectives (quality, speed, dependability, flexibility and cost) are really composites of many smaller measures.

Example: operation's cost is derived from factors such as purchasing efficiency, productivity of the staff, efficiency with which it converts materials.

All of these measures individually give a partial view of the operation's cost performance, and many of them overlap in terms of the information they include. However, each of them does give a perspective on the cost performance of an operation that could be useful either to identify areas for improvement or to monitor the extent of improvement.

| Performance objective | Some typical measures                                                                                                                                                                                                          |
|-----------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Quality               | Number of defects per unit<br>Level of customer complaints<br>Scrap level<br>Warranty claims<br>Mean time between failures<br>Customer satisfaction score                                                                      |
| Speed                 | Customer query time<br>Order lead time<br>Frequency of delivery<br>Actual versus theoretical throughput time<br>Cycle time                                                                                                     |
| Dependability         | Percentage of orders delivered late<br>Average lateness of orders<br>Proportion of products in stock<br>Mean deviation from promised arrival<br>Schedule adherence                                                             |
| Flexibility           | Time needed to develop new products/services<br>Range of products/services<br>Machine changeover time<br>Average batch size<br>Time to increase activity rate<br>Average capacity/maximum capacity<br>Time to change schedules |
| Cost                  | Minimum delivery time/average delivery time<br>Variance against budget<br>Utilisation of resources<br>Labour productivity<br>Added value<br>Efficiency                                                                         |

## BALANCED SCORECARD APPROACH

Balanced scorecard approach is an approach to performance measurement. It attempts to provide the important information that is required to allow the overall strategy of an organization to be reflected adequately in specific performance measures.

Operations performance measures have been broadening in their scope. It's now widely accepted that the scope of measurement should, at some level, include external and internal, long-term and short-term, and soft as well as hard measures.

In addition to financial measures of performance, it also includes more operational measures of customer satisfaction, internal processes, innovation and other improvement activities.

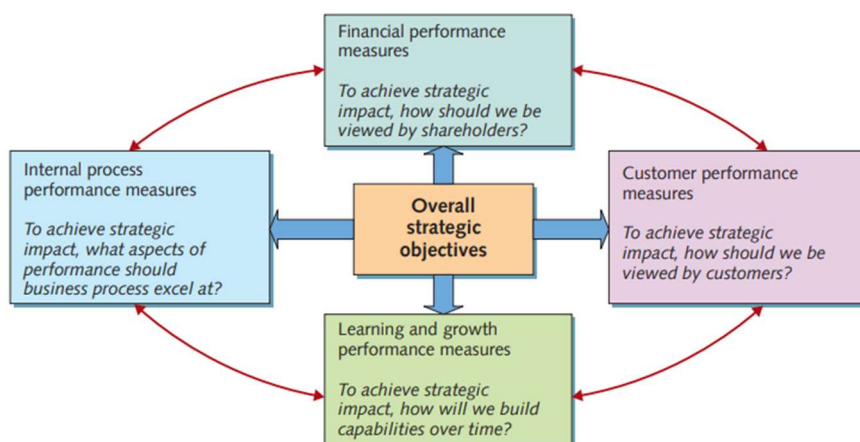
A balanced range of measures enables managers to address the following questions:

- How do we look to our shareholders? (financial perspective)
- What must we excel at? (internal process perspective)
- How do our customers see us? (the customer perspective)
- How can we continue to improve and build capabilities? (the learning and growth perspective)

The balanced scorecard attempts to bring together the elements that reflect a business's strategic position, including product or service quality measures, product and service development times, customer complaints, labor productivity, and so on.

Advantages:

- It presents an overall picture of the organization's performance in a single report
- Encourages companies to take decisions in the interests of the whole organization, rather than sub-optimizing around narrow measures
- But it's a complex process



# SETTING TARGET PERFORMANCE

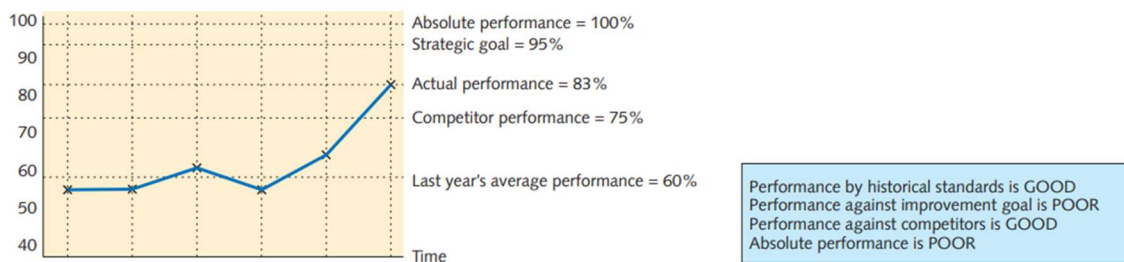
A performance measure means relatively little until it is compared against some kind of target. We know if it is better or worse than we were achieving previously, and if it is better or worse than other similar operations (especially competitors) are achieving.

Setting performance targets transforms performance measures into performance judgement.

There are different APPROACHES to setting targets:

- **Historically based targets:** targets that compare current against previous performance
- **Strategic targets:** targets set to reflect the level of performance that is regarded as appropriate to achieve strategic objectives
- **External performance-based targets:** targets set to reflect the performance that is achieved by similar, or competitor, external operations
- **Absolute performance targets:** based on theoretical upper limit of performance

One of the problems in setting targets is that different targets can give very different messages regarding the improvement being achieved. Different standards of comparison give different messages.



# BENCHMARKING

Benchmarking is the process of learning from others and involves comparing one's own performance or methods against other comparable operations.

It is more than setting performance targets, because it includes investigating other organizations' operations practice in order to derive ideas that could contribute to performance improvement.

It's based on the idea that:

- problems in managing processes are almost certainly shared by processes elsewhere
- there is probably another operation that has developed a better way of doing things.

## TYPES OF BENCHMARKING

There are different types of benchmarking, which are not necessarily mutually exclusive:

- Internal benchmarking  
Comparison between operations that are within the same total organization
- External benchmarking  
Comparison between operation and other operations that are part of a different organization
- Non-competitive benchmarking  
Comparison against external organizations that don't compete directly in the same markets
- Competitive benchmarking  
Direct comparison between competitors in the same, or similar, markets
- Performance benchmarking  
Comparison between the levels of achieved performance in different operations
- Practice benchmarking  
Comparison between an organization's operations practices, or way of doing things, and those adopted by another operation.

## IMPROVEMENT TOOL

Some businesses have failed to derive the maximum benefit from benchmarking:

- It's best practiced as a continuous process of comparison
- It doesn't provide solutions: it provides ideas and information that can lead to solutions
- It doesn't involve copying or imitating other operations: it's a process of learning and adapting
- It means devoting resources to the activity.

There are also some basic rules about how benchmarking can be organized:

- A prerequisite for benchmarking success is to understand thoroughly your own processes. Without this it is difficult to compare your processes against those of other companies
- Look at the information that is available in the public domain (published accounts, journals)
- Do not discard information because it seems irrelevant: small pieces of information only make sense in the context of other pieces of information that may emerge subsequently

## DEPLOYING EXTERNAL IDEAS

It's important to take into account the improvements that other companies are deploying, because they are a potentially huge source of innovation.

Whether they are competitors, suppliers, customers or simply other firms with similar challenges, firms in the wider external business environment can provide solutions to internal problems. Benchmarking is related to the idea of finding inspiration from outside the organization. But some commentators on innovation go further and argue that (legally) copying from outsiders can be an effective, if underused, approach to improvement.

## STRATEGIC TYPES OF IMITATORS

- Pioneer importer

An imitator that is the pioneer in another place (another country, industry or product market). Pioneer importer imitators must move relatively slowly, especially if the original innovator is unlikely to compete directly in the same market.

Example: what Ryanair did in Europe when it imported the South west model.

- The fast second

A rapid mover arriving quickly after an innovator or pioneer, but before it has had an opportunity to establish an unassailable lead, and before other potentially rival imitators take a large share of the market. This strategy basically lets the pioneer take much of the risk of innovation in the hope that the follower can learn from the pioneer's experience.

- The come from behind

A late entrant or adopter that has deliberately delayed adopting a new idea. When it does adopt the idea, it may rely on differentiating itself from the original pioneers.

# ASSESS GAP ACTUAL-TARGET PERFORMANCE

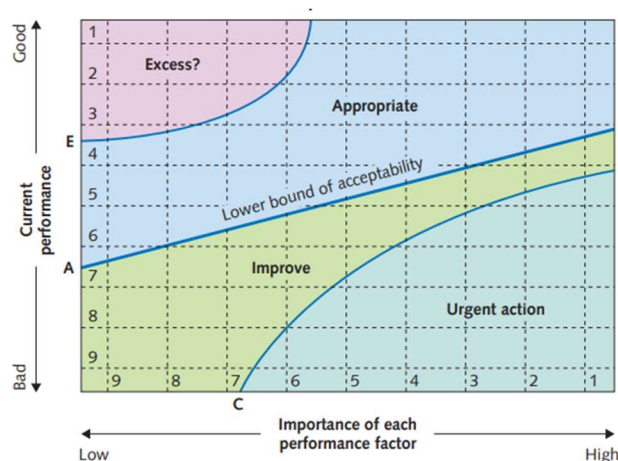
A comparison of actual and target performance should guide the relative priorities for improvement. Both the relative importance of the various performance measures, and their performance against target, need to be brought together in order to prioritize for improvement.

## IMPORTANCE – PERFORMANCE MATRIX

The importance–performance matrix positions each aspect of performance on a matrix according to its scores or ratings on how important each aspect of relative performance is, and what performance it is currently achieving.

Lower bound of acceptability: boundary between acceptable and unacceptable current performance.

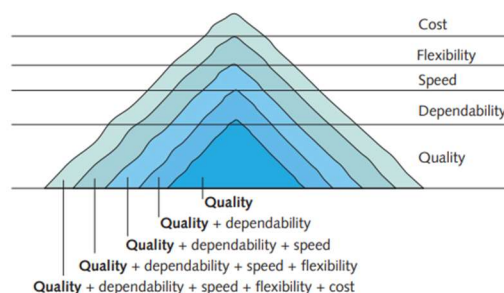
- Appropriate: satisfactory (above the lower bound of acceptability)
- Improve: candidates for improvement (below the lower bound of acceptability)
- Urgent action: these performance factors are important to customers but current performance is unacceptable. Need immediate improvement
- Excess: factors with high performing performance but not important



## SANDCONE THEORY

The sandcone theory give the generic best sequence of improvement. Building a stable sandcone needs a stable foundation of quality, dependability, speed, flexibility and cost.

Building up improvement is a cumulative process, not a sequential one. Moving on to the second priority for improvement doesn't mean dropping the first.



## 12.3 IMPROVEMENT PATH

Once the priority of improvement has been determined, has been determined, an operation must consider the approach or path it wishes to take to reach its improvement goals.

There are 2 improvement paths: breakthrough improvement and continuous improvement. Although they represent different philosophies of improvement, they are not mutually exclusive.

### BREAKTHROUGH IMPROVEMENT

Breakthrough improvement focuses on **major and dramatic changes** that are intended to result in dramatic increases in performance. Such improvements **can be expensive** and frequently **involve changes in the product/service or process technology**.

Example: total reorganization of an operation's process structure or the introduction of a fully integrated information system.

### RE-ENGINEERING

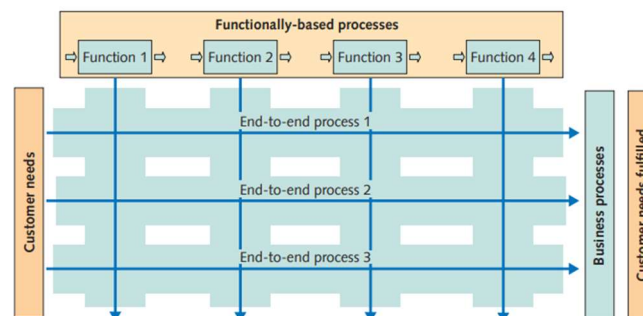
The business process re-engineering approach is typical of breakthrough improvement. It's a **blend of several ideas** such as fast throughput, waste elimination through process flow charting, **customer-focused operations**.

BPR is the fundamental rethinking and radical redesign of business processes to achieve dramatic improvements in critical contemporary measures of performance (cost, quality, service and speed).

Operations should be organized around the total process that adds value for customers.

Principles of BPR:

- Rethink business processes in a cross-functional manner that organizes work around the natural flow of information (or materials or customers)
- Strive for dramatic improvements in the performance by radically rethinking and redesigning the process
- The one that use the output from a process have to perform the process
- Put decision points where the work is performed. Do not separate those who do the work from those who control and manage the work



# CONTINUOUS IMPROVEMENT

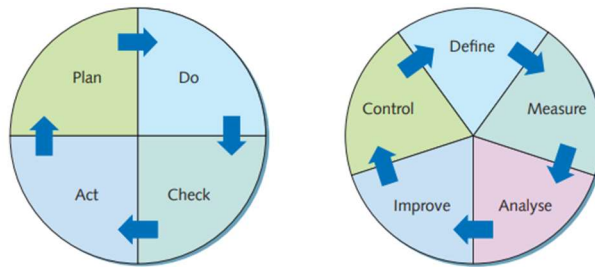
Continuous improvement focuses on small but never-ending improvements that become part of normal operations life. Its objective is to make improvement part of the culture of the organization.

Kaizen = continuous improvement involving everyone at every level of the organization.

It isn't the rate of improvement that is important; it is the momentum of improvement. It does not matter if successive improvements are small; what does matter is that every month some kind of improvement has actually taken place.

## IMPROVEMENT CYCLE MODELS

An important element of continuous improvement is the idea that improvement can be represented by a never-ending process of repeatedly questioning and re-questioning the detailed working of a process. CI implies a never-ending cycle of analysis and action.



## PDCA CYCLE

- P. Plan  
Collect and analyze data for formulate a plan that is intended to improve performance
- D. Do  
Implementation stage: plan is tried out in the operation
- C. Check  
The newly implemented solution is evaluated to see if it resulted in expected improvement
- A. Act  
The change is consolidated and standardized, if it has been successful

## DMAIC CYCLE

- D. Defining the problem
- M. Measurement: validating the problem, using data to refine the problem and measuring exactly what is happening
- A. Analysis: develop hypotheses about what are the root causes of the problem
- I. Improving: develop ideas, the solutions are tested and those solutions that seem to work are implemented, formalised and results measured
- C. Check: continually monitored and controlled

# DIFFERENCES

## BREAKTHROUGH IMPROVEMENT

- Radical philosophy
- It places a high value on creative solutions, and encourages free thinking and individualism

Exploration = exploration of new possibilities.

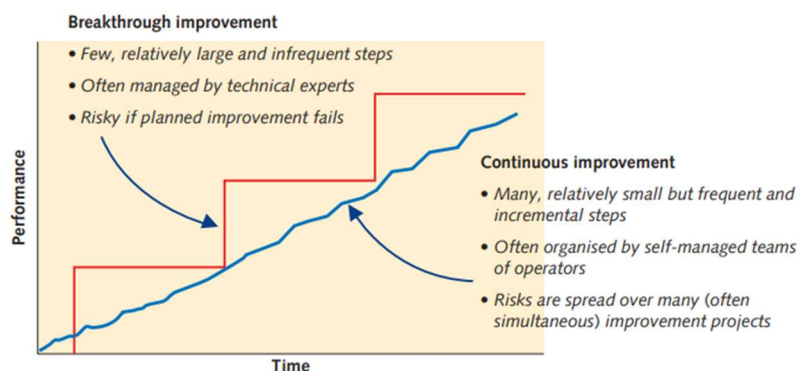
- Searching for and recognizing new mindsets and ways of doing things
- Involves experimentation, taking risks, simulation of possible consequences, flexibility and innovation
- Benefits are long-term and difficult to predict

## CONTINUOUS IMPROVEMENT

- No radical, it builds upon the wealth of accumulated experience within the operation
- Less ambitious in the short term
- It stresses adaptability, teamwork and attention to detail

Exploitation = enhancing processes that already exist in a firm.

- Focus on creating efficiencies
- Emphasis on tight control of the improvement process, standardizing processes, clear organizational structures and organizational stability
- Benefits from exploitation tend to be relatively immediate, incremental and predictable



# SIX SIGMA

Six Sigma is an approach to improvement that combines breakthrough and continuous philosophies.

Six Sigma is a comprehensive and flexible system for achieving, sustaining and maximizing business success. It's uniquely driven by close understanding of customer needs, data, statistical analysis, and diligent attention to managing, improving, and reinventing business processes.

Example: Motorola, the electronics company, decided that true customer satisfaction would only be achieved when its products were delivered when promised, with no defects, with no early-life failures and no excessive failure in service. To achieve this, it focused initially on removing manufacturing defects, but it soon realized that many problems were caused by latent defects, hidden within the design of its products. The only way to eliminate these defects was to make sure that design specifications were tight and its processes very capable. The specification range of any part of a product or service should be  $\pm 6$  the standard deviation of the process.

The Six Sigma approach also used the measure of defects per million opportunities: the number of defects that the process will produce if there were 1 million opportunities to do so.

## BELTS IN $6\sigma$

Improvement initiatives can be successful only if significant resources and training are devoted to their management.

- Master Black Belts
  - Experts in the use and implementation of Six Sigma tools and techniques
  - They guide improvement projects and coach and mentor Black Belts and Green Belts
  - Given their responsibilities, they are employed full time
- Black Belts
  - Take a direct hand in organizing improvement teams
  - They have undertaken a minimum of 20–25 days training and carried out at least one major improvement project
  - They develop their quantitative analytical skills and act as coaches for Green Belts
  - They are dedicated full time to improvement (1 Black Belt for every 100 employees)
- Green Belts
  - Work within improvement teams, as team leaders
  - They have undertaken around 10–15 days training
  - They are not full-time positions. They have normal day-to-day process responsibilities but are expected to spend at least 20% of their time on improvement projects

Devoting such large amounts of training and time to improvement is a significant investment. But if operated well, Six Sigma improvement projects can save far more than they cost.

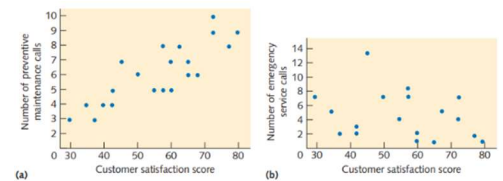
# 12.4 TECHNIQUE FOR FACILITATE IMPROVEMENT

Almost all techniques in operations management contribute directly or indirectly to the performance improvement. However, some more general techniques have become associated with improvement.

## SCATTER DIAGRAMS

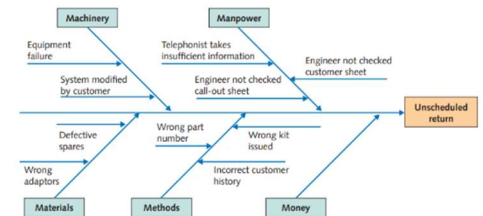
Provide a quick and simple method of identifying whether there is evidence of a connection between two sets of data.

They can be treated in a far more sophisticated manner by quantifying how strong is the relationship. But this type of graph only identifies the existence of a relationship.



## CAUSE EFFECT DIAGRAMS

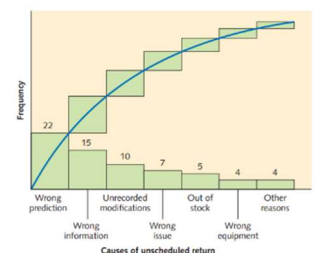
Helpful for search the root causes of problem. They do this by asking what, when, where, how and why questions, but also add some possible 'answers' in an explicit way.



## PARETO DIAGRAMS

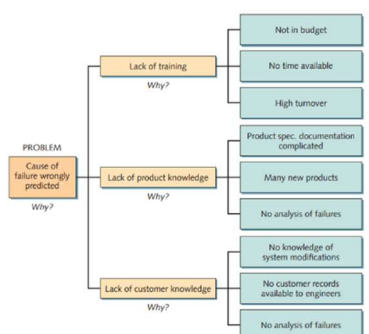
The purpose of Pareto diagram is to distinguish between the vital few issues and the trivial many. It involves arranging items of information on the types or causes of problems into their order of importance (measured by the frequency of occurrence).

It is based on the phenomenon of relatively few causes explaining the majority of effects.



## WHY-WHY ANALYSIS

Starts by stating the problem and asking why that problem has occurred. Once the major reasons for the problem occurring have been identified, each of the major reasons is taken in turn, and again the question is asked why those reasons have occurred, and so on.



## **12.5 PRESERVE IMPROVEMENT**

One of the biggest problems in improvement is to preserve improvement momentum over time. One factor that inhibits improvement becoming accepted as a regular part of operations activity is the emphasis on the fashionability of each new improvement approach.

### **VICTIM OF IMPROVEMENT FASHION**

Improvement has, to some extent, become a fashion industry with new ideas and concepts continually being introduced as offering a novel way to improve business performance.

The problem lies not with new improvement ideas, but rather with some managers becoming victims of the process, where some new idea will entirely displace whatever went before. Jumping from one fad to another will not only generate a backlash against any new idea, but also destroy the ability to accumulate the experience that comes from experimenting with each one.

### **AVOID BECOMING A VICTIM**

Avoiding becoming an improvement fashion victim is not easy. It requires:

- They must take responsibility for improvement as an ongoing activity, rather than becoming champions for only one specific improvement initiative
- They must take responsibility for understanding the underlying ideas behind each new concept: necessary to understand how they can be made to work properly
- They must take responsibility for understanding the antecedents to a new improvement idea, because it helps to understand it and judge it
- They must be prepared to adapt a new idea so that it makes sense within the context of their own operation
- They must take responsibility for the (often significant) education and learning effort
- They must avoid the over-exaggeration and hype that many new ideas attract

# MANAGING IMPROVEMENT PROCESS

There is no universal approach to improvement: any improvement process should reflect the uniqueness of each operation's characteristics.

Most new improvement ideas contain some worthwhile elements, but none will provide the ultimate answer. There must be management of the improvement process that absorb the best of each idea.

## IMPROVEMENT STRATEGY

Should an improvement strategy be defined? Yes, because without thinking through the overall goals of the improvement process it's difficult for any operation to know where it is going.

An improvement strategy should say about:

- Competitive priorities of the organization, and how the improvement process is expected to contribute to achieving increased strategic impact
- Roles and responsibilities of the parts of the organization in the improvement process
- Resources that will be available for the improvement process
- General approach to, and philosophy of, improvement in the organization

The strategy should not be too rigid because it could be inappropriate if the business's competitive circumstances change.

## TOP MANAGEMENT SUPPORT

What degree of top management support is required? A significant amount because without support, improvement cannot succeed. Top management support means that personnel must:

- understand the link between improvement and the business's overall strategic impact
- understand the practicalities of the improvement process and be able to communicate its principles and techniques to the rest of the organization
- be able to participate in the total problem-solving process to improve performance
- formulate and maintain a clear idea of the operation's improvement philosophy

## SUPERVISION OF IMPROVEMENT PROCESS

Should the improvement process be formally supervised? Some improvement processes fail because they develop an unwieldy bureaucracy to run them. But any process needs to be managed, so they need some kind of group to design, plan and control its efforts. Also there's a need for some sort of repository of knowledge to ensure that the learning and experience accumulated from the improvement process is not lost.

However, a goal for many improvement processes is to make them self-governing over time.

## **IMPROVEMENT GROUP**

- To what extent should improvement be group based?

The people who operate it are the only one that really know a process: they have access to informal and formal information networks that contain the way processes really work.

But working alone, individuals can't pool their experience or learn from one another, so improvement processes are almost always based on teams.

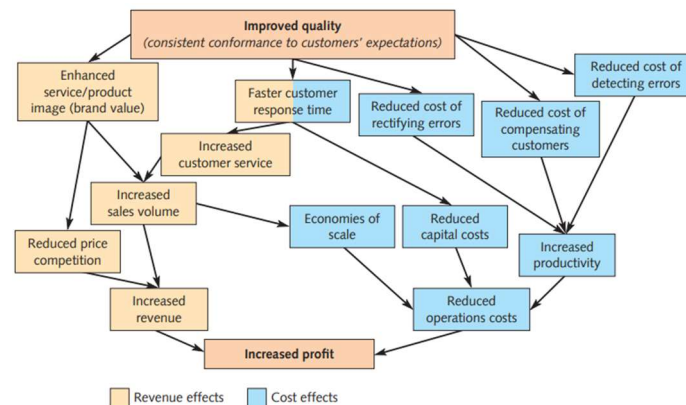
- How much training is required?

Training is necessary because it provides the necessary skills that will allow staff to solve process problems and implement improvements and also because it provides an understanding of the appropriate interpersonal, group and organizational skills that are needed to lubricate the improvement process.

# 13. QUALITY MANAGEMENT

Quality is important because it has such a significant impact on both the revenues and costs of any business. It can build a company's reputation, allowing it to preserve its margins and save the operational costs of poor quality and providing high quality brings significant competitive advantage.

Quality management is an approach to the way processes should be managed and improved, focused on the ability to produce and deliver the products and services that the market requires.



## 13.1 QUALITY MANAGEMENT

- QUALITY = consistent conformance to customers' expectations.  
Specification (what the product can do) + conformance (always does what it's supposed to)
- QOS = quality of service is used to describe how an operation serves its customers by combining what we call quality with some or all of speed, dependability and flexibility.  
Includes aspects of a service that are under the control of the operation creating the service.
- QOE = quality of experience is the overall acceptability of the service, as perceived subjectively by the end user. It's the degree of delight or annoyance of the user of an application or service resulting from the fulfilment of his expectations.  
Involves both the aspects governed by the operation and those that are a function of the individual customer, and the context in which the service is consumed.

Moving from quality, through quality of service to quality of experience shifts focus from the offering, through the provider, to the user.

- TOTAL QUALITY MANAGEMENT = act of managing all activities and tasks that must be completed in order to maintain a degree of perfection

# QUALITY AND ALL ORGANIZATION

Managing quality means ensuring that an understanding of its importance, and the way in which it can be improved, is spread throughout the business.

Quality management can be universally applied throughout a business and that also is the responsibility of all managers in the business. It's seen as applying to all parts of the organization. In fact, if quality management is to be effective, all processes must work properly together.

A total approach to quality should include every individual in the business. People are the source of both good and bad quality, and it is everyone's personal responsibility to get quality right. This applies also to those who are less directly involved in producing products and services.

If everyone has the ability to impair quality, they also have the ability to improve, because everyone is capable of improving the way in which they do their own jobs.

## INTERNAL CUSTOMER

It's recognition that every part of the organization is both an internal customer and, at the same time, an internal supplier for other parts of the organization. This means that errors in the service provided within an organization will eventually affect the product/service that reaches the external customer.

So, one of the best ways of satisfying external customers is to satisfy internal customers. Each process has a responsibility to manage its own internal customer-supplier relationships by clearly defining its own and its customers' exact requirements.

## SLA AND OLA

Service-level agreements can be used to operationalize the internal customer concept: they are a form of bringing a degree of formality.

- SLA = the service that an internal customer should receive
- OLA = how the parts of the operation that produce that service will go about producing the service in practice

## 13.2 QUALITY

Quality is consistent conformance to customers' expectations.

Quality needs to be understood from the customers' point of view because the quality is defined by the **customers' perceptions** and **expectations** (what he perceives). However individual customers' expectations may be different.

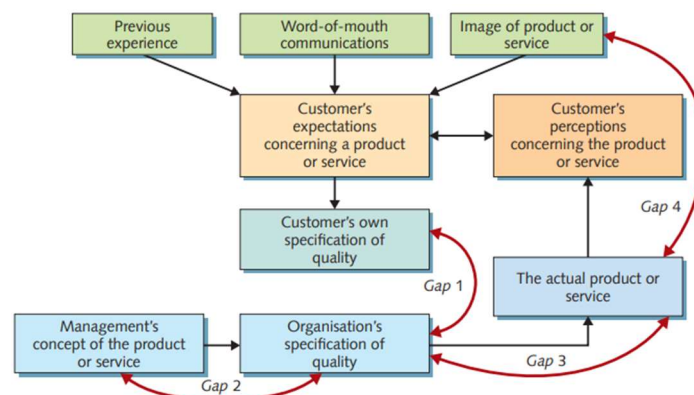
### CLOSING THE GAP

One way of understand quality is to use a **quality gap model**. This starts from the fundamental potential gap between **customers' expectations and perceptions** and **deconstructs the various influences on those perceptions and expectations**. Gaps between these factors can then be used to diagnose possible root causes of quality problems.

Gap:

- Expectations > perceptions → perceived quality is poor
- Expectations = perceptions → perceived quality is acceptable
- Expectations < perceptions → perceived quality is high

4 other gaps can explain a perceived quality gap between customers' perceptions and expectations.



#### 1) CUSTOMER'S SPECIFICATION – OPERATION'S SPECIFICATION

Perceived quality could be poor because there may be a mismatch between the organization's own internal quality specification and the specification expected by customer. Example: a car may be **designed to need servicing every 10,000km**, but the customer may expect 15,000km service intervals

#### 2) CONCEPT – SPECIFICATION

Perceived quality could be poor because there is a mismatch between the **product/service concept** and the way the organization has specified the quality of product/service internally. Example: the concept of a car might have been for inexpensive and energy-efficient, but the inclusion of a catalytic converter may have added costs and made it less energy efficient.

### 3) QUALITY SPECIFICATION – ACTUAL QUALITY

Perceived quality could be poor because there is a mismatch between the actual quality of the service or product provided by the operation and its internal quality specification. This may be the result of an inappropriate specification or of poorly trained personnel.

Example: if, despite an airline's policy of charging for drinks, some flight crews provide free drinks, they add unexpected costs to the airline and influence customers' expectations for the next flight, when they may be disappointed

### 4) ACTUAL QUALITY – COMMUNICATED IMAGE

Perceived quality could also be poor because there is a gap between the organization's external communications or market image and the actual quality of the service or product delivered to the customer. This may be the result of market positioning setting unachievable expectations in the minds of customers, or operations not providing the level of quality expected by the customer.

Example: the advertising campaign for an airline might show a cabin attendant offering to replace a customer's shirt on which food has been spilt, but this service may not be available.

## QUALITY CHARACTERISTICS

A further development is to define the quality characteristics of products or service in terms of their functionality, appearance, reliability, durability, recovery and contact.

Much of the quality of a product or service will have been specified in its design. It is the consequences of the design that are perceived by customers. These consequences of the design are called quality characteristics.

| Quality characteristics                                                                                                                                                      | Car                                                                     | Flight                                                                                   |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------|------------------------------------------------------------------------------------------|
| <b>Functionality</b> – how well the product or service does its job, including its performance and features                                                                  | Speed, acceleration, fuel consumption, ride quality, road holding, etc. | Safety and duration of journey, onboard meals and drinks, car and hotel booking services |
| <b>Appearance</b> – the sensory characteristics of the product or service: its aesthetic appeal, look, feel, sound and smell                                                 | Aesthetics, shape, finish, door gaps, etc.                              | Décor and cleanliness of aircraft, look of waiting lounges and crew                      |
| <b>Reliability</b> – the consistency of the product's or service's performance over time, or the average time for which it performs within its tolerated band of performance | Mean time to failure                                                    | Keeping to the published flight times                                                    |
| <b>Durability</b> – the total useful life of the product or service assuming occasional repair or modification                                                               | Useful life (with repair)                                               | Keeping up with trends in the industry                                                   |
| <b>Recovery</b> – the ease with which problems with the product or service can be rectified or resolved                                                                      | Ease of repair                                                          | Resolution of service failures                                                           |
| <b>Contact</b> – the nature of the person-to-person contact that might take place; could include the courtesy, empathy, sensitivity and knowledge of contact staff           | Knowledge and courtesy of sales and service staff                       | Knowledge, courtesy and sensitivity of airline staff                                     |

## 13.3 MEASURE QUALITY

Without measuring quality, it is difficult to control it. Some quality characteristics are easy to measure, other more difficult-to-measure quality characteristics (appearance) need to be deconstructed into their constituent elements (color, n° of visible scratches), all of which are capable of being measured in a relatively objective manner. However, deconstructing can result in loss of meaning.

The measures used to describe quality characteristics are 2 types:

- Variables = can be measured on a continuously variable scale (length)
- Attribute = a binary “acceptable or not acceptable” judgement (right or wrong)

## MEASURING THE COSTS OF QUALITY

One approach to measuring quality is to express all quality-related issues in cost terms.

Quality costs are usually categorized into 4 categories: prevention costs, appraisal costs, internal failure costs and external failure costs.

| Category of quality-related cost                                                                                                                                                     | Factors in these categories                                                                                                                                                                                                                                                                                                                                                                   |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Prevention costs</b> – those costs incurred in trying to prevent problems, failures and errors from occurring in the first place                                                  | <ul style="list-style-type: none"><li>• Identifying potential problems and putting the process right before poor quality occurs</li><li>• Designing and improving the design of products and services and processes to reduce quality problems</li><li>• Training and development of personnel in the best way to perform their jobs</li><li>• Process control</li></ul>                      |
| <b>Appraisal costs</b> – those costs associated with controlling quality to check to see if problems or errors have occurred during and after the creation of the product or service | <ul style="list-style-type: none"><li>• The setting up of statistical acceptance sampling plans</li><li>• The time and effort required to inspect inputs, processes and outputs</li><li>• Obtaining processing inspection and test data</li><li>• Investigating quality problems and providing quality reports</li><li>• Conducting customer surveys and quality audits</li></ul>             |
| <b>Internal failure costs</b> – failure costs that are associated with errors dealt with inside the operation                                                                        | <ul style="list-style-type: none"><li>• The cost of scrapped parts and materials</li><li>• Reworked parts and materials</li><li>• The lost production time as a result of coping with errors</li><li>• Lack of concentration due to time spent on troubleshooting rather than improvement</li></ul>                                                                                           |
| <b>External failure costs</b> – failure costs that are associated with errors being experienced by customers                                                                         | <ul style="list-style-type: none"><li>• Loss of customer goodwill affecting future business</li><li>• Aggrieved customers who may take up time</li><li>• Litigation (or payments to avoid litigation)</li><li>• Guarantee and warranty costs</li><li>• The cost to the company of providing excessive capability (too much coffee in the pack, or too much information to a client)</li></ul> |

## OPTIMUM QUALITY EFFORT

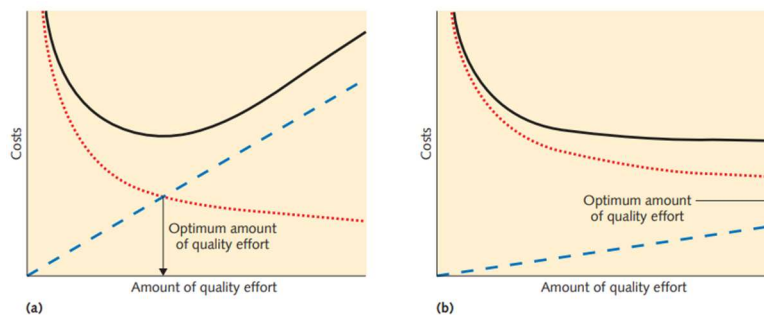
- TRADITIONAL

It was assumed that failure costs reduce as the money spent on appraisal and prevention increases. So there must be an optimum amount of quality effort to be applied in any situation that minimize the total costs of quality.

- MODERN

The optimum-quality-effort approach has been challenged:

- Why should any operation accept the inevitability of errors? Some occupations seem to be able to accept a zero-defect standard (no one accepts pilots crashing)
- Failure costs are generally underestimated. They include cost of reworking defective products, reserving customers but they must include also all the management time wasted in organizing rework, the loss of concentration
- Prevention costs are inevitably high
- By accepting compromise, this approach does little to challenge operations managers and staff to find ways of improving quality

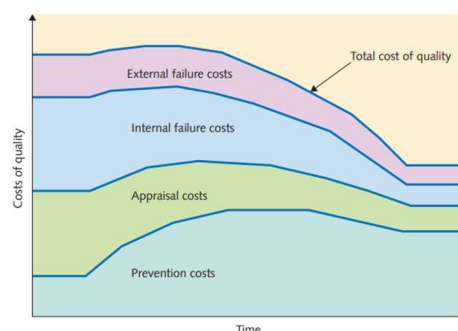


## TQM AND QUALITY COST MODEL

TQM rejected the optimum-quality level concept.

It concentrated on how to reduce all known and unknown failure costs. There is the shift of the emphasis from a reactive approach to quality (waiting for errors to happen, then screening them out), to a more proactive getting it right first time approach (doing something before errors happen).

This has a significant positive effect on internal failure costs, followed by reductions in both external failure costs and, once confidence has been firmly established, in appraisal costs.



## 13.4 CONTROLL QUALITY

After quality has been defined and measured, processes will need to check that their quality conforms to whatever quality standards are deemed appropriate.

### CHECK EVERYTHING OR SAMPLING?

Controlling doesn't necessarily mean checking everything; sampling may be more appropriate.

Checking everything:

- It could be dangerous (a blood test is just a small sample)
- Might destroy the product or interfere with the service
- May be too costly
- Even 100% checking won't always guarantee that all defects or problems will be identified, because inspections can miss mistakes
- Sometimes is necessary if the product is so critical that its failure would result in death or injury (aircrafts, healthcare services)

### TYPE I-II ERRORS

Checking quality by sampling, although requiring less time than checking everything, does have its own problems.

- Type I  
Occur when a decision was made to do something and the situation did not warrant it
- Type II  
Occur when nothing was done, yet a decision to do something should have been taken, as the situation did indeed warrant it.

| <i>Road conditions</i>          |                                      |                                            |
|---------------------------------|--------------------------------------|--------------------------------------------|
| <i>Decision</i>                 | <i>Safe (action was appropriate)</i> | <i>Unsafe (action was not appropriate)</i> |
| <i>Cross (take some action)</i> | Correct decision                     | <i>Type I error</i>                        |
| <i>Wait (take no action)</i>    | <i>Type II error</i>                 | Correct decision                           |

# STATISTICAL PROCESS CONTROL

The most common method of checking the quality of a sampled product or service is SPC. This technique attempts to **reduce the variation in quality performance**, so as to enhance process knowledge and also is used to **detect deviations outside the normal range of quality variation**.

Statistical process control is concerned with:

- Sampling the process during the production of goods or the delivery of service
- Based on this sample, decisions are made as to **whether the process is in control**: it is operating as it should be
- If there seems to be a problem with the process, then it can be stopped and the problem identified and rectified.

Example: an airport may regularly ask a sample of customers if the cleanliness of its restaurants is satisfactory. If an unacceptable n° is unhappy, they have to consider improving cleaning procedures.

## CONTROL CHARTS

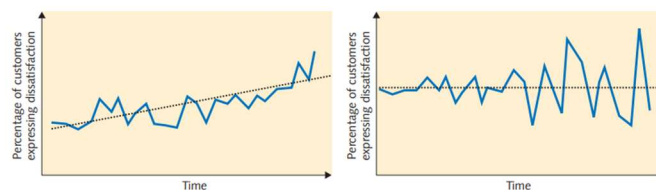
The value of SPC is not just to make checks of a single sample but to **monitor the results of many samples over a period of time**. It does this by using control charts.

Control charts **record some aspect of quality** (or performance generally) over time to see if the process seems to be performing as it should (in control), or not (out of control). If the process seem to be going out of control, steps can be taken before there is a major problem.

Any aspect of the performance of a process is measured over time and may **show trends in average performance** and/or **changes in the variation of performance over time**.

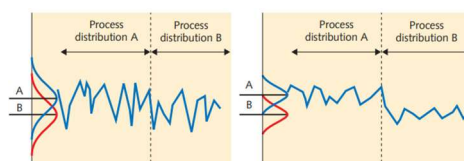
Looking for trends is an important use of control charts:

- If the trend suggests the process is getting steadily worse, then it will be worth investigating the process
- If the trend is steadily improving, it may still be worthy of investigation to try to identify what is happening that is making the process better



## VARIATION

Variation is a problem because it masks any changes in process behavior. High levels of variation reduce the ability to detect changes in process performance and so it will be difficult to intervene.



## PROCESS CONTROL, LEARNING AND KNOWLEDGE

In recent years, the role of SPC is seen not just as a convenient method of keeping processes in control, but also as an activity that is fundamental to the acquisition of competitive advantage.

Traditionally it was seen as one of the most operational and immediate operations management techniques, but now is seen as contributing to an operation's strategic capabilities.

- SPC is based on the idea that process variability indicates if a process is in control or not
- Processes are brought into control and improved by progressively reducing process variability. This involves eliminating the assignable causes of variation
- It's important to learn about the process for eliminate assignable causes of variation
- This learning means that process knowledge is enhanced, which in turn means that operations managers are able to predict how the process will perform under different circumstances. It also means that the process has a greater capability to carry out its tasks at a higher level of performance
- This increased process capability is particularly difficult for competitors to copy: process capability leads to strategic advantage

Statistical process control (SPC) gives the potential to enhance process knowledge.

## **13.5 IMPROVEMENT AND QUALITY MANAGEMENT**

Quality improvements often may not sustain because there is no set of systems and procedures to support and embed them within the operation's day-to-day routines.

- **QUALITY SYSTEM** = the organizational structure for implementing quality management. It should cover all facets of a business's operations and processes, and define the responsibilities, procedures and processes.

### **ISO 9000**

It's the best-known quality system, adopted now throughout the world. It's a set of worldwide standards that establishes requirements for companies' quality management systems. It provides a framework for quality assurance.

It stresses 4 other principles:

- Quality management should be customer focused and improvement against customer standards should be documented
- Quality performance should be measured and analyzed, and relate to products and services themselves, the processes that created them and customer satisfaction
- Quality management should be improvement driven
- Top management must demonstrate their commitment to maintaining and continually improving management systems

### **EQM**

Of the other systems, one of the most widely known is the EFQM Excellence Model. Once known only as the basis of the European Quality Award, it is now used extensively as a self-assessment tool that allows organizations to assess their own quality systems.

The model is an overarching framework for developing sustainable quality excellence. It is based on the idea that it is important to understand the cause-and-effect relationships between what an organization does (the enablers) and the results it achieves.

Results of EQM are assessed using 4 criteria:

- Customer results: meeting or exceeding the needs and expectations of customers
- People results: meeting or exceeding the needs and expectations of employees
- Society results: achieving and sustaining results that meet or exceed the needs and expectations of the relevant stakeholders within society
- Business results: achieving and sustaining results that meet or exceed the needs and expectations of business stakeholders.

## 5 ENABLERS

- 1) Leadership  
Looks to the future, acts as a role model for values and ethics, inspires trust, is flexible, enables anticipation and so can react in a timely manner
- 2) Strategy  
Implements the organization's mission and vision by developing and deploying a stakeholder-focused strategy
- 3) People  
Organizations should value their people, creating a culture, develops the capabilities, promotes fairness and equality, cares for, communicates
- 4) Partnership and resources  
Organizations should plan and manage external partnerships, suppliers and internal resources in order to support strategy and policies, and the effective operation of processes
- 5) Processes, products and services  
Organizations should design, manage and improve processes to ensure value for customers and other stakeholders.

